THE CENTRALIZED COLLEGE STUDY MATERIAL AND PREVIOUS QUESTION PAPER PORTAL

A Full-Stack Responsive Academic Repository with Role-Based Moderation & Offline Local Area Network (LAN) Access

Project Abstract:
The Centralized College Study Material and Previous Question Paper Portal is a web based application that allows anyone to access, upload, share, and download academic resources. Users can easily find and contribute study materials, lecture notes, assignments, and previous years' question papers. The system provides a centralized platform where resources can be organized by department, semester, and subject. It includes file upload functionality, database management, and an admin panel to manage users and uploaded content. The responsive website supports both online access and offline access through a local network. The main objective of this project is to encourage knowledge sharing, simplify access to educational resources, save time, and make learning materials available to everyone.

Keywords: Study Materials, Previous Question Papers, File Sharing, Upload and Download, Database, Online and Offline Access, Progressive Web App (PWA).

Domain: Educational Tech / DBMS / Networks	**Tech Stack:** Python 3, Flask 3, SQLite WAL, Tailwind CSS
Architecture: MVC / Blueprint Modular	**Network Engine:** Socket LAN discovery & PWA Service Worker
Database: Relational SQLite with Foreign Keys	**Submission Status:** Complete Codebase & Verified Test Suite

Pre-Configured Test Credentials:

Role	Username	Password	Email	Permissions / Access Level
Admin	admin	admin123	admin@college.edu	Full portal control, moderation queue, user management, academic CI
Faculty	prof_sharma	faculty123	sharma@college.edu	Direct document publishing, departmental moderation, curriculum over
Student	student1	student123	aarav@college.edu	Browse, download, bookmark, rate, and submit uploads (sent to review

Document Outline & Code Index:

Section 1: Project Overview & Architectural Specifications
Section 2: Database Schema & Relational Tables (departments, subjects, users, resources, ratings, bookmarks, logs)
Section 3: Core Backend Application Files (app.py, config.py, database.py, helpers.py, seed_data.py, test_app.py)

Section 4: Blueprint Controllers (routes/auth.py, routes/main.py, routes/upload.py, routes/admin.py, routes/api.py)
Section 5: Frontend Templates (base.html, index.html, browse.html, resource_detail.html, upload.html, my_uploads.html, bookmarks.html, offline.html, error.html, auth/, admin/)
Section 6: Progressive Web App & Client Scripts (static/js/main.js, static/js/sw.js, manifest.json, custom.css)
Section 7: Configuration & Setup Scripts (start.bat, requirements.txt, test verification suite)

1. SYSTEM ARCHITECTURE & DATABASE SPECIFICATIONS

The application is architected around a modular MVC (Model-View-Controller) structure built on Flask 3.1. The backend handles authentication, content ingestion, MIME verification, full-text database querying, and administrative workflows. The database uses SQLite with Write-Ahead Logging (WAL) and enforced Foreign Keys to ensure high-performance concurrent reads without database locking. To satisfy the offline requirement, the web server binds on **0.0.0.0:5000** and dynamically queries the local network interface using Python sockets. A Progressive Web App (PWA) Service Worker intercepts GET requests, caching critical academic pages and static assets for seamless operation without internet data.

Table Name	Columns & Datatypes	Constraints / Relationships	Functional Purpose
departments	id (INT PK), name (TEXT), code (TEXT), description, icon	UNIQUE(code, name)	Academic branches (CSE, IT, AI-DS, ECE, MECH, etc.)
subjects	id (INT PK), name (TEXT), code (TEXT), department_id (INT FK), semester (INT)	FOREIGN KEY -> departments.id, UNIQUE(code, dept)	Course directory mapped to Dept and Semesters 1-8
users	id (INT PK), username (TEXT), email (TEXT), password_hash (TEXT), role (TEXT)	UNIQUE(username), UNIQUE(email), role IN ('student', 'admin')	Authentication, session management, and role-based acces
resources	id (INT PK), title, description, resource_type, dept_id, semester, subject_id, exam_type, file_path, original_filename, file_size, mime_type, uploaded_by, upload_date, is_approved, download_count, rejection_reason	FOREIGN KEYS -> departments.id, subjects.id, users.id; CHECK(resource_type IN ('Question Paper', 'Notes', 'Lab Manuals', a	Core file storage and metadata tracking for all uploaded documents
resource_ratings	id (INT PK), resource_id (INT FK), user_id (INT FK), rating (1-5), review	UNIQUE(resource_id, user_id)	Peer review and 5-star rating system for study resources
bookmarks	id (INT PK), resource_id (INT FK), user_id (INT FK), created_at	UNIQUE(resource_id, user_id)	Student personalized pre-exam study list
activity_logs	id (INT PK), user_id (INT FK), action, details, ip_address, created_at	FOREIGN KEY -> users.id	Security audit trail for moderation, uploads, and logins

Automated Verification & Test Execution Results:

Test ID	Scope / Component	Execution Command	Result	Status
TC-01	Homepage & Metadata	GET /	Status 200 OK, Stats rendered	PASS
TC-02	Repository Explorer	GET /browse?type=Question Paper	Status 200 OK, Filtered results returned	PASS
TC-03	Document Detail & Ratings	GET /resource/1	Status 200 OK, Rating stats calculated	PASS
TC-04	In-Browser PDF Viewer	GET /preview/1	Status 200 OK, File stream inline	PASS
TC-05	Download Tracking	GET /download/1	Status 200 OK, download_count incremented	PASS
TC-06	Dynamic Cascading API	GET /api/subjects?dept_id=1&semester=3	Status 200 OK, Correct subjects JSON returned	PASS
TC-07	Live Autocomplete API	GET /api/quick-search?q=data	Status 200 OK, Search matches returned	PASS
TC-08	PWA Offline Endpoint	GET /offline	Status 200 OK, Fallback instructions rendered	PASS
TC-09	Student File Upload	POST /upload (with multipart PDF)	File saved, record status = 'pending'	PASS
TC-10	Admin Review Queue	POST /admin/approve/<id>	Record status updated to 'approved'	PASS

2. COMPLETE SOURCE CODE REPOSITORY

Below is the complete, unabridged source code for all backend controllers, configuration files, database setup modules, frontend Jinja2 templates, JavaScript utilities, PWA Service Worker, and runner scripts. Every file is labeled with its repository path and technical explanation.

■ File: app.py

Main Application Entry Point, Socket LAN IP Discovery, Jinja Custom Filters, Context Processors

```
import os
import socket
from flask import Flask, render_template
from config import Config
from database import close_db, init_db, query_db
from helpers import get_current_user

def get_lan_ip():
    """Discover host's local IPv4 address for local network access."""
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
        s.settimeout(0.5)
        # Doesn't need to actually send packets, just determines the outbound interface
        s.connect(('8.8.8.8', 80))
        ip = s.getsockname()[0]
        s.close()
        return ip
    except Exception:
        try:
            return socket.gethostbyname(socket.gethostname())
        except Exception:
            return '127.0.0.1'

def create_app():
    app = Flask(__name__)
    app.config.from_object(Config)

    # Initialize database if not already existing
    if not os.path.exists(Config.DATABASE):
        with app.app_context():
            from seed_data import seed_database
            seed_database()

    # Register Database teardown
    app.teardown_appcontext(close_db)

    # Custom Jinja filters
    @app.template_filter('str_date')
    def filter_str_date(val):
        if not val:
            return ""
        if hasattr(val, 'strftime'):
            return val.strftime('%Y-%m-%d')
        return str(val)[:10]

    @app.template_filter('str_time')
    def filter_str_time(val):
        if not val:
            return ""
        if hasattr(val, 'strftime'):
            return val.strftime('%Y-%m-%d %H:%M')
        return str(val)[:19]

    # Context processors
    lan_ip = get_lan_ip()

    @app.context_processor
    def inject_global_data():
        user = get_current_user()
        pending_count = 0
        if user and user['role'] == 'admin':
            res = query_db("SELECT COUNT(*) as c FROM resources WHERE status = 'pending'", o...
            pending_count = res['c'] if res else 0

        return {
            'current_user': user,
            'pending_review_count': pending_count,
            'lan_ip': lan_ip,
            'server_port': Config.PORT
        }

    # Register Blueprints
    from routes.main import main_bp
    from routes.auth import auth_bp
    from routes.upload import upload_bp
```

```
    from routes.admin import admin_bp
    from routes.api import api_bp

    app.register_blueprint(main_bp)
    app.register_blueprint(auth_bp)
    app.register_blueprint(upload_bp)
    app.register_blueprint(admin_bp)
    app.register_blueprint(api_bp)

    # Error handlers

    @app.errorhandler(404)
    def page_not_found(e):
        return render_template('error.html', error_code=404, message="The page or study mate...

    @app.errorhandler(500)
    def internal_server_error(e):
        return render_template('error.html', error_code=500, message="An internal server err...

    @app.errorhandler(413)
    def request_entity_too_large(e):
        return render_template('error.html', error_code=413, message="The uploaded file exce...

    return app

if __name__ == '__main__':
    application = create_app()
    lan_ip = get_lan_ip()
    port = Config.PORT

    print("\n" + "=" * 68)
    print("  CENTRALIZED COLLEGE STUDY MATERIAL & QUESTION PAPER PORTAL")
    print("=" * 68)
    print(f"  * Local Access:     http://localhost:{port}")
    print(f"  * LAN/Offline WiFi: http://{lan_ip}:{port}")
    print("  * Status:           Online & Local Network Ready")
    print("----------------------------------------------------------------------")
    print("  Default Credentials:")
    print("    - Admin:   username: admin       | password: admin123")
    print("    - Faculty: username: prof_sharma | password: faculty123")
    print("    - Student: username: student1    | password: student123")
    print("=" * 68 + "\n")

    application.run(host=Config.HOST, port=port, debug=True)
```

■ File: config.py

Central Configuration (Database path, Upload folder, Max file size 64MB, Allowed Extensions)

```
import os

BASE_DIR = os.path.abspath(os.path.dirname(__file__))

class Config:
    BASE_DIR = BASE_DIR
    SECRET_KEY = os.environ.get('SECRET_KEY', 'college-study-portal-secret-key-2026-supersec...
    DATABASE = os.path.join(BASE_DIR, 'portal.db')
    UPLOAD_FOLDER = os.path.join(BASE_DIR, 'static', 'uploads')
    MAX_CONTENT_LENGTH = 64 * 1024 * 1024  # 64 MB maximum upload size
    ALLOWED_EXTENSIONS = {
        'pdf', 'doc', 'docx', 'ppt', 'pptx', 'xls', 'xlsx',
        'txt', 'zip', 'rar', 'jpg', 'jpeg', 'png'
    }
    PORT = int(os.environ.get('PORT', 5000))
    HOST = os.environ.get('HOST', '0.0.0.0')  # Listen on all interfaces for LAN & offline a...
```

■ File: database.py

SQLite Schema Initialization, Table Definitions, Performance Indexes, Connection Management

```
import sqlite3
import os
from flask import g, current_app
from config import Config

def get_db():
    """Get database connection for current request context."""
    if 'db' not in g:
        g.db = sqlite3.connect(Config.DATABASE)
        g.db.row_factory = sqlite3.Row
        g.db.execute("PRAGMA foreign_keys = ON;")
        g.db.execute("PRAGMA journal_mode = WAL;")
    return g.db

def close_db(e=None):
    """Close database connection at end of request."""
    db = g.pop('db', None)
    if db is not None:
        db.close()

def query_db(query, args=(), one=False):
    """Execute a query and fetch results."""
    cur = get_db().execute(query, args)
    rv = cur.fetchall()
    cur.close()
    return (rv[0] if rv else None) if one else rv

def execute_db(query, args=(), commit=True):
    """Execute an INSERT/UPDATE/DELETE query and return the lastrowid or rowcount."""
    db = get_db()
    cur = db.execute(query, args)
    if commit:
        db.commit()
    last_id = cur.lastrowid
    cur.close()
    return last_id

def init_db():
    """Create database tables and indexes if they do not exist."""
    os.makedirs(os.path.dirname(Config.DATABASE), exist_ok=True)
    os.makedirs(Config.UPLOAD_FOLDER, exist_ok=True)

    conn = sqlite3.connect(Config.DATABASE)
    conn.execute("PRAGMA foreign_keys = ON;")
    cur = conn.cursor()

    # 1. Departments Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS departments (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            name TEXT NOT NULL UNIQUE,
            code TEXT NOT NULL UNIQUE,
            description TEXT,
            icon TEXT DEFAULT 'fas fa-graduation-cap',
            is_active INTEGER DEFAULT 1,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
        );
    ''')

    # 2. Subjects Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS subjects (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            name TEXT NOT NULL,
            code TEXT NOT NULL,
            department_id INTEGER NOT NULL,
            semester INTEGER NOT NULL,
            description TEXT,
            is_active INTEGER DEFAULT 1,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            FOREIGN KEY (department_id) REFERENCES departments(id) ON DELETE CASCADE,
            UNIQUE(code, department_id)
        );
    ''')

```

```
    # 3. Users Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS users (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            username TEXT NOT NULL UNIQUE,
            email TEXT NOT NULL UNIQUE,
            password_hash TEXT NOT NULL,
            full_name TEXT NOT NULL,
            roll_number TEXT,
            department_id INTEGER,
            semester INTEGER,
            role TEXT DEFAULT 'student', -- 'student', 'faculty', 'admin'
            is_active INTEGER DEFAULT 1,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            FOREIGN KEY (department_id) REFERENCES departments(id) ON DELETE SET NULL
        );
    ''')

    # 4. Resources Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS resources (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            title TEXT NOT NULL,
            description TEXT,
            resource_type TEXT NOT NULL, -- 'Question Paper', 'Lecture Notes', 'Syllabus', '...
            department_id INTEGER NOT NULL,
            semester INTEGER NOT NULL,
            subject_id INTEGER NOT NULL,
            academic_year TEXT DEFAULT '2025-2026',
            exam_type TEXT DEFAULT 'End-Semester', -- 'Mid-Term 1', 'Mid-Term 2', 'End-Semes...
            file_name TEXT NOT NULL,
            original_file_name TEXT NOT NULL,
            file_size INTEGER DEFAULT 0, -- in bytes
            file_ext TEXT NOT NULL,
            uploader_id INTEGER,
            status TEXT DEFAULT 'pending', -- 'pending', 'approved', 'rejected'
            rejection_reason TEXT,
            download_count INTEGER DEFAULT 0,
            view_count INTEGER DEFAULT 0,
            is_featured INTEGER DEFAULT 0,
            tags TEXT,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            FOREIGN KEY (department_id) REFERENCES departments(id) ON DELETE CASCADE,
            FOREIGN KEY (subject_id) REFERENCES subjects(id) ON DELETE CASCADE,
            FOREIGN KEY (uploader_id) REFERENCES users(id) ON DELETE SET NULL
        );
    ''')

    # 5. Resource Ratings & Reviews Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS resource_ratings (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            resource_id INTEGER NOT NULL,
            user_id INTEGER NOT NULL,
            rating INTEGER NOT NULL CHECK(rating >= 1 AND rating <= 5),
            review TEXT,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            FOREIGN KEY (resource_id) REFERENCES resources(id) ON DELETE CASCADE,
            FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
            UNIQUE(resource_id, user_id)
        );
    ''')

    # 6. Bookmarks Table
    cur.execute('''
        CREATE TABLE IF NOT EXISTS bookmarks (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            resource_id INTEGER NOT NULL,
            user_id INTEGER NOT NULL,
            created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
            FOREIGN KEY (resource_id) REFERENCES resources(id) ON DELETE CASCADE,
            FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
            UNIQUE(resource_id, user_id)
        );
    ''')

    # 7. Activity Logs Table
```

```
154 |     cur.execute('''
155 |         CREATE TABLE IF NOT EXISTS activity_logs (
156 |             id INTEGER PRIMARY KEY AUTOINCREMENT,
157 |             user_id INTEGER,
158 |             action TEXT NOT NULL,
159 |             details TEXT,
160 |             ip_address TEXT,
161 |             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
162 |             FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
163 |         );
164 |     ''')
165 |
166 |     # Performance Indexes
167 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_dept ON resources(department_id);')
168 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_subj ON resources(subject_id);')
169 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_sem ON resources(semester);')
170 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_type ON resources(resource_type);')

171 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_status ON resources(status);')
172 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_resources_created ON resources(created_at DE...
173 |     cur.execute('CREATE INDEX IF NOT EXISTS idx_subjects_dept_sem ON subjects(department_id,...
174 |
175 |     conn.commit()
176 |     conn.close()
177 |     print("Database schema successfully initialized.")
```

■ File: helpers.py

Role Decorators (login_required, admin_required), Formatters, Audit Logger

```
import os
from functools import wraps
from flask import session, redirect, url_for, flash, request, g
from config import Config
from database import query_db, execute_db

def get_current_user():
    """Retrieve logged in user information."""
    user_id = session.get('user_id')
    if not user_id:
        return None
    user = query_db(
        """SELECT u.*, d.name as department_name, d.code as department_code
           FROM users u
           LEFT JOIN departments d ON u.department_id = d.id
           WHERE u.id = ? AND u.is_active = 1""",
        (user_id,),
        one=True
    )
    return user

def login_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if not session.get('user_id'):
            flash('Please log in to access this page.', 'warning')
            return redirect(url_for('auth.login', next=request.url))
        return f(*args, **kwargs)
    return decorated_function

def admin_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if not session.get('user_id'):
            flash('Please log in to access the administrator panel.', 'warning')
            return redirect(url_for('auth.login', next=request.url))
        if session.get('role') != 'admin':
            flash('Access denied. Administrator privileges required.', 'danger')
            return redirect(url_for('main.index'))
        return f(*args, **kwargs)
    return decorated_function

def faculty_or_admin_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if not session.get('user_id'):
            flash('Please log in to continue.', 'warning')
            return redirect(url_for('auth.login', next=request.url))
        if session.get('role') not in ('admin', 'faculty'):
            flash('Access restricted to faculty and administrators.', 'danger')
            return redirect(url_for('main.index'))
        return f(*args, **kwargs)
    return decorated_function

def allowed_file(filename):
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in Config.ALLOWED_EXTENSIONS

def format_file_size(size_bytes):
    """Format file size in human readable format."""
    if not size_bytes or size_bytes <= 0:
        return "0 KB"
    for unit in ['B', 'KB', 'MB', 'GB']:
        if size_bytes < 1024.0:
            return f"{size_bytes:.1f} {unit}" if unit != 'B' else f"{int(size_bytes)} B"
        size_bytes /= 1024.0
    return f"{size_bytes:.1f} TB"

def log_activity(user_id, action, details):
    """Record audit log into the activity_logs table."""
    try:
        ip = request.remote_addr if request else '127.0.0.1'
        execute_db(
            "INSERT INTO activity_logs (user_id, action, details, ip_address) VALUES (?, ?, ...
            (user_id, action, details, ip)
        )
```

```
76 |     except Exception as e:
77 |         print(f"Failed to log activity: {e}")
```

■ File: seed_data.py

Comprehensive Seed Script: Departments, Subjects, Users, Sample Academic Question Papers

```
import os
import sqlite3
from werkzeug.security import generate_password_hash
from config import Config
from database import init_db

def create_sample_pdf(filepath, title, subject, content_lines):
    """Generate a clean, readable PDF-like or styled document."""
    # We can write a clean text/pdf formatted file
    content = f"""%PDF-1.4
1 0 obj
<<
  /Type /Catalog
  /Pages 2 0 R
>>
endobj
2 0 obj
<<
  /Type /Pages
  /Kids [3 0 R]
  /Count 1
>>
endobj
3 0 obj
<<
  /Type /Page
  /Parent 2 0 R
  /MediaBox [0 0 612 792]
  /Contents 4 0 R
  /Resources <<
    /Font <<
      /F1 5 0 R
    >>
  >>
>>
endobj
4 0 obj
<<
  /Length {len(title) + 500}
>>
stream
BT
/F1 18 Tf
50 720 Td
({title}) Tj
/F1 12 Tf
0 -30 Td
(Subject: {subject}) Tj
0 -20 Td
(Centralized College Academic Repository) Tj
0 -40 Td
"""
    for line in content_lines[:15]:
        safe_line = line.replace('(', '[').replace(')', ']')
        content += f"({safe_line}) Tj\n0 -18 Td\n"

    content += """ET
endstream
endobj
5 0 obj
<<
  /Type /Font
  /Subtype /Type1
  /BaseFont /Helvetica
>>
endobj
xref
0 6
0000000000 65535 f
0000000010 00000 n
0000000060 00000 n
0000000117 00000 n
0000000262 00000 n
0000000850 00000 n
trailer
```

```
<<
  /Size 6
  /Root 1 0 R
>>
startxref
940
%%EOF
"""
    with open(filepath, 'wb') as f:
        f.write(content.encode('latin1', errors='replace'))

def seed_database():
    init_db()
    conn = sqlite3.connect(Config.DATABASE)
    cur = conn.cursor()

    # 1. Insert Departments
    departments_data = [
        ("Computer Science & Engineering", "CSE", "Core computing, software engineering, alg...
        ("Information Technology", "IT", "Applied information systems, networking, cloud, an...
        ("Artificial Intelligence & Data Science", "AI-DS", "Modern AI, machine learning, ne...
        ("Electronics & Communication", "ECE", "Circuits, embedded systems, signal processin...
        ("Electrical & Electronics", "EEE", "Power systems, control systems, electrical driv...
        ("Mechanical Engineering", "MECH", "Thermodynamics, mechanics, CAD/CAM, fluid machin...
        ("Civil Engineering", "CIVIL", "Structural engineering, surveying, geo-tech, and con...
        ("Basic Sciences & Humanities", "BSH", "Foundational mathematics, physics, chemistry...
    ]

    dept_map = {}
    for name, code, desc, icon in departments_data:
        cur.execute("""
            INSERT INTO departments (name, code, description, icon)
            VALUES (?, ?, ?, ?)
            ON CONFLICT(code) DO UPDATE SET name=excluded.name, description=excluded.descrip...
        """, (name, code, desc, icon))
        cur.execute("SELECT id FROM departments WHERE code = ?", (code,))
        dept_map[code] = cur.fetchone()[0]

    # 2. Insert Subjects
    subjects_data = [
        # CSE
        ("Data Structures and Algorithms", "CS301", dept_map["CSE"], 3, "Arrays, Stacks, Que...
        ("Database Management Systems", "CS401", dept_map["CSE"], 4, "Relational algebra, SQ...
        ("Operating Systems", "CS402", dept_map["CSE"], 4, "Process scheduling, Threads, Dea...
        ("Computer Networks", "CS501", dept_map["CSE"], 5, "OSI model, TCP/IP, Routing algor...
        ("Software Engineering", "CS502", dept_map["CSE"], 5, "Agile methodology, SDLC, Requ...
        ("Machine Learning", "CS601", dept_map["CSE"], 6, "Supervised & Unsupervised learnin...
        ("Cloud Computing", "CS602", dept_map["CSE"], 6, "Virtualization, AWS, Azure, Docker...
        ("Compiler Design", "CS603", dept_map["CSE"], 6, "Lexical analysis, Parsing, Syntax ...

        # IT
        ("Web Technologies", "IT501", dept_map["IT"], 5, "HTML5, CSS3, JavaScript, RESTful A...
        ("Information Security", "IT601", dept_map["IT"], 6, "Symmetric/Asymmetric cryptogra...
        ("Mobile App Development", "IT701", dept_map["IT"], 7, "Android & iOS architectures,...

        # AI & DS
        ("Artificial Intelligence & Expert Systems", "AI501", dept_map["AI-DS"], 5, "State s...
        ("Deep Learning & Computer Vision", "AI601", dept_map["AI-DS"], 6, "CNNs, RNNs, Tran...
        ("Big Data Analytics", "AI701", dept_map["AI-DS"], 7, "Hadoop ecosystem, MapReduce, ...

        # ECE
        ("Digital Logic & Microprocessors", "EC301", dept_map["ECE"], 3, "Boolean algebra, F...
        ("Signals and Systems", "EC401", dept_map["ECE"], 4, "Continuous and discrete signal...
        ("VLSI Design", "EC601", dept_map["ECE"], 6, "MOSFET theory, CMOS inverters, Layout ...

        # MECH
        ("Engineering Thermodynamics", "ME301", dept_map["MECH"], 3, "Laws of thermodynamics...
        ("Fluid Mechanics and Machinery", "ME401", dept_map["MECH"], 4, "Fluid statics, Bern...

        # BSH
        ("Engineering Mathematics I", "MA101", dept_map["BSH"], 1, "Differential calculus, M...
        ("Engineering Physics", "PH101", dept_map["BSH"], 1, "Quantum mechanics, Lasers, Fib...
    ]

    subj_map = {}
    for name, code, dept_id, sem, desc in subjects_data:
        cur.execute("""
            INSERT INTO subjects (name, code, department_id, semester, description)
```

```
            VALUES (?, ?, ?, ?, ?)
            ON CONFLICT(code, department_id) DO UPDATE SET name=excluded.name, semester=excl...
        """, (name, code, dept_id, sem, desc))
        cur.execute("SELECT id FROM subjects WHERE code = ? AND department_id = ?", (code, d...
        subj_map[code] = cur.fetchone()[0]

    # 3. Insert Users
    users_data = [
        ("admin", "admin@college.edu", generate_password_hash("admin123"), "System Administr...
        ("prof_sharma", "sharma@college.edu", generate_password_hash("faculty123"), "Dr. R. ...
        ("prof_priya", "priya@college.edu", generate_password_hash("faculty123"), "Prof. Pri...
        ("student1", "aarav@college.edu", generate_password_hash("student123"), "Aarav Patel...
        ("student2", "sneha@college.edu", generate_password_hash("student123"), "Sneha Rao",...
    ]

    user_map = {}
    for uname, email, pwd_hash, fname, roll, dept_id, sem, role in users_data:

        cur.execute("""
            INSERT INTO users (username, email, password_hash, full_name, roll_number, depar...
            VALUES (?, ?, ?, ?, ?, ?, ?, ?)
            ON CONFLICT(username) DO UPDATE SET password_hash=excluded.password_hash, role=e...
        """, (uname, email, pwd_hash, fname, roll, dept_id, sem, role))
        cur.execute("SELECT id FROM users WHERE username = ?", (uname,))
        user_map[uname] = cur.fetchone()[0]

    # 4. Generate Sample PDF files in uploads folder
    os.makedirs(Config.UPLOAD_FOLDER, exist_ok=True)

    sample_resources = [
        {
            "title": "Dec 2024 End-Semester Question Paper - Data Structures & Algorithms",
            "desc": "Official college end-semester exam paper with answer pattern, section-w...
            "type": "Question Paper",
            "dept": dept_map["CSE"],
            "sem": 3,
            "subj": subj_map["CS301"],
            "year": "2024-2025",
            "exam": "End-Semester",
            "file": "CS301_EndSem_Dec2024.pdf",
            "uploader": user_map["prof_sharma"],
            "status": "approved",
            "downloads": 142,
            "views": 430,
            "featured": 1,
            "tags": "data structures, exam paper, pyq, end sem, algorithms",
            "pdf_title": "End-Semester Exam - Data Structures (CS301)",
            "pdf_subject": "Data Structures & Algorithms - CSE Sem 3",
            "content": [
                "Part A (10 x 2 = 20 Marks): Answer all questions.",
                "1. Define Time Complexity and Big-O notation.",
                "2. What is an AVL Tree and how does it balance?",
                "3. Differentiate between BFS and DFS traversal.",
                "4. Explain Infix to Postfix conversion rules.",
                "Part B (5 x 16 = 80 Marks): Comprehensive questions.",
                "5. (a) Construct a Red-Black Tree for given keys.",
                "5. (b) Explain Dijkstra's shortest path algorithm with example.",
                "6. Write quicksort algorithm and derive its average time complexity."
            ]
        },
        {
            "title": "Operating Systems Complete Unit 1 to 5 Lecture Notes",
            "desc": "Comprehensive lecture notes prepared by Dr. R. K. Sharma covering Proce...
            "type": "Lecture Notes",
            "dept": dept_map["CSE"],
            "sem": 4,
            "subj": subj_map["CS402"],
            "year": "2025-2026",
            "exam": "N/A",
            "file": "CS402_OS_Lecture_Notes_Unit1_5.pdf",
            "uploader": user_map["prof_sharma"],
            "status": "approved",
            "downloads": 289,
            "views": 750,
            "featured": 1,
            "tags": "operating systems, lecture notes, study material, processes, memory man...
            "pdf_title": "Operating Systems - Complete Course Notes",
            "pdf_subject": "Operating Systems (CS402)",
            "content": [
```

```
                "Unit 1: Introduction to OS, Dual-mode operation, System calls.",
                "Unit 2: Process Management, CPU Scheduling (FCFS, SJF, RR, Priority).",
                "Unit 3: Process Synchronization, Critical Section, Semaphores, Monitors.",
                "Unit 4: Deadlocks: Characterization, Prevention, Avoidance (Banker's Algori...
                "Unit 5: Memory Management: Paging, Segmentation, Virtual Memory, Page Faults."
            ]
        },
        {
            "title": "Database Management Systems Mid-Term Question Paper with Answer Key",
            "desc": "Mid-Term 1 question paper including ER modeling, Relational Algebra, BC...
            "type": "Question Paper",
            "dept": dept_map["CSE"],
            "sem": 4,
            "subj": subj_map["CS401"],
            "year": "2024-2025",
            "exam": "Mid-Term 1",
            "file": "CS401_MidTerm1_DBMS_2024.pdf",
            "uploader": user_map["student1"],
            "status": "approved",
            "downloads": 95,
            "views": 210,
            "featured": 0,
            "tags": "dbms, mid-term, sql, normalization, question paper",
            "pdf_title": "DBMS Mid-Term Exam Paper",

            "pdf_subject": "Database Management Systems (CS401)",
            "content": [
                "Q1. Draw an ER diagram for a Hospital Management System.",
                "Q2. Explain 1NF, 2NF, 3NF, and BCNF with concrete counterexamples.",
                "Q3. Write SQL queries for nested subqueries and joins.",
                "Q4. Discuss ACID properties and serializability."
            ]
        },
        {
            "title": "Machine Learning Lab Manual & Practical Experiments",
            "desc": "Step-by-step practical lab experiments using Python, Scikit-Learn, Pand...
            "type": "Lab Manual",
            "dept": dept_map["AI-DS"],
            "sem": 6,
            "subj": subj_map["CS601"],
            "year": "2025-2026",
            "exam": "N/A",
            "file": "AI_DS_Machine_Learning_Lab_Manual.pdf",
            "uploader": user_map["prof_priya"],
            "status": "approved",
            "downloads": 178,
            "views": 412,
            "featured": 1,
            "tags": "machine learning, lab manual, python, scikit-learn, experiments",
            "pdf_title": "Machine Learning Laboratory Manual",
            "pdf_subject": "Machine Learning (CS601 / AI601)",
            "content": [
                "Experiment 1: Implementation of Linear and Polynomial Regression.",
                "Experiment 2: K-Nearest Neighbor (KNN) Classification.",
                "Experiment 3: ID3 Decision Tree Induction algorithm.",
                "Experiment 4: Support Vector Machine (SVM) for image classification.",
                "Experiment 5: K-Means Clustering on customer segmentation dataset."
            ]
        },
        {
            "title": "Computer Networks Nov 2024 End-Semester Model Paper",
            "desc": "Model question paper prepared according to latest university syllabus c...
            "type": "Question Paper",
            "dept": dept_map["CSE"],
            "sem": 5,
            "subj": subj_map["CS501"],
            "year": "2024-2025",
            "exam": "Model Exam",
            "file": "CS501_Model_Paper_Networks_2024.pdf",
            "uploader": user_map["student1"],
            "status": "approved",
            "downloads": 110,
            "views": 290,
            "featured": 0,
            "tags": "computer networks, model paper, routing, tcp ip, semester 5",
            "pdf_title": "Model Exam - Computer Networks",
            "pdf_subject": "Computer Networks (CS501)",
            "content": [
                "1. Explain OSI 7-Layer Reference Model with functions of each layer.",
```

```
                "2. Perform CIDR Subnetting for a class B network with 6 subnets.",
                "3. Differentiate between Distance Vector and Link State Routing.",
                "4. Explain 3-Way Handshake in TCP connection establishment and termination."
            ]
        },
        {
            "title": "Engineering Mathematics I Solved Question Bank & Formula Sheet",
            "desc": "Quick revision formula sheet, solved calculus problems, eigenvalues & e...
            "type": "Lecture Notes",
            "dept": dept_map["BSH"],
            "sem": 1,
            "subj": subj_map["MA101"],
            "year": "2025-2026",
            "exam": "End-Semester",
            "file": "MA101_Maths1_Formula_Sheet_Solved.pdf",
            "uploader": user_map["prof_sharma"],
            "status": "approved",
            "downloads": 315,
            "views": 890,
            "featured": 1,
            "tags": "mathematics, formula sheet, semester 1, solved problems, calculus",
            "pdf_title": "Engineering Mathematics I - Solved Guide",
            "pdf_subject": "Engineering Mathematics I (MA101)",
            "content": [
                "1. Matrices: Cayley-Hamilton Theorem and Diagonalization.",
                "2. Differential Calculus: Curvature, Evolutes, and Envelopes.",
                "3. Functions of Several Variables: Maxima & Minima, Lagrange Multipliers.",
                "4. Multiple Integrals: Double & Triple integrals in polar/spherical coordin...
            ]
        },
        {

            "title": "Cloud Computing Assignment 1 - AWS Architectures & Docker",
            "desc": "Assignment problems covering microservices deployment, Docker container...
            "type": "Assignment",
            "dept": dept_map["CSE"],
            "sem": 6,
            "subj": subj_map["CS602"],
            "year": "2025-2026",
            "exam": "N/A",
            "file": "CS602_Cloud_Assignment_1.pdf",
            "uploader": user_map["student2"],
            "status": "pending",
            "downloads": 12,
            "views": 45,
            "featured": 0,
            "tags": "cloud computing, assignment, docker, aws, microservices",
            "pdf_title": "Cloud Computing - Assignment 1",
            "pdf_subject": "Cloud Computing (CS602)",
            "content": [
                "Task 1: Create a multi-container Docker application using docker-compose.",
                "Task 2: Design high-availability architecture on AWS using EC2, ALB, and RD...
                "Task 3: Compare cost and latency between Serverless (AWS Lambda) vs Contain...
            ]
        }
    ]

    for item in sample_resources:
        filepath = os.path.join(Config.UPLOAD_FOLDER, item["file"])
        create_sample_pdf(filepath, item["pdf_title"], item["pdf_subject"], item["content"])
        filesize = os.path.getsize(filepath)

        cur.execute("""
            INSERT INTO resources (
                title, description, resource_type, department_id, semester, subject_id,
                academic_year, exam_type, file_name, original_file_name, file_size,
                file_ext, uploader_id, status, download_count, view_count, is_featured, tags
            ) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
        """, (
            item["title"], item["desc"], item["type"], item["dept"], item["sem"],
            item["subj"], item["year"], item["exam"], item["file"], item["file"],
            filesize, "pdf", item["uploader"], item["status"], item["downloads"],
            item["views"], item["featured"], item["tags"]
        ))

    # 5. Insert Ratings & Bookmarks
    cur.execute("SELECT id FROM resources WHERE status = 'approved'")
    approved_ids = [r[0] for r in cur.fetchall()]

```

```
    if approved_ids:
        cur.execute("""
            INSERT OR IGNORE INTO resource_ratings (resource_id, user_id, rating, review)
            VALUES (?, ?, ?, ?)
        """, (approved_ids[0], user_map["student1"], 5, "Extremely useful! Exactly the forma...

        cur.execute("""
            INSERT OR IGNORE INTO resource_ratings (resource_id, user_id, rating, review)
            VALUES (?, ?, ?, ?)
        """, (approved_ids[1], user_map["student2"], 5, "Crystal clear notes on Banker's alg...

        cur.execute("""
            INSERT OR IGNORE INTO bookmarks (resource_id, user_id)
            VALUES (?, ?)
        """, (approved_ids[0], user_map["student1"]))

        cur.execute("""
            INSERT OR IGNORE INTO bookmarks (resource_id, user_id)
            VALUES (?, ?)
        """, (approved_ids[1], user_map["student1"]))

    # 6. Insert Activity Log
    cur.execute("""
        INSERT INTO activity_logs (user_id, action, details, ip_address)
        VALUES (?, 'SYSTEM_INIT', 'System initialized with departments, subjects, and sample...
    """, (user_map["admin"],))

    conn.commit()
    conn.close()
    print("Database successfully seeded with realistic academic resources!")

if __name__ == "__main__":
    seed_database()
```

■ File: test_app.py

Automated End-to-End Verification Suite for Upload, Pending Moderation, and Approval

```
import io
from app import create_app
from database import query_db, execute_db

def test_full_lifecycle():
    app = create_app()
    client = app.test_client()

    with app.app_context():
        # 1. Clean up old test data if exists
        execute_db("DELETE FROM resources WHERE title = 'Test Mid-Term Paper 2026 - Algorith...

        # 2. Student uploads paper
        with client.session_transaction() as sess:
            sess['user_id'] = 4
            sess['username'] = 'student1'
            sess['full_name'] = 'Aarav Patel'
            sess['role'] = 'student'

        data = {
            'title': 'Test Mid-Term Paper 2026 - Algorithms',
            'description': 'Student uploaded trial document for verification',
            'resource_type': 'Question Paper',
            'department_id': '1',
            'semester': '3',
            'subject_id': '1',
            'academic_year': '2025-2026',
            'exam_type': 'Mid-Term 1',
            'tags': 'algorithms, test',
            'file': (io.BytesIO(b'%PDF-1.4 sample content'), 'test_paper.pdf')
        }

        res = client.post('/upload', data=data, content_type='multipart/form-data', follow_r...
        assert res.status_code == 200
        print("[TEST 1/3] Student file upload POST -> 200 OK")

        # 3. Check DB for pending status
        res_db = query_db("SELECT id, status FROM resources WHERE title = 'Test Mid-Term Pap...
        assert res_db is not None
        assert res_db['status'] == 'pending'
        test_id = res_db['id']
        print(f"[TEST 2/3] Resource ID {test_id} verified in DB with status = 'pending'")

        # 4. Admin approves
        with client.session_transaction() as sess:
            sess['user_id'] = 1
            sess['username'] = 'admin'
            sess['full_name'] = 'System Administrator'
            sess['role'] = 'admin'

        res_appr = client.post(f'/admin/approve/{test_id}', follow_redirects=True)
        assert res_appr.status_code == 200

        # 5. Check DB for approved status
        res_db_after = query_db("SELECT status FROM resources WHERE id = ?", (test_id,), one...
        assert res_db_after['status'] == 'approved'
        print(f"[TEST 3/3] Resource ID {test_id} successfully approved and status = 'approve...

        # Clean up test row
        execute_db("DELETE FROM resources WHERE id = ?", (test_id,))

        print("\n========================================================")
        print(" ALL END-TO-END WORKFLOW TESTS VERIFIED SUCCESSFULLY!   ")
        print("========================================================")

if __name__ == '__main__':
    test_full_lifecycle()
```

■ File: routes/__init__.py

Modular Blueprint Package Init

```
# Blueprint package
```

■ File: routes/auth.py

Authentication Blueprint: Login, Register, Profile Management, Password Hashing

```
from flask import Blueprint, render_template, request, redirect, url_for, flash, session
from werkzeug.security import generate_password_hash, check_password_hash
from database import query_db, execute_db
from helpers import login_required, log_activity, get_current_user

auth_bp = Blueprint('auth', __name__)

@auth_bp.route('/login', methods=['GET', 'POST'])
def login():
    if session.get('user_id'):
        return redirect(url_for('main.index'))

    if request.method == 'POST':
        identifier = request.form.get('identifier', '').strip()
        password = request.form.get('password', '')

        if not identifier or not password:
            flash('Please enter both username/email and password.', 'warning')
            return render_template('auth/login.html')

        user = query_db(
            "SELECT * FROM users WHERE (username = ? OR email = ?)",
            (identifier, identifier),
            one=True
        )

        if not user or not check_password_hash(user['password_hash'], password):
            flash('Invalid username/email or password.', 'danger')
            return render_template('auth/login.html')

        if not user['is_active']:
            flash('Your account has been deactivated. Please contact the administrator.', 'd...
            return render_template('auth/login.html')

        # Set session
        session['user_id'] = user['id']
        session['username'] = user['username']
        session['full_name'] = user['full_name']
        session['role'] = user['role']
        session['department_id'] = user['department_id']

        log_activity(user['id'], 'LOGIN', f"User {user['username']} logged in successfully")

        flash(f"Welcome back, {user['full_name']}!", 'success')
        next_page = request.args.get('next')
        if next_page and next_page.startswith('/'):
            return redirect(next_page)
        return redirect(url_for('main.index'))

    return render_template('auth/login.html')

@auth_bp.route('/register', methods=['GET', 'POST'])
def register():
    if session.get('user_id'):
        return redirect(url_for('main.index'))

    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")

    if request.method == 'POST':
        full_name = request.form.get('full_name', '').strip()
        username = request.form.get('username', '').strip().lower()
        email = request.form.get('email', '').strip().lower()
        password = request.form.get('password', '')
        confirm_password = request.form.get('confirm_password', '')
        roll_number = request.form.get('roll_number', '').strip().upper()
        department_id = request.form.get('department_id')
        semester = request.form.get('semester')
        role = request.form.get('role', 'student')

        # Validation
        if not full_name or not username or not email or not password:
            flash('All required fields must be filled.', 'warning')
            return render_template('auth/register.html', departments=departments)

        if password != confirm_password:
```

```
            flash('Passwords do not match.', 'danger')
            return render_template('auth/register.html', departments=departments)

        if len(password) < 6:
            flash('Password must be at least 6 characters long.', 'warning')
            return render_template('auth/register.html', departments=departments)

        # Check existing
        existing_user = query_db(
            "SELECT * FROM users WHERE username = ? OR email = ?",
            (username, email),
            one=True
        )
        if existing_user:
            if existing_user['username'] == username:
                flash('Username is already taken. Please choose another.', 'warning')
            else:
                flash('Email is already registered. Please log in.', 'warning')
            return render_template('auth/register.html', departments=departments)

        # Allow student or faculty registration (faculty requires admin verification or defa...
        assigned_role = 'faculty' if role == 'faculty' else 'student'

        pwd_hash = generate_password_hash(password)
        user_id = execute_db(
            """INSERT INTO users (username, email, password_hash, full_name, roll_number, de...
               VALUES (?, ?, ?, ?, ?, ?, ?, ?)""",
            (username, email, pwd_hash, full_name, roll_number or None, department_id or Non...
        )

        log_activity(user_id, 'REGISTER', f"New account registered: {username} ({assigned_ro...

        # Auto login
        session['user_id'] = user_id
        session['username'] = username
        session['full_name'] = full_name
        session['role'] = assigned_role
        session['department_id'] = department_id

        flash('Registration successful! Welcome to the Centralized College Portal.', 'success')
        return redirect(url_for('main.index'))

    return render_template('auth/register.html', departments=departments)

@auth_bp.route('/logout')
def logout():
    user_id = session.get('user_id')
    if user_id:
        log_activity(user_id, 'LOGOUT', "User logged out")
    session.clear()
    flash('You have been logged out successfully.', 'info')
    return redirect(url_for('main.index'))

@auth_bp.route('/profile', methods=['GET', 'POST'])
@login_required
def profile():
    user = get_current_user()
    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")

    # Fetch user statistics
    stats = {
        'uploads_count': query_db("SELECT COUNT(*) as cnt FROM resources WHERE uploader_id =...
        'approved_count': query_db("SELECT COUNT(*) as cnt FROM resources WHERE uploader_id ...
        'pending_count': query_db("SELECT COUNT(*) as cnt FROM resources WHERE uploader_id =...
        'bookmarks_count': query_db("SELECT COUNT(*) as cnt FROM bookmarks WHERE user_id = ?...
        'total_downloads': query_db("SELECT COALESCE(SUM(download_count), 0) as cnt FROM res...
    }

    if request.method == 'POST':
        action = request.form.get('action')

        if action == 'update_info':
            full_name = request.form.get('full_name', '').strip()
            email = request.form.get('email', '').strip().lower()
            roll_number = request.form.get('roll_number', '').strip().upper()
            department_id = request.form.get('department_id')
            semester = request.form.get('semester')

```

```
154 |             if not full_name or not email:
155 |                 flash('Full Name and Email are required.', 'warning')
156 |                 return render_template('auth/profile.html', user=user, stats=stats, departme...
157 |
158 |             # Check email collision
159 |             existing = query_db("SELECT id FROM users WHERE email = ? AND id != ?", (email, ...
160 |             if existing:
161 |                 flash('This email is already associated with another account.', 'danger')
162 |                 return render_template('auth/profile.html', user=user, stats=stats, departme...
163 |
164 |             execute_db(
165 |                 """UPDATE users SET full_name = ?, email = ?, roll_number = ?, department_id...
166 |                    WHERE id = ?""",
167 |                 (full_name, email, roll_number or None, department_id or None, semester or N...
168 |             )
169 |             session['full_name'] = full_name
170 |             session['department_id'] = department_id

171 |             log_activity(user['id'], 'UPDATE_PROFILE', "Profile details updated")
172 |             flash('Profile updated successfully!', 'success')
173 |             return redirect(url_for('auth.profile'))
174 |
175 |         elif action == 'change_password':
176 |             current_password = request.form.get('current_password', '')
177 |             new_password = request.form.get('new_password', '')
178 |             confirm_new_password = request.form.get('confirm_new_password', '')
179 |
180 |             if not check_password_hash(user['password_hash'], current_password):
181 |                 flash('Current password entered is incorrect.', 'danger')
182 |                 return render_template('auth/profile.html', user=user, stats=stats, departme...
183 |
184 |             if len(new_password) < 6:
185 |                 flash('New password must be at least 6 characters long.', 'warning')
186 |                 return render_template('auth/profile.html', user=user, stats=stats, departme...
187 |
188 |             if new_password != confirm_new_password:
189 |                 flash('New passwords do not match.', 'danger')
190 |                 return render_template('auth/profile.html', user=user, stats=stats, departme...
191 |
192 |             new_hash = generate_password_hash(new_password)
193 |             execute_db("UPDATE users SET password_hash = ? WHERE id = ?", (new_hash, user['i...
194 |             log_activity(user['id'], 'CHANGE_PASSWORD', "Password changed successfully")
195 |             flash('Password updated successfully!', 'success')
196 |             return redirect(url_for('auth.profile'))
197 |
198 |     return render_template('auth/profile.html', user=user, stats=stats, departments=departme...
```

■ File: routes/main.py

Core Controller: Home, Repository Browse with Multi-Tier Filters, Preview, Download, Ratings

```
import os
from flask import Blueprint, render_template, request, redirect, url_for, flash, session, se...
from config import Config
from database import query_db, execute_db
from helpers import get_current_user, login_required, log_activity, format_file_size

main_bp = Blueprint('main', __name__)

@main_bp.route('/')
def index():
    # Metrics
    stats = {
        'total_resources': query_db("SELECT COUNT(*) as c FROM resources WHERE status = 'app...
        'total_downloads': query_db("SELECT COALESCE(SUM(download_count), 0) as c FROM resou...
        'total_departments': query_db("SELECT COUNT(*) as c FROM departments WHERE is_active...
        'total_users': query_db("SELECT COUNT(*) as c FROM users WHERE is_active = 1", one=T...
    }

    # Featured Resources
    featured = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
               (SELECT COUNT(*) FROM resource_ratings WHERE resource_id = r.id) as rating_count
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.status = 'approved' AND r.is_featured = 1
        ORDER BY r.download_count DESC, r.created_at DESC
        LIMIT 6
    """)

    # Recent Uploads
    recent = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.status = 'approved'
        ORDER BY r.created_at DESC
        LIMIT 6
    """)

    # Departments with resource counts
    departments = query_db("""
        SELECT d.*,
               (SELECT COUNT(*) FROM resources r WHERE r.department_id = d.id AND r.status =...
        FROM departments d
        WHERE d.is_active = 1
        ORDER BY d.name ASC
    """)

    # Top Question Papers
    question_papers = query_db("""
        SELECT r.*, d.code as department_code, s.name as subject_name
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.status = 'approved' AND r.resource_type = 'Question Paper'
        ORDER BY r.download_count DESC, r.created_at DESC
        LIMIT 4
    """)

    return render_template(
        'index.html',
        stats=stats,
        featured=featured,
        recent=recent,
        departments=departments,
        question_papers=question_papers,
        format_file_size=format_file_size
    )

@main_bp.route('/browse')
def browse():
```

```
    q = request.args.get('q', '').strip()
    dept_id = request.args.get('dept', '')
    sem = request.args.get('sem', '')
    subj_id = request.args.get('subj', '')
    res_type = request.args.get('type', '')
    year = request.args.get('year', '')
    exam = request.args.get('exam', '')
    sort_by = request.args.get('sort', 'newest')
    page = int(request.args.get('page', 1))
    per_page = 12

    offset = (page - 1) * per_page

    query_conditions = ["r.status = 'approved'"]
    params = []

    if q:
        query_conditions.append("(r.title LIKE ? OR r.description LIKE ? OR r.tags LIKE ? OR...
        search_term = f"%{q}%"
        params.extend([search_term, search_term, search_term, search_term])

    if dept_id and dept_id.isdigit():
        query_conditions.append("r.department_id = ?")
        params.append(int(dept_id))

    if sem and sem.isdigit():
        query_conditions.append("r.semester = ?")
        params.append(int(sem))

    if subj_id and subj_id.isdigit():
        query_conditions.append("r.subject_id = ?")
        params.append(int(subj_id))

    if res_type:
        query_conditions.append("r.resource_type = ?")
        params.append(res_type)

    if year:
        query_conditions.append("r.academic_year = ?")
        params.append(year)

    if exam:
        query_conditions.append("r.exam_type = ?")
        params.append(exam)

    where_clause = " WHERE " + " AND ".join(query_conditions)

    # Sorting
    if sort_by == 'downloads':
        order_clause = "ORDER BY r.download_count DESC, r.created_at DESC"
    elif sort_by == 'rating':
        order_clause = "ORDER BY avg_rating DESC, r.download_count DESC"
    elif sort_by == 'title':
        order_clause = "ORDER BY r.title ASC"
    else:
        order_clause = "ORDER BY r.created_at DESC"

    # Count total matching
    count_sql = f"""
        SELECT COUNT(*) as total
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        {where_clause}
    """
    total_resources = query_db(count_sql, params, one=True)['total']
    total_pages = max(1, (total_resources + per_page - 1) // per_page)

    # Select resources
    select_sql = f"""
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
               u.full_name as uploader_name, u.role as uploader_role,
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
               (SELECT COUNT(*) FROM resource_ratings WHERE resource_id = r.id) as rating_count
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        LEFT JOIN users u ON r.uploader_id = u.id
        {where_clause}
```

```
        {order_clause}
        LIMIT ? OFFSET ?
    """
    query_params = list(params) + [per_page, offset]
    resources = query_db(select_sql, query_params)

    # Master dropdown options
    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")

    # If a department is selected, filter subjects list accordingly
    if dept_id and dept_id.isdigit():
        subjects = query_db("SELECT * FROM subjects WHERE department_id = ? AND is_active = ...
    else:
        subjects = query_db("SELECT * FROM subjects WHERE is_active = 1 ORDER BY name ASC")

    # Distinct academic years available
    available_years = [r['academic_year'] for r in query_db("SELECT DISTINCT academic_year F...

    if not available_years:
        available_years = ['2025-2026', '2024-2025', '2023-2024', '2022-2023']

    return render_template(
        'browse.html',
        resources=resources,
        total_resources=total_resources,
        departments=departments,
        subjects=subjects,
        available_years=available_years,
        page=page,
        total_pages=total_pages,
        q=q,
        selected_dept=dept_id,
        selected_sem=sem,
        selected_subj=subj_id,
        selected_type=res_type,
        selected_year=year,
        selected_exam=exam,
        selected_sort=sort_by,
        format_file_size=format_file_size
    )

@main_bp.route('/resource/<int:resource_id>')
def resource_detail(resource_id):
    resource = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code,
               s.name as subject_name, s.code as subject_code,
               u.full_name as uploader_name, u.role as uploader_role, u.username as uploader...
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
               (SELECT COUNT(*) FROM resource_ratings WHERE resource_id = r.id) as rating_count
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        LEFT JOIN users u ON r.uploader_id = u.id
        WHERE r.id = ?
    """, (resource_id,), one=True)

    if not resource:
        flash('Requested resource not found.', 'danger')
        return redirect(url_for('main.browse'))

    # If pending or rejected, only uploader or admin can view
    if resource['status'] != 'approved':
        current_uid = session.get('user_id')
        current_role = session.get('role')
        if current_role != 'admin' and current_uid != resource['uploader_id']:
            flash('This resource is currently under review.', 'warning')
            return redirect(url_for('main.browse'))

    # Increment view count
    execute_db("UPDATE resources SET view_count = view_count + 1 WHERE id = ?", (resource_id,))

    # Check bookmark status for current user
    user_id = session.get('user_id')
    is_bookmarked = False
    user_rating = None
    if user_id:
        bm = query_db("SELECT id FROM bookmarks WHERE resource_id = ? AND user_id = ?", (res...
        is_bookmarked = (bm is not None)
        ur = query_db("SELECT rating, review FROM resource_ratings WHERE resource_id = ? AND...
```

```
        if ur:
            user_rating = ur

    # Fetch reviews
    reviews = query_db("""
        SELECT rr.*, u.full_name, u.username, u.role
        FROM resource_ratings rr
        JOIN users u ON rr.user_id = u.id
        WHERE rr.resource_id = ?
        ORDER BY rr.created_at DESC
    """, (resource_id,))

    # Related resources
    related = query_db("""
        SELECT r.*, d.code as department_code, s.name as subject_name,
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.id != ? AND r.status = 'approved' AND (r.subject_id = ? OR r.department_id = ?)
        ORDER BY r.download_count DESC
        LIMIT 4
    """, (resource_id, resource['subject_id'], resource['department_id']))


    # Split tags
    tag_list = [t.strip() for t in resource['tags'].split(',')] if resource['tags'] else []

    return render_template(
        'resource_detail.html',
        resource=resource,
        is_bookmarked=is_bookmarked,
        user_rating=user_rating,
        reviews=reviews,
        related=related,
        tag_list=tag_list,
        format_file_size=format_file_size
    )

@main_bp.route('/download/<int:resource_id>')
def download_resource(resource_id):
    resource = query_db("SELECT * FROM resources WHERE id = ?", (resource_id,), one=True)
    if not resource:
        flash('Resource not found.', 'danger')
        return redirect(url_for('main.browse'))

    filepath = os.path.join(Config.UPLOAD_FOLDER, resource['file_name'])
    if not os.path.isfile(filepath):
        flash('File not found on server.', 'danger')
        return redirect(url_for('main.resource_detail', resource_id=resource_id))

    # Increment download count
    execute_db("UPDATE resources SET download_count = download_count + 1 WHERE id = ?", (res...

    user_id = session.get('user_id')
    if user_id:
        log_activity(user_id, 'DOWNLOAD', f"Downloaded resource: {resource['title']} (ID: {r...

    return send_from_directory(
        Config.UPLOAD_FOLDER,
        resource['file_name'],
        as_attachment=True,
        download_name=resource['original_file_name']
    )

@main_bp.route('/preview/<int:resource_id>')
def preview_resource(resource_id):
    resource = query_db("SELECT * FROM resources WHERE id = ?", (resource_id,), one=True)
    if not resource:
        abort(404)

    filepath = os.path.join(Config.UPLOAD_FOLDER, resource['file_name'])
    if not os.path.isfile(filepath):
        abort(404)

    return send_from_directory(
        Config.UPLOAD_FOLDER,
        resource['file_name'],
        as_attachment=False
```

```
    )

@main_bp.route('/bookmark/<int:resource_id>', methods=['POST'])
@login_required
def toggle_bookmark(resource_id):
    user_id = session['user_id']
    existing = query_db("SELECT id FROM bookmarks WHERE resource_id = ? AND user_id = ?", (r...

    if existing:
        execute_db("DELETE FROM bookmarks WHERE id = ?", (existing['id'],))
        is_bookmarked = False
        message = "Removed from your bookmarks."
    else:
        execute_db("INSERT INTO bookmarks (resource_id, user_id) VALUES (?, ?)", (resource_i...
        is_bookmarked = True
        message = "Added to your bookmarks."

    if request.headers.get('X-Requested-With') == 'XMLHttpRequest' or request.is_json:
        return jsonify({'status': 'success', 'is_bookmarked': is_bookmarked, 'message': mess...

    flash(message, 'info')
    return redirect(request.referrer or url_for('main.resource_detail', resource_id=resource...

@main_bp.route('/rate/<int:resource_id>', methods=['POST'])
@login_required
def rate_resource(resource_id):
    user_id = session['user_id']
    rating_val = request.form.get('rating')
    review_text = request.form.get('review', '').strip()

    if not rating_val or not rating_val.isdigit():

        flash('Please select a valid rating between 1 and 5.', 'warning')
        return redirect(url_for('main.resource_detail', resource_id=resource_id))

    rating_int = int(rating_val)
    if rating_int < 1 or rating_int > 5:
        flash('Rating must be between 1 and 5.', 'warning')
        return redirect(url_for('main.resource_detail', resource_id=resource_id))

    execute_db("""
        INSERT INTO resource_ratings (resource_id, user_id, rating, review)
        VALUES (?, ?, ?, ?)
        ON CONFLICT(resource_id, user_id) DO UPDATE SET rating=excluded.rating, review=exclu...
    """, (resource_id, user_id, rating_int, review_text))

    log_activity(user_id, 'RATE_RESOURCE', f"Rated resource ID {resource_id} with {rating_in...
    flash('Thank you for rating and reviewing this study material!', 'success')
    return redirect(url_for('main.resource_detail', resource_id=resource_id))

@main_bp.route('/bookmarks')
@login_required
def bookmarks():
    user_id = session['user_id']
    bookmarks_list = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code,
               s.name as subject_name, b.created_at as bookmarked_at,
               COALESCE((SELECT AVG(rating) FROM resource_ratings WHERE resource_id = r.id),...
        FROM bookmarks b
        JOIN resources r ON b.resource_id = r.id
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE b.user_id = ?
        ORDER BY b.created_at DESC
    """, (user_id,))

    return render_template('bookmarks.html', bookmarks=bookmarks_list, format_file_size=form...

@main_bp.route('/offline')
def offline():
    return render_template('offline.html')

@main_bp.route('/presentation')
def presentation():
    return render_template('presentation.html')

@main_bp.route('/download-video')
def download_video():
    video_path = os.path.join(Config.BASE_DIR, 'project_presentation_video.mp4')
```

```
    if not os.path.isfile(video_path):
        flash('Video file is being generated or was moved.', 'warning')
        return redirect(url_for('main.presentation'))
    return send_from_directory(
        Config.BASE_DIR,
        'project_presentation_video.mp4',
        as_attachment=True,
        download_name='College_Study_Material_Portal_Video_Presentation.mp4'
    )

@main_bp.route('/download-codebook')
def download_codebook_pdf():
    pdf_path = os.path.join(Config.BASE_DIR, 'College_Study_Material_Portal_Complete_Code.pdf')
    if not os.path.isfile(pdf_path):
        flash('Codebook PDF not found.', 'warning')
        return redirect(url_for('main.presentation'))
    return send_from_directory(
        Config.BASE_DIR,
        'College_Study_Material_Portal_Complete_Code.pdf',
        as_attachment=True,
        download_name='College_Study_Material_Portal_Complete_Code.pdf'
    )
```

■ File: routes/upload.py

Resource Ingestion Controller: Drag-and-Drop Handler, Validation, 'My Uploads' Status Dashboard

```
import os
import uuid
from flask import Blueprint, render_template, request, redirect, url_for, flash, session
from werkzeug.utils import secure_filename
from config import Config
from database import query_db, execute_db
from helpers import login_required, log_activity, allowed_file, format_file_size

upload_bp = Blueprint('upload', __name__)

@upload_bp.route('/upload', methods=['GET', 'POST'])
@login_required
def upload():
    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")

    if request.method == 'POST':
        title = request.form.get('title', '').strip()
        description = request.form.get('description', '').strip()
        resource_type = request.form.get('resource_type', '').strip()
        department_id = request.form.get('department_id')
        semester = request.form.get('semester')
        subject_id = request.form.get('subject_id')
        academic_year = request.form.get('academic_year', '2025-2026').strip()
        exam_type = request.form.get('exam_type', 'End-Semester').strip()
        tags = request.form.get('tags', '').strip()

        # Validation
        if not title or not resource_type or not department_id or not semester or not subjec...
            flash('Please complete all mandatory academic fields.', 'warning')
            return render_template('upload.html', departments=departments)

        if 'file' not in request.files:
            flash('No file was selected for upload.', 'danger')
            return render_template('upload.html', departments=departments)

        file = request.files['file']
        if file.filename == '':
            flash('No file selected.', 'warning')
            return render_template('upload.html', departments=departments)

        if not allowed_file(file.filename):
            allowed_list = ', '.join(Config.ALLOWED_EXTENSIONS)
            flash(f'File format not supported. Allowed formats: {allowed_list}', 'danger')
            return render_template('upload.html', departments=departments)

        # Secure file saving
        original_name = secure_filename(file.filename)
        ext = original_name.rsplit('.', 1)[1].lower() if '.' in original_name else 'dat'
        unique_prefix = uuid.uuid4().hex[:12]
        saved_filename = f"{unique_prefix}_{original_name}"
        destination = os.path.join(Config.UPLOAD_FOLDER, saved_filename)
        file.save(destination)

        file_size = os.path.getsize(destination)

        # Role-based approval: Faculty and Admin uploads are automatically approved; Student...
        user_role = session.get('role', 'student')
        initial_status = 'approved' if user_role in ('admin', 'faculty') else 'pending'

        res_id = execute_db("""
            INSERT INTO resources (
                title, description, resource_type, department_id, semester, subject_id,
                academic_year, exam_type, file_name, original_file_name, file_size,
                file_ext, uploader_id, status, tags
            ) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
        """, (
            title, description, resource_type, int(department_id), int(semester), int(subjec...
            academic_year, exam_type, saved_filename, original_name, file_size,
            ext, session['user_id'], initial_status, tags
        ))

        log_activity(
            session['user_id'],
            'UPLOAD_RESOURCE',
            f"Uploaded resource '{title}' (ID: {res_id}, Status: {initial_status})"
```

```
        )

        if initial_status == 'approved':
            flash('Resource uploaded and published successfully!', 'success')
            return redirect(url_for('main.resource_detail', resource_id=res_id))
        else:
            flash('Resource uploaded successfully! It is now in the review queue and will be...
            return redirect(url_for('upload.my_uploads'))

    return render_template('upload.html', departments=departments)

@upload_bp.route('/my-uploads')
@login_required
def my_uploads():
    user_id = session['user_id']
    resources = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.uploader_id = ?
        ORDER BY r.created_at DESC
    """, (user_id,))

    counts = {
        'total': len(resources),
        'approved': sum(1 for r in resources if r['status'] == 'approved'),
        'pending': sum(1 for r in resources if r['status'] == 'pending'),
        'rejected': sum(1 for r in resources if r['status'] == 'rejected')
    }

    return render_template('my_uploads.html', resources=resources, counts=counts, format_fil...

@upload_bp.route('/delete-my-upload/<int:resource_id>', methods=['POST'])
@login_required
def delete_my_upload(resource_id):
    user_id = session['user_id']
    res = query_db("SELECT * FROM resources WHERE id = ? AND uploader_id = ?", (resource_id,...
    if not res:
        flash('Resource not found or unauthorized.', 'danger')
        return redirect(url_for('upload.my_uploads'))

    # Remove physical file if exists
    filepath = os.path.join(Config.UPLOAD_FOLDER, res['file_name'])
    if os.path.isfile(filepath):
        try:
            os.remove(filepath)
        except Exception as e:
            print(f"Error deleting file {filepath}: {e}")

    execute_db("DELETE FROM resources WHERE id = ?", (resource_id,))
    log_activity(user_id, 'DELETE_MY_RESOURCE', f"Deleted resource: {res['title']}")
    flash('Resource deleted successfully.', 'success')
    return redirect(url_for('upload.my_uploads'))
```

■ File: routes/admin.py

Administrator Control Panel: Review Queue, Resource Manager, Users, Departments, Subjects, Logs

```
import os
from flask import Blueprint, render_template, request, redirect, url_for, flash, session
from config import Config
from database import query_db, execute_db
from helpers import admin_required, log_activity, format_file_size

admin_bp = Blueprint('admin', __name__, url_prefix='/admin')

@admin_bp.route('/')
@admin_required
def dashboard():
    stats = {
        'total_resources': query_db("SELECT COUNT(*) as c FROM resources", one=True)['c'],
        'approved_resources': query_db("SELECT COUNT(*) as c FROM resources WHERE status = '...
        'pending_resources': query_db("SELECT COUNT(*) as c FROM resources WHERE status = 'p...
        'total_downloads': query_db("SELECT COALESCE(SUM(download_count), 0) as c FROM resou...
        'total_users': query_db("SELECT COUNT(*) as c FROM users", one=True)['c'],
        'total_departments': query_db("SELECT COUNT(*) as c FROM departments", one=True)['c'],
        'total_subjects': query_db("SELECT COUNT(*) as c FROM subjects", one=True)['c']
    }

    # Pending uploads for immediate action
    pending_items = query_db("""
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
               u.full_name as uploader_name, u.username as uploader_username, u.role as uplo...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        LEFT JOIN users u ON r.uploader_id = u.id
        WHERE r.status = 'pending'
        ORDER BY r.created_at ASC
        LIMIT 6
    """)

    # Breakdown by resource type
    type_breakdown = query_db("""
        SELECT resource_type, COUNT(*) as count
        FROM resources
        GROUP BY resource_type
        ORDER BY count DESC
    """)

    # Breakdown by department
    dept_breakdown = query_db("""
        SELECT d.code, d.name, COUNT(r.id) as count
        FROM departments d
        LEFT JOIN resources r ON d.id = r.department_id
        GROUP BY d.id
        ORDER BY count DESC
        LIMIT 6
    """)

    # Recent Audit Logs
    recent_logs = query_db("""
        SELECT l.*, u.username, u.full_name, u.role
        FROM activity_logs l
        LEFT JOIN users u ON l.user_id = u.id
        ORDER BY l.created_at DESC
        LIMIT 8
    """)

    return render_template(
        'admin/dashboard.html',
        stats=stats,
        pending_items=pending_items,
        type_breakdown=type_breakdown,
        dept_breakdown=dept_breakdown,
        recent_logs=recent_logs,
        format_file_size=format_file_size
    )

@admin_bp.route('/pending')
@admin_required
def pending_queue():
    items = query_db("""
```

```
        SELECT r.*, d.name as department_name, d.code as department_code, s.name as subject_...
               u.full_name as uploader_name, u.username as uploader_username, u.email as upl...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        LEFT JOIN users u ON r.uploader_id = u.id
        WHERE r.status = 'pending'
        ORDER BY r.created_at ASC
    """)
    return render_template('admin/pending.html', items=items, format_file_size=format_file_s...

@admin_bp.route('/approve/<int:resource_id>', methods=['POST'])
@admin_required
def approve_resource(resource_id):
    res = query_db("SELECT * FROM resources WHERE id = ?", (resource_id,), one=True)
    if not res:
        flash('Resource not found.', 'danger')
        return redirect(url_for('admin.pending_queue'))

    execute_db("UPDATE resources SET status = 'approved', rejection_reason = NULL, updated_a...
    log_activity(session['user_id'], 'APPROVE_RESOURCE', f"Approved resource '{res['title']}...
    flash(f"Resource '{res['title']}' has been approved and published to the portal.", 'succ...
    return redirect(request.referrer or url_for('admin.pending_queue'))

@admin_bp.route('/reject/<int:resource_id>', methods=['POST'])
@admin_required
def reject_resource(resource_id):
    res = query_db("SELECT * FROM resources WHERE id = ?", (resource_id,), one=True)
    if not res:
        flash('Resource not found.', 'danger')
        return redirect(url_for('admin.pending_queue'))

    reason = request.form.get('reason', 'Content does not meet academic quality guidelines o...
    execute_db("UPDATE resources SET status = 'rejected', rejection_reason = ?, updated_at =...
    log_activity(session['user_id'], 'REJECT_RESOURCE', f"Rejected resource '{res['title']}'...
    flash(f"Resource '{res['title']}' marked as rejected with feedback.", 'warning')
    return redirect(request.referrer or url_for('admin.pending_queue'))

@admin_bp.route('/resources')
@admin_required
def manage_resources():
    q = request.args.get('q', '').strip()
    status_filter = request.args.get('status', '')
    dept_filter = request.args.get('dept', '')

    query_conditions = ["1=1"]
    params = []

    if q:
        query_conditions.append("(r.title LIKE ? OR s.name LIKE ?)")
        params.extend([f"%{q}%", f"%{q}%"])

    if status_filter:
        query_conditions.append("r.status = ?")
        params.append(status_filter)

    if dept_filter and dept_filter.isdigit():
        query_conditions.append("r.department_id = ?")
        params.append(int(dept_filter))

    where_clause = " WHERE " + " AND ".join(query_conditions)

    resources = query_db(f"""
        SELECT r.*, d.code as department_code, s.name as subject_name, u.full_name as upload...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        LEFT JOIN users u ON r.uploader_id = u.id
        {where_clause}
        ORDER BY r.created_at DESC
    """, params)

    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")

    return render_template(
        'admin/resources.html',
        resources=resources,
        departments=departments,
```

```
        q=q,
        status_filter=status_filter,
        dept_filter=dept_filter,
        format_file_size=format_file_size
    )

@admin_bp.route('/resource/toggle-featured/<int:resource_id>', methods=['POST'])
@admin_required
def toggle_featured(resource_id):
    res = query_db("SELECT is_featured, title FROM resources WHERE id = ?", (resource_id,), ...
    if not res:
        flash('Resource not found.', 'danger')
        return redirect(url_for('admin.manage_resources'))

    new_val = 0 if res['is_featured'] else 1
    execute_db("UPDATE resources SET is_featured = ? WHERE id = ?", (new_val, resource_id))
    state_str = "featured" if new_val else "unfeatured"

    log_activity(session['user_id'], 'TOGGLE_FEATURED', f"Marked '{res['title']}' as {state_...
    flash(f"Resource '{res['title']}' is now {state_str}.", 'success')
    return redirect(request.referrer or url_for('admin.manage_resources'))

@admin_bp.route('/resource/delete/<int:resource_id>', methods=['POST'])
@admin_required
def delete_resource(resource_id):
    res = query_db("SELECT * FROM resources WHERE id = ?", (resource_id,), one=True)
    if not res:
        flash('Resource not found.', 'danger')
        return redirect(url_for('admin.manage_resources'))

    # Delete physical file
    filepath = os.path.join(Config.UPLOAD_FOLDER, res['file_name'])
    if os.path.isfile(filepath):
        try:
            os.remove(filepath)
        except Exception as e:
            print(f"Error removing file: {e}")

    execute_db("DELETE FROM resources WHERE id = ?", (resource_id,))
    log_activity(session['user_id'], 'ADMIN_DELETE_RESOURCE', f"Admin deleted resource '{res...
    flash(f"Resource '{res['title']}' deleted successfully.", 'success')
    return redirect(request.referrer or url_for('admin.manage_resources'))

@admin_bp.route('/users')
@admin_required
def manage_users():
    users = query_db("""
        SELECT u.*, d.name as department_name, d.code as department_code,
               (SELECT COUNT(*) FROM resources WHERE uploader_id = u.id) as upload_count
        FROM users u
        LEFT JOIN departments d ON u.department_id = d.id
        ORDER BY u.created_at DESC
    """)
    departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")
    return render_template('admin/users.html', users=users, departments=departments)

@admin_bp.route('/user/update-role/<int:user_id>', methods=['POST'])
@admin_required
def update_user_role(user_id):
    if user_id == session['user_id']:
        flash('You cannot change your own administrative role.', 'danger')
        return redirect(url_for('admin.manage_users'))

    new_role = request.form.get('role')
    if new_role not in ('student', 'faculty', 'admin'):
        flash('Invalid role specified.', 'warning')
        return redirect(url_for('admin.manage_users'))

    execute_db("UPDATE users SET role = ? WHERE id = ?", (new_role, user_id))
    log_activity(session['user_id'], 'UPDATE_USER_ROLE', f"Changed user ID {user_id} role to...
    flash('User role updated successfully.', 'success')
    return redirect(url_for('admin.manage_users'))

@admin_bp.route('/user/toggle-active/<int:user_id>', methods=['POST'])
@admin_required
def toggle_user_active(user_id):
    if user_id == session['user_id']:
        flash('You cannot deactivate your own administrative account.', 'danger')
        return redirect(url_for('admin.manage_users'))
```

```
232 |
233 |     user = query_db("SELECT is_active, username FROM users WHERE id = ?", (user_id,), one=True)
234 |     if not user:
235 |         flash('User not found.', 'danger')
236 |         return redirect(url_for('admin.manage_users'))
237 |
238 |     new_state = 0 if user['is_active'] else 1
239 |     execute_db("UPDATE users SET is_active = ? WHERE id = ?", (new_state, user_id))
240 |     action_text = "activated" if new_state else "deactivated"
241 |     log_activity(session['user_id'], 'TOGGLE_USER_STATUS', f"User {user['username']} {action...
242 |     flash(f"User account '{user['username']}' has been {action_text}.", 'info')
243 |     return redirect(url_for('admin.manage_users'))
244 |
245 | @admin_bp.route('/departments', methods=['GET', 'POST'])
246 | @admin_required
247 | def manage_departments():
248 |     if request.method == 'POST':
249 |         name = request.form.get('name', '').strip()
250 |         code = request.form.get('code', '').strip().upper()
251 |         description = request.form.get('description', '').strip()
252 |         icon = request.form.get('icon', 'fas fa-graduation-cap').strip()
253 |
254 |         if not name or not code:
255 |             flash('Department Name and Department Code are required.', 'warning')

256 |         else:
257 |             try:
258 |                 execute_db(
259 |                     "INSERT INTO departments (name, code, description, icon) VALUES (?, ?, ?...
260 |                     (name, code, description, icon)
261 |                 )
262 |                 log_activity(session['user_id'], 'ADD_DEPARTMENT', f"Created department '{co...
263 |                 flash(f"Department '{code}' created successfully!", 'success')
264 |             except Exception as e:
265 |                 flash(f"Error adding department (possibly duplicate code or name): {e}", 'da...
266 |         return redirect(url_for('admin.manage_departments'))
267 |
268 |     departments = query_db("""
269 |         SELECT d.*,
270 |                (SELECT COUNT(*) FROM subjects WHERE department_id = d.id) as subject_count,
271 |                (SELECT COUNT(*) FROM resources WHERE department_id = d.id) as resource_count
272 |         FROM departments d
273 |         ORDER BY d.name ASC
274 |     """)
275 |     return render_template('admin/departments.html', departments=departments)
276 |
277 | @admin_bp.route('/subjects', methods=['GET', 'POST'])
278 | @admin_required
279 | def manage_subjects():
280 |     dept_id = request.args.get('dept')
281 |
282 |     if request.method == 'POST':
283 |         name = request.form.get('name', '').strip()
284 |         code = request.form.get('code', '').strip().upper()
285 |         department_id = request.form.get('department_id')
286 |         semester = request.form.get('semester')
287 |         description = request.form.get('description', '').strip()
288 |
289 |         if not name or not code or not department_id or not semester:
290 |             flash('All subject fields must be filled.', 'warning')
291 |         else:
292 |             try:
293 |                 execute_db(
294 |                     "INSERT INTO subjects (name, code, department_id, semester, description)...
295 |                     (name, code, int(department_id), int(semester), description)
296 |                 )
297 |                 log_activity(session['user_id'], 'ADD_SUBJECT', f"Created subject '{code} - ...
298 |                 flash(f"Subject '{code}' added successfully!", 'success')
299 |             except Exception as e:
300 |                 flash(f"Error adding subject (possibly duplicate code): {e}", 'danger')
301 |         return redirect(url_for('admin.manage_subjects', dept=department_id))
302 |
303 |     departments = query_db("SELECT * FROM departments WHERE is_active = 1 ORDER BY name ASC")
304 |
305 |     if dept_id and dept_id.isdigit():
306 |         subjects = query_db("""
307 |             SELECT s.*, d.name as department_name, d.code as department_code,
308 |                    (SELECT COUNT(*) FROM resources WHERE subject_id = s.id) as resource_count
309 |             FROM subjects s
```

```
310 |             JOIN departments d ON s.department_id = d.id
311 |             WHERE s.department_id = ?
312 |             ORDER BY s.semester, s.name
313 |         """, (int(dept_id),))
314 |     else:
315 |         subjects = query_db("""
316 |             SELECT s.*, d.name as department_name, d.code as department_code,
317 |                    (SELECT COUNT(*) FROM resources WHERE subject_id = s.id) as resource_count
318 |             FROM subjects s
319 |             JOIN departments d ON s.department_id = d.id
320 |             ORDER BY d.name, s.semester, s.name
321 |         """)
322 |
323 |     return render_template('admin/subjects.html', subjects=subjects, departments=departments...
324 |
325 | @admin_bp.route('/logs')
326 | @admin_required
327 | def activity_logs():
328 |     logs = query_db("""
329 |         SELECT l.*, u.username, u.full_name, u.role
330 |         FROM activity_logs l
331 |         LEFT JOIN users u ON l.user_id = u.id
332 |         ORDER BY l.created_at DESC
333 |         LIMIT 100
334 |     """)
335 |     return render_template('admin/logs.html', logs=logs)
```

■ File: routes/api.py

REST API Endpoints: Dynamic Cascading Subject Loader & Live Autocomplete Search

```
from flask import Blueprint, request, jsonify
from database import query_db

api_bp = Blueprint('api', __name__, url_prefix='/api')

@api_bp.route('/subjects')
def get_subjects():
    dept_id = request.args.get('dept_id')
    semester = request.args.get('semester')

    query_str = "SELECT id, name, code, semester FROM subjects WHERE is_active = 1"
    params = []

    if dept_id and dept_id.isdigit():
        query_str += " AND department_id = ?"
        params.append(int(dept_id))

    if semester and semester.isdigit():
        query_str += " AND semester = ?"
        params.append(int(semester))

    query_str += " ORDER BY semester ASC, name ASC"
    subjects = query_db(query_str, params)

    return jsonify([
        {
            'id': s['id'],
            'name': s['name'],
            'code': s['code'],
            'semester': s['semester']
        } for s in subjects
    ])

@api_bp.route('/quick-search')
def quick_search():
    q = request.args.get('q', '').strip()
    if not q or len(q) < 2:
        return jsonify([])

    search_term = f"%{q}%"
    results = query_db("""
        SELECT r.id, r.title, r.resource_type, r.academic_year, d.code as dept_code, s.name ...
        FROM resources r
        JOIN departments d ON r.department_id = d.id
        JOIN subjects s ON r.subject_id = s.id
        WHERE r.status = 'approved' AND (r.title LIKE ? OR r.tags LIKE ? OR s.name LIKE ?)
        ORDER BY r.download_count DESC
        LIMIT 6
    """, (search_term, search_term, search_term))

    return jsonify([
        {
            'id': r['id'],
            'title': r['title'],
            'type': r['resource_type'],
            'year': r['academic_year'],
            'dept': r['dept_code'],
            'subject': r['subject_name']
        } for r in results
    ])
```

■ File: templates/base.html

Master Responsive Template: Navbar, Offline LAN Notification Bar, Footer, PWA Registration

```
<!DOCTYPE html>
<html lang="en" class="h-full">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{% block title %}College Study Material & Previous Question Paper Portal{% endblo...

    <!-- PWA & Mobile Meta -->
    <meta name="theme-color" content="#2563eb">
    <link rel="manifest" href="{{ url_for('static', filename='manifest.json') }}">
    <link rel="icon" type="image/png" href="https://cdn-icons-png.flaticon.com/512/2997/2997...

    <!-- Tailwind CSS CDN -->
    <script src="https://cdn.tailwindcss.com"></script>
    <script>
        tailwind.config = {
            darkMode: 'class',
            theme: {
                extend: {
                    colors: {
                        primary: {
                            50: '#eff6ff',
                            100: '#dbeafe',
                            200: '#bfdbfe',
                            500: '#3b82f6',
                            600: '#2563eb',
                            700: '#1d4ed8',
                            800: '#1e40af',
                            900: '#1e3a8a',
                        },
                        academic: {
                            blue: '#1e40af',
                            amber: '#d97706',
                            emerald: '#059669',
                            purple: '#7c3aed'
                        }
                    }
                }
            }
        }
    </script>

    <!-- FontAwesome 6 -->
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/c...

    <!-- Google Fonts: Inter -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&...

    <style>
        body { font-family: 'Inter', sans-serif; }
        .glass-header {
            background: rgba(255, 255, 255, 0.95);
            backdrop-filter: blur(8px);
        }
        .dark .glass-header {
            background: rgba(15, 23, 42, 0.95);
        }
    </style>
    {% block extra_head %}{% endblock %}
</head>
<body class="flex flex-col min-h-screen bg-slate-50 text-slate-800 antialiased selection:bg-...

    <!-- Top Local Network / Offline Access Notification Bar -->
    <aside aria-label="Local network access status" class="bg-gradient-to-r from-blue-900 vi...
        <div class="max-w-7xl mx-auto flex flex-wrap items-center justify-between gap-2">
            <div class="flex items-center space-x-2">
                <span class="inline-flex items-center px-2 py-0.5 rounded text-[11px] font-s...
                    <span class="w-1.5 h-1.5 mr-1.5 bg-emerald-400 rounded-full animate-puls...
                </span>
                <span class="hidden sm:inline text-slate-300">College Intranet / Local Wi-Fi...
                <code class="bg-black/40 px-2 py-0.5 rounded text-amber-300 font-mono font-m...
                    http://{{ lan_ip }}:{{ server_port }}
                </code>
```

```
            </div>
            <div class="flex items-center space-x-4 text-slate-300">
                <span class="hidden md:inline"><i class="fas fa-wifi text-emerald-400 mr-1">...
                <button onclick="navigator.clipboard.writeText('http://{{ lan_ip }}:{{ serve...
                        class="hover:text-white underline text-[11px] flex items-center">
                    <i class="far fa-copy mr-1"></i> Copy Link
                </button>
            </div>
        </div>
    </aside>

    <!-- Main Navigation Bar -->
    <header class="sticky top-0 z-40 glass-header border-b border-slate-200 shadow-sm transi...
        <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
            <div class="flex items-center justify-between h-16">
                <!-- Brand / Logo -->
                <div class="flex items-center space-x-3">
                    <a href="{{ url_for('main.index') }}" class="flex items-center space-x-3...
                        <div class="w-10 h-10 rounded-xl bg-gradient-to-tr from-blue-700 to-...
                            <i class="fas fa-book-open-reader text-lg"></i>
                        </div>
                        <div>
                            <span class="text-lg font-bold tracking-tight text-slate-900 gro...
                                Academix<span class="text-blue-600 font-extrabold ml-0.5">Po...
                            </span>
                            <span class="block text-[10px] text-slate-500 uppercase tracking...
                                Study Materials & PYQ Hub
                            </span>
                        </div>
                    </a>
                </div>

                <!-- Live Quick Search Bar (Desktop) -->
                <div class="hidden lg:block flex-1 max-w-md mx-8 relative">
                    <form action="{{ url_for('main.browse') }}" method="GET" class="relative">
                        <div class="relative">
                            <input type="text" name="q" id="global-search-input"
                                   placeholder="Search previous papers, notes, subjects (e.g...
                                   class="w-full bg-slate-100 hover:bg-slate-200/70 focus:bg...
                            <div class="absolute inset-y-0 left-0 pl-3 flex items-center poi...
                                <i class="fas fa-search text-xs"></i>
                            </div>
                        </div>
                    </form>
                    <!-- Quick Search Autocomplete Dropdown -->
                    <div id="quick-search-results" class="hidden absolute left-0 right-0 top...
                    </div>
                </div>

                <!-- Navigation Links -->
                <nav class="hidden md:flex items-center space-x-1 lg:space-x-2">
                    <a href="{{ url_for('main.index') }}"
                       class="px-3 py-2 rounded-lg text-sm font-medium text-slate-700 hover:...
                        <i class="fas fa-house-chimney text-xs mr-1.5 text-slate-400"></i> Home
                    </a>
                    <a href="{{ url_for('main.browse') }}"
                       class="px-3 py-2 rounded-lg text-sm font-medium text-slate-700 hover:...
                        <i class="fas fa-layer-group text-xs mr-1.5 text-slate-400"></i> All...
                    </a>
                    <a href="{{ url_for('main.browse', type='Question Paper') }}"
                       class="px-3 py-2 rounded-lg text-sm font-medium text-slate-700 hover:...
                        <i class="fas fa-file-invoice text-xs mr-1.5 text-amber-500"></i> PYQs
                    </a>
                    <a href="{{ url_for('main.browse', type='Lecture Notes') }}"
                       class="px-3 py-2 rounded-lg text-sm font-medium text-slate-700 hover:...
                        <i class="fas fa-file-lines text-xs mr-1.5 text-emerald-500"></i> Notes
                    </a>
                    <a href="{{ url_for('main.presentation') }}"
                       class="px-3 py-2 rounded-lg text-sm font-medium text-slate-700 hover:...
                        <i class="fas fa-play-circle text-xs mr-1.5 text-rose-500"></i> Vide...
                    </a>

                    <!-- Upload Action Button -->
                    <a href="{{ url_for('upload.upload') }}"
                       class="ml-2 inline-flex items-center px-3.5 py-2 rounded-xl text-sm f...
                        <i class="fas fa-cloud-arrow-up mr-1.5"></i> Upload Resource
                    </a>
                </nav>
```

```
154 |
155 |                 <!-- User Session Menu / Login Actions -->
156 |                 <div class="flex items-center space-x-2">
157 |                     {% if current_user %}
158 |                         <!-- Admin Notification Badge -->
159 |                         {% if current_user.role == 'admin' %}
160 |                             <a href="{{ url_for('admin.pending_queue') }}"
161 |                                class="relative p-2 text-slate-600 hover:text-blue-600 rounde...
162 |                                title="Pending Approvals">
163 |                                 <i class="fas fa-bell text-base"></i>
164 |                                 {% if pending_review_count > 0 %}
165 |                                     <span class="absolute top-1 right-1 inline-flex items-ce...
166 |                                         {{ pending_review_count }}
167 |                                     </span>
168 |                                 {% endif %}
169 |                             </a>
170 |                         {% endif %}

171 |
172 |                         <!-- User Profile Dropdown -->
173 |                         <div class="relative" id="user-menu-container">
174 |                             <button id="user-menu-button" type="button"
175 |                                     class="flex items-center space-x-2 p-1.5 rounded-xl hove...
176 |                                 <div class="w-8 h-8 rounded-lg bg-gradient-to-br from-indigo...
177 |                                     {{ current_user.full_name[:2].upper() }}
178 |                                 </div>
179 |                                 <span class="hidden sm:inline text-xs font-semibold text-sla...
180 |                                     {{ current_user.full_name }}
181 |                                 </span>
182 |                                 <i class="fas fa-chevron-down text-[10px] text-slate-400"></i>
183 |                             </button>
184 |
185 |                             <!-- Dropdown Menu -->
186 |                             <div id="user-menu-dropdown" class="hidden absolute right-0 mt-2...
187 |                                 <div class="px-4 py-2 border-b border-slate-100">
188 |                                     <p class="text-xs text-slate-500">Signed in as</p>
189 |                                     <p class="text-sm font-bold text-slate-800 truncate">{{ ...
190 |                                     <span class="inline-block mt-1 px-2 py-0.5 text-[10px] f...
191 |                                         {% if current_user.role == 'admin' %}bg-rose-100 tex...
192 |                                         {% elif current_user.role == 'faculty' %}bg-amber-10...
193 |                                         {% else %}bg-blue-100 text-blue-700{% endif %}">
194 |                                         {{ current_user.role }}
195 |                                     </span>
196 |                                 </div>
197 |
198 |                                 {% if current_user.role == 'admin' %}
199 |                                     <a href="{{ url_for('admin.dashboard') }}" class="flex i...
200 |                                         <i class="fas fa-gauge-high w-5 text-slate-400"></i>...
201 |                                     </a>
202 |                                     <a href="{{ url_for('admin.pending_queue') }}" class="fl...
203 |                                         <span class="flex items-center"><i class="fas fa-inb...
204 |                                         {% if pending_review_count > 0 %}
205 |                                             <span class="bg-rose-500 text-white text-[10px] ...
206 |                                         {% endif %}
207 |                                     </a>
208 |                                     <a href="{{ url_for('admin.manage_resources') }}" class=...
209 |                                         <i class="fas fa-folder-open w-5 text-slate-400"></i...
210 |                                     </a>
211 |                                     <div class="border-t border-slate-100 my-1"></div>
212 |                                 {% endif %}
213 |
214 |                                 <a href="{{ url_for('upload.my_uploads') }}" class="flex ite...
215 |                                     <i class="fas fa-cloud-arrow-up w-5 text-slate-400"></i>...
216 |                                 </a>
217 |                                 <a href="{{ url_for('main.bookmarks') }}" class="flex items-...
218 |                                     <i class="fas fa-bookmark w-5 text-slate-400"></i> Saved...
219 |                                 </a>
220 |                                 <a href="{{ url_for('auth.profile') }}" class="flex items-ce...
221 |                                     <i class="fas fa-user-gear w-5 text-slate-400"></i> Acco...
222 |                                 </a>
223 |                                 <div class="border-t border-slate-100 my-1"></div>
224 |                                 <a href="{{ url_for('auth.logout') }}" class="flex items-cen...
225 |                                     <i class="fas fa-arrow-right-from-bracket w-5"></i> Sign...
226 |                                 </a>
227 |                             </div>
228 |                         </div>
229 |                     {% else %}
230 |                         <a href="{{ url_for('auth.login') }}"
231 |                            class="px-3.5 py-1.5 rounded-xl text-sm font-medium text-slate-70...
```

```
                            Sign In
                        </a>
                        <a href="{{ url_for('auth.register') }}"
                           class="hidden sm:inline-flex items-center px-3.5 py-1.5 rounded-x...
                            Register
                        </a>
                    {% endif %}

                    <!-- Mobile Menu Button -->
                    <button id="mobile-menu-btn" type="button" class="md:hidden p-2 rounded-...
                        <i class="fas fa-bars text-lg"></i>
                    </button>
                </div>
            </div>
        </div>

        <!-- Mobile Navigation Menu -->
        <div id="mobile-menu" class="hidden md:hidden border-t border-slate-200 bg-white px-...
            <form action="{{ url_for('main.browse') }}" method="GET" class="pt-2 pb-1">
                <input type="text" name="q" placeholder="Search papers, notes..." class="w-f...
            </form>
            <a href="{{ url_for('main.index') }}" class="block px-3 py-2 rounded-lg text-sm ...
            <a href="{{ url_for('main.browse') }}" class="block px-3 py-2 rounded-lg text-sm...
            <a href="{{ url_for('main.browse', type='Question Paper') }}" class="block px-3 ...

            <a href="{{ url_for('main.browse', type='Lecture Notes') }}" class="block px-3 p...
            <a href="{{ url_for('main.presentation') }}" class="block px-3 py-2 rounded-lg t...
            <a href="{{ url_for('upload.upload') }}" class="block px-3 py-2 rounded-lg text-...
        </div>
    </header>

    <!-- Flash Messages Container -->
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 mt-4 w-full">
        {% with messages = get_flashed_messages(with_categories=true) %}
            {% if messages %}
                {% for category, message in messages %}
                    {% set cat_color = 'blue' %}
                    {% if category == 'danger' %}{% set cat_color = 'rose' %}
                    {% elif category == 'success' %}{% set cat_color = 'emerald' %}
                    {% elif category == 'warning' %}{% set cat_color = 'amber' %}
                    {% elif category == 'info' %}{% set cat_color = 'blue' %}{% endif %}

                    <div class="mb-3 flex items-center justify-between p-4 rounded-xl border...
                        <div class="flex items-center space-x-3">
                            <i class="fas {% if category == 'success' %}fa-circle-check text...
                                          {% elif category == 'danger' %}fa-circle-exclamat...
                                          {% elif category == 'warning' %}fa-triangle-excla...
                                          {% else %}fa-circle-info text-blue-600{% endif %}...
                            <span class="text-sm font-medium">{{ message }}</span>
                        </div>
                        <button type="button" onclick="this.parentElement.remove()" class="t...
                            <i class="fas fa-xmark"></i>
                        </button>
                    </div>
                {% endfor %}
            {% endif %}
        {% endwith %}
    </div>

    <!-- Main Body Content Block -->
    <main class="flex-1 w-full">
        {% block content %}{% endblock %}
    </main>

    <!-- Footer -->
    <footer class="bg-slate-900 text-slate-300 border-t border-slate-800 mt-16 pt-12 pb-8">
        <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
            <div class="grid grid-cols-1 md:grid-cols-4 gap-8 mb-8">
                <!-- Portal Info -->
                <div class="space-y-3">
                    <div class="flex items-center space-x-2">
                        <div class="w-8 h-8 rounded-lg bg-blue-600 flex items-center justify...
                            <i class="fas fa-book-open-reader text-sm"></i>
                        </div>
                        <span class="text-lg font-bold text-white">Academix Portal</span>
                    </div>
                    <p class="text-xs text-slate-400 leading-relaxed">
                        Centralized repository for college study materials, previous year ex...
                    </p>
```

```
                    <div class="flex items-center space-x-2 text-xs text-emerald-400">
                        <span class="w-2 h-2 rounded-full bg-emerald-400 animate-ping"></span>
                        <span>Campus LAN Active at http://{{ lan_ip }}:{{ server_port }}</span>
                    </div>
                </div>

                <!-- Academic Quick Links -->
                <div>
                    <h3 class="text-xs font-bold text-white uppercase tracking-wider mb-3">R...
                    <ul class="space-y-2 text-xs">
                        <li><a href="{{ url_for('main.browse', type='Question Paper') }}" cl...
                        <li><a href="{{ url_for('main.browse', type='Lecture Notes') }}" cla...
                        <li><a href="{{ url_for('main.browse', type='Lab Manual') }}" class=...
                        <li><a href="{{ url_for('main.browse', type='Assignment') }}" class=...
                        <li><a href="{{ url_for('main.browse', type='Syllabus') }}" class="h...
                    </ul>
                </div>

                <!-- Popular Engineering Departments -->
                <div>
                    <h3 class="text-xs font-bold text-white uppercase tracking-wider mb-3">T...
                    <ul class="space-y-2 text-xs">
                        <li><a href="{{ url_for('main.browse', dept=1) }}" class="hover:text...
                        <li><a href="{{ url_for('main.browse', dept=2) }}" class="hover:text...
                        <li><a href="{{ url_for('main.browse', dept=3) }}" class="hover:text...
                        <li><a href="{{ url_for('main.browse', dept=4) }}" class="hover:text...
                        <li><a href="{{ url_for('main.browse', dept=6) }}" class="hover:text...
                    </ul>
                </div>

                <!-- Offline Access & Project Info -->

                <div class="space-y-3">
                    <h3 class="text-xs font-bold text-white uppercase tracking-wider">Offlin...
                    <p class="text-xs text-slate-400 leading-relaxed">
                        Students can connect their smartphones and laptops to the host compu...
                    </p>
                    <div class="p-2.5 rounded-lg bg-slate-800/80 border border-slate-700/80 ...
                        <i class="fas fa-qrcode mr-1 text-amber-400"></i> Open browser & vis...
                        <span class="text-amber-300 font-mono font-bold select-all">http://{...
                    </div>
                </div>
            </div>

            <div class="border-t border-slate-800 pt-6 flex flex-col sm:flex-row items-cente...
                <p>&copy; 2026 Centralized College Study Material & Previous Question Paper ...
                <div class="flex items-center space-x-4 mt-2 sm:mt-0">
                    <span>Designed for Academic Excellence</span>
                    <span>•</span>
                    <a href="{{ url_for('main.offline') }}" class="hover:text-blue-400">Offl...
                </div>
            </div>
        </div>
    </footer>

    <!-- Core Scripts -->
    <script src="{{ url_for('static', filename='js/main.js') }}"></script>
    <script>
        // Register Service Worker for offline PWA capabilities
        if ('serviceWorker' in navigator) {
            window.addEventListener('load', () => {
                navigator.serviceWorker.register("{{ url_for('static', filename='js/sw.js') ...
                    .then(reg => console.log('Service Worker Registered for Offline Access',...
                    .catch(err => console.log('Service Worker Registration error:', err));
            });
        }

        // Toggle user dropdown
        const userMenuBtn = document.getElementById('user-menu-button');
        const userMenuDropdown = document.getElementById('user-menu-dropdown');
        if (userMenuBtn && userMenuDropdown) {
            userMenuBtn.addEventListener('click', (e) => {
                e.stopPropagation();
                userMenuDropdown.classList.toggle('hidden');
            });
            document.addEventListener('click', () => {
                if (!userMenuDropdown.classList.contains('hidden')) {
                    userMenuDropdown.classList.add('hidden');
                }
```

```
            });
        }

        // Toggle mobile menu
        const mobileMenuBtn = document.getElementById('mobile-menu-btn');
        const mobileMenu = document.getElementById('mobile-menu');
        if (mobileMenuBtn && mobileMenu) {
            mobileMenuBtn.addEventListener('click', () => {
                mobileMenu.classList.toggle('hidden');
            });
        }
    </script>
    {% block extra_scripts %}{% endblock %}
</body>
</html>
```

■ File: templates/index.html

Homepage: Hero Search, Statistics Counter, Department Grid, Featured Materials

```
{% extends 'base.html' %}

{% block title %}Academix Portal | Centralized College Study Materials & Previous Question P...

{% block content %}
<!-- Hero Section with Academic Background -->
<div class="relative bg-gradient-to-b from-blue-900 via-indigo-900 to-slate-900 text-white o...
    <!-- Background subtle academic pattern decoration -->
    <div class="absolute inset-0 opacity-10 bg-[radial-gradient(#ffffff_1px,transparent_1px)...
    <div class="absolute -top-24 -right-24 w-96 h-96 bg-blue-500/20 rounded-full blur-3xl po...
    <div class="absolute -bottom-24 -left-24 w-96 h-96 bg-indigo-500/20 rounded-full blur-3x...

    <div class="relative max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 text-center">
        <!-- Badge -->
        <div class="inline-flex items-center space-x-2 px-3 py-1 rounded-full bg-blue-500/10...
            <i class="fas fa-graduation-cap"></i>
            <span>Centralized Academic Knowledge Sharing</span>
        </div>

        <h1 class="text-3xl sm:text-5xl lg:text-6xl font-extrabold tracking-tight text-white...
            One Stop Portal for <br class="hidden sm:inline">
            <span class="text-transparent bg-clip-text bg-gradient-to-r from-blue-400 via-sk...
                Study Materials & Previous Question Papers
            </span>
        </h1>

        <p class="max-w-2xl mx-auto text-base sm:text-lg text-slate-300 mb-10 leading-relaxed">
            Easily search, upload, share, and download syllabus, lecture notes, lab manuals,...
        </p>

        <!-- Search Bar Card -->
        <div class="max-w-3xl mx-auto bg-white/10 backdrop-blur-md p-2 sm:p-3 rounded-2xl bo...
            <form action="{{ url_for('main.browse') }}" method="GET" class="flex flex-col sm...
                <div class="flex-1 relative">
                    <i class="fas fa-search absolute left-4 top-3.5 text-slate-400"></i>
                    <input type="text" name="q" placeholder="Enter subject name, paper topic...
                           class="w-full pl-11 pr-4 py-3 bg-white text-slate-800 text-sm rou...
                </div>
                <div class="w-full sm:w-48">
                    <select name="type" class="w-full py-3 px-3 bg-white text-slate-700 text...
                        <option value="">All Categories</option>
                        <option value="Question Paper">Previous Question Papers</option>
                        <option value="Lecture Notes">Lecture Notes</option>
                        <option value="Lab Manual">Lab Manuals</option>
                        <option value="Assignment">Assignments</option>
                        <option value="Syllabus">Curriculum / Syllabus</option>
                    </select>
                </div>
                <button type="submit"
                        class="px-6 py-3 bg-blue-600 hover:bg-blue-500 text-white font-semib...
                    <span>Search</span>
                    <i class="fas fa-arrow-right text-xs"></i>
                </button>
            </form>
        </div>

        <!-- Quick Filters Chips -->
        <div class="mt-6 flex flex-wrap items-center justify-center gap-2 text-xs">
            <span class="text-slate-400 font-medium">Quick Filters:</span>
            <a href="{{ url_for('main.browse', type='Question Paper', exam='End-Semester') }...
                <i class="fas fa-file-invoice mr-1 text-amber-400"></i> End-Semester PYQs
            </a>
            <a href="{{ url_for('main.browse', type='Question Paper', exam='Mid-Term 1') }}"...
                <i class="fas fa-file-lines mr-1 text-sky-400"></i> Mid-Term Papers
            </a>
            <a href="{{ url_for('main.browse', type='Lecture Notes') }}" class="px-3 py-1 ro...
                <i class="fas fa-chalkboard-user mr-1 text-emerald-400"></i> Faculty Notes
            </a>
            <a href="{{ url_for('main.browse', type='Lab Manual') }}" class="px-3 py-1 round...
                <i class="fas fa-flask mr-1 text-purple-400"></i> Lab Manuals
            </a>
        </div>
    </div>
</div>

```

```
<!-- Statistics Counter Ribbon -->
<div class="relative -mt-8 max-w-6xl mx-auto px-4 sm:px-6 lg:px-8 z-10">
    <div class="bg-white rounded-2xl shadow-xl border border-slate-200/80 p-6 grid grid-cols...
        <div class="pt-2 md:pt-0">
            <div class="text-3xl font-extrabold text-blue-600 mb-1">{{ stats.total_resources...
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Verif...
        </div>
        <div class="pt-2 md:pt-0">
            <div class="text-3xl font-extrabold text-indigo-600 mb-1">{{ stats.total_downloa...
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Total...

        </div>
        <div class="pt-2 md:pt-0">
            <div class="text-3xl font-extrabold text-emerald-600 mb-1">{{ stats.total_depart...
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Acade...
        </div>
        <div class="pt-2 md:pt-0">
            <div class="text-3xl font-extrabold text-purple-600 mb-1">{{ stats.total_users }...
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Activ...
        </div>
    </div>
</div>

<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 mt-16 space-y-16">

    <!-- Academic Departments Section -->
    <section>
        <div class="flex flex-col sm:flex-row sm:items-end justify-between mb-8">
            <div>
                <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Curri...
                <h2 class="text-2xl font-bold text-slate-900 mt-1">Explore by Department</h2>
                <p class="text-sm text-slate-500 mt-1">Select your engineering or science de...
            </div>
            <a href="{{ url_for('main.browse') }}" class="mt-4 sm:mt-0 text-sm font-semibold...
                <span>View all departments</span>
                <i class="fas fa-chevron-right text-xs"></i>
            </a>
        </div>

        <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">
            {% for dept in departments %}
                <a href="{{ url_for('main.browse', dept=dept.id) }}"
                   class="group bg-white rounded-2xl p-6 border border-slate-200 hover:borde...
                    <div class="w-12 h-12 rounded-xl bg-blue-50 text-blue-600 flex items-cen...
                        <i class="{{ dept.icon }}"></i>
                    </div>
                    <div class="flex items-center justify-between mb-1">
                        <span class="text-xs font-bold px-2 py-0.5 rounded bg-slate-100 text...
                        <span class="text-xs font-semibold text-emerald-600">{{ dept.resourc...
                    </div>
                    <h3 class="text-base font-bold text-slate-900 group-hover:text-blue-600 ...
                        {{ dept.name }}
                    </h3>
                    <p class="text-xs text-slate-500 mt-2 line-clamp-2 leading-relaxed">
                        {{ dept.description }}
                    </p>
                </a>
            {% endfor %}
        </div>
    </section>

    <!-- Featured High-Yield Study Materials -->
    {% if featured %}
    <section>
        <div class="flex flex-col sm:flex-row sm:items-end justify-between mb-8">
            <div>
                <span class="text-xs font-bold text-amber-500 uppercase tracking-wider">Top ...
                <h2 class="text-2xl font-bold text-slate-900 mt-1">Featured Study Materials<...
                <p class="text-sm text-slate-500 mt-1">Hand-picked previous papers and notes...
            </div>
            <a href="{{ url_for('main.browse') }}" class="mt-4 sm:mt-0 text-sm font-semibold...
                <span>Browse all</span>
                <i class="fas fa-chevron-right text-xs"></i>
            </a>
        </div>

        <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {% for item in featured %}
                <div class="bg-white rounded-2xl border border-slate-200 hover:border-slate-...
```

```
                    <div>
                        <!-- Category Badge & Year -->
                        <div class="flex items-center justify-between text-xs mb-3">
                            <span class="inline-flex items-center px-2.5 py-1 rounded-md fon...
                                {% if item.resource_type == 'Question Paper' %}bg-amber-100 ...
                                {% elif item.resource_type == 'Lecture Notes' %}bg-emerald-1...
                                {% elif item.resource_type == 'Lab Manual' %}bg-purple-100 t...
                                {% else %}bg-blue-100 text-blue-800{% endif %}">
                                <i class="fas fa-file-lines mr-1.5 text-xs"></i> {{ item.res...
                            </span>
                            <span class="text-slate-400 font-medium font-mono text-[11px]">{...
                        </div>

                        <!-- Department & Subject -->
                        <div class="text-xs font-semibold text-blue-600 mb-1">
                            {{ item.department_code }} • Sem {{ item.semester }} • {{ item.s...
                        </div>


                        <!-- Title -->
                        <h3 class="text-base font-bold text-slate-900 line-clamp-2 hover:tex...
                            <a href="{{ url_for('main.resource_detail', resource_id=item.id)...
                                {{ item.title }}
                            </a>
                        </h3>

                        <!-- Description -->
                        <p class="text-xs text-slate-500 line-clamp-2 mb-4 leading-relaxed">
                            {{ item.description }}
                        </p>
                    </div>

                    <!-- Card Footer: Stats & Download -->
                    <div class="border-t border-slate-100 pt-4 flex items-center justify-bet...
                        <div class="flex items-center space-x-3">
                            <span title="Downloads"><i class="fas fa-download mr-1 text-slat...
                            {% if item.rating_count > 0 %}
                                <span class="flex items-center text-amber-500 font-bold">
                                    <i class="fas fa-star mr-1"></i> {{ "%.1f"|format(item.a...
                                </span>
                            {% endif %}
                        </div>

                        <div class="flex items-center space-x-2">
                            <a href="{{ url_for('main.resource_detail', resource_id=item.id)...
                               class="px-3 py-1.5 rounded-lg text-slate-700 bg-slate-100 hov...
                                Preview
                            </a>
                            <a href="{{ url_for('main.download_resource', resource_id=item.i...
                               class="px-3 py-1.5 rounded-lg text-white bg-blue-600 hover:bg...
                                <i class="fas fa-download text-[10px]"></i>
                                <span>Get File</span>
                            </a>
                        </div>
                    </div>
                </div>
            {% endfor %}
        </div>
    </section>
    {% endif %}

    <!-- Recent Question Papers Highlight Section -->
    {% if question_papers %}
    <section class="bg-gradient-to-br from-amber-50/70 to-orange-50/50 rounded-3xl p-6 sm:p-...
        <div class="flex flex-col sm:flex-row sm:items-end justify-between mb-8">
            <div>
                <span class="inline-flex items-center px-2.5 py-0.5 rounded-full text-xs fon...
                    <i class="fas fa-clock-rotate-left mr-1"></i> Exam Ready
                </span>
                <h2 class="text-2xl font-bold text-slate-900">Latest Examination Question Pa...
                <p class="text-sm text-slate-600 mt-1">Past university semester question pap...
            </div>
            <a href="{{ url_for('main.browse', type='Question Paper') }}" class="mt-4 sm:mt-...
                View All Question Papers
            </a>
        </div>

        <div class="grid grid-cols-1 md:grid-cols-2 gap-4">
            {% for qp in question_papers %}
```

```
                <div class="bg-white rounded-2xl p-5 border border-amber-200/70 shadow-sm fl...
                    <div class="flex items-start space-x-3">
                        <div class="w-10 h-10 rounded-xl bg-amber-100 text-amber-700 flex it...
                            <i class="fas fa-file-pdf"></i>
                        </div>
                        <div>
                            <div class="flex items-center space-x-2 text-xs mb-1">
                                <span class="font-bold text-slate-700">{{ qp.department_code...
                                <span class="text-slate-300">•</span>
                                <span class="px-2 py-0.5 rounded bg-amber-50 text-amber-700 ...
                                <span class="text-slate-300">•</span>
                                <span class="text-slate-500 font-mono text-[11px]">{{ qp.aca...
                            </div>
                            <h4 class="text-sm font-bold text-slate-900 hover:text-blue-600 ...
                                <a href="{{ url_for('main.resource_detail', resource_id=qp.i...
                            </h4>
                            <p class="text-xs text-slate-500 mt-1">{{ qp.subject_name }}</p>
                        </div>
                    </div>
                    <div class="flex-shrink-0 flex flex-col items-end space-y-1">
                        <a href="{{ url_for('main.download_resource', resource_id=qp.id) }}"
                           class="p-2.5 rounded-xl bg-amber-50 hover:bg-amber-100 text-amber...
                           title="Download Paper">
                            <i class="fas fa-download"></i>

                        </a>
                        <span class="text-[10px] text-slate-400 font-medium">{{ qp.download_...
                    </div>
                </div>
            {% endfor %}
        </div>
    </section>
    {% endif %}

    <!-- Key Advantages & Offline Mode Callout -->
    <section class="grid grid-cols-1 md:grid-cols-3 gap-8 py-4">
        <div class="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-3">
            <div class="w-12 h-12 rounded-xl bg-blue-100 text-blue-600 flex items-center jus...
                <i class="fas fa-sitemap"></i>
            </div>
            <h3 class="text-lg font-bold text-slate-900">Centralized Organization</h3>
            <p class="text-xs text-slate-500 leading-relaxed">
                No more scattered WhatsApp PDFs or lost Google Drive links. Filter strictly ...
            </p>
        </div>

        <div class="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-3">
            <div class="w-12 h-12 rounded-xl bg-emerald-100 text-emerald-600 flex items-cent...
                <i class="fas fa-wifi"></i>
            </div>
            <h3 class="text-lg font-bold text-slate-900">Offline & Campus LAN</h3>
            <p class="text-xs text-slate-500 leading-relaxed">
                Zero internet required. The portal operates smoothly across local offline co...
            </p>
        </div>

        <div class="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-3">
            <div class="w-12 h-12 rounded-xl bg-purple-100 text-purple-600 flex items-center...
                <i class="fas fa-shield-halved"></i>
            </div>
            <h3 class="text-lg font-bold text-slate-900">Moderated Academic Content</h3>
            <p class="text-xs text-slate-500 leading-relaxed">
                Every study material and question paper uploaded by students undergoes facul...
            </p>
        </div>
    </section>

</div>
{% endblock %}
```

■ File: templates/browse.html

Centralized Academic Repository: Filters by Dept, Sem, Subject, Category, Exam Type, Year

```
{% extends 'base.html' %}

{% block title %}Browse Academic Study Materials & Question Papers | Academix Portal{% endbl...

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

    <!-- Page Header & Search Bar -->
    <div class="bg-white rounded-2xl p-6 sm:p-8 border border-slate-200 shadow-sm mb-8">
        <div class="max-w-3xl">
            <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Repositor...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-1">Browse Study...
            <p class="text-xs sm:text-sm text-slate-500 mt-2">Filter through our centralized...
        </div>

        <!-- Filter Form -->
        <form action="{{ url_for('main.browse') }}" method="GET" class="mt-6 space-y-4">
            <!-- Search Keyword Bar -->
            <div class="flex flex-col sm:flex-row gap-3">
                <div class="flex-1 relative">
                    <i class="fas fa-search absolute left-4 top-3.5 text-slate-400"></i>
                    <input type="text" name="q" value="{{ q }}"
                           placeholder="Search by keywords, subject name, paper topic, tags ...
                           class="w-full pl-11 pr-4 py-2.5 bg-slate-50 focus:bg-white text-s...
                </div>
                <div class="w-full sm:w-48">
                    <select name="sort" onchange="this.form.submit()"
                            class="w-full py-2.5 px-3 bg-slate-50 text-slate-700 text-sm rou...
                        <option value="newest" {% if selected_sort == 'newest' %}selected{% ...
                        <option value="downloads" {% if selected_sort == 'downloads' %}selec...
                        <option value="rating" {% if selected_sort == 'rating' %}selected{% ...
                        <option value="title" {% if selected_sort == 'title' %}selected{% en...
                    </select>
                </div>
                <button type="submit"
                        class="px-5 py-2.5 bg-blue-600 hover:bg-blue-700 text-white font-sem...
                    <i class="fas fa-filter text-xs mr-1"></i> Apply Filters
                </button>
                {% if q or selected_dept or selected_sem or selected_subj or selected_type o...
                    <a href="{{ url_for('main.browse') }}"
                       class="px-4 py-2.5 bg-slate-100 hover:bg-slate-200 text-slate-600 fon...
                        <i class="fas fa-rotate-left mr-1"></i> Reset
                    </a>
                {% endif %}
            </div>

            <!-- Cascade Dropdowns: Department, Semester, Subject, Resource Type, Year, Exam...
            <div class="grid grid-cols-2 sm:grid-cols-3 lg:grid-cols-6 gap-3 pt-2">
                <!-- Department -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="dept" id="dept-select" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Departments</option>
                        {% for d in departments %}
                            <option value="{{ d.id }}" {% if selected_dept|string == d.id|st...
                                {{ d.code }} - {{ d.name }}
                            </option>
                        {% endfor %}
                    </select>
                </div>

                <!-- Semester -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="sem" id="sem-select" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Semesters</option>
                        {% for s in range(1, 9) %}
                            <option value="{{ s }}" {% if selected_sem|string == s|string %}...
                                Semester {{ s }}
                            </option>
                        {% endfor %}
                    </select>
                </div>
```

```
                <!-- Subject -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="subj" id="subj-select" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Subjects</option>
                        {% for subj in subjects %}
                            <option value="{{ subj.id }}" {% if selected_subj|string == subj...
                                {{ subj.code }} - {{ subj.name }}

                            </option>
                        {% endfor %}
                    </select>
                </div>

                <!-- Resource Type -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="type" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Types</option>
                        <option value="Question Paper" {% if selected_type == 'Question Pape...
                        <option value="Lecture Notes" {% if selected_type == 'Lecture Notes'...
                        <option value="Lab Manual" {% if selected_type == 'Lab Manual' %}sel...
                        <option value="Assignment" {% if selected_type == 'Assignment' %}sel...
                        <option value="Syllabus" {% if selected_type == 'Syllabus' %}selecte...
                        <option value="Reference Book" {% if selected_type == 'Reference Boo...
                    </select>
                </div>

                <!-- Academic Year -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="year" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Years</option>
                        {% for y in available_years %}
                            <option value="{{ y }}" {% if selected_year == y %}selected{% en...
                        {% endfor %}
                    </select>
                </div>

                <!-- Exam Type -->
                <div>
                    <label class="block text-[11px] font-bold text-slate-500 uppercase track...
                    <select name="exam" onchange="this.form.submit()"
                            class="w-full py-2 px-2.5 bg-white text-xs text-slate-700 rounde...
                        <option value="">All Exams</option>
                        <option value="End-Semester" {% if selected_exam == 'End-Semester' %...
                        <option value="Mid-Term 1" {% if selected_exam == 'Mid-Term 1' %}sel...
                        <option value="Mid-Term 2" {% if selected_exam == 'Mid-Term 2' %}sel...
                        <option value="Model Exam" {% if selected_exam == 'Model Exam' %}sel...
                        <option value="Supplementary" {% if selected_exam == 'Supplementary'...
                    </select>
                </div>
            </div>
        </form>
    </div>

    <!-- Results Header -->
    <div class="flex items-center justify-between mb-6">
        <div class="text-sm text-slate-600">
            Showing <span class="font-bold text-slate-900">{{ resources|length }}</span> of ...
        </div>
        <div class="text-xs text-slate-500">
            Page {{ page }} of {{ total_pages }}
        </div>
    </div>

    <!-- Results Grid -->
    {% if resources %}
        <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {% for item in resources %}
                <div class="bg-white rounded-2xl border border-slate-200 hover:border-blue-3...
                    <div>
                        <!-- Header badge and tags -->
                        <div class="flex items-center justify-between text-xs mb-3">
                            <span class="inline-flex items-center px-2.5 py-1 rounded-md fon...
```

```
                                {% if item.resource_type == 'Question Paper' %}bg-amber-100 ...
                                {% elif item.resource_type == 'Lecture Notes' %}bg-emerald-1...
                                {% elif item.resource_type == 'Lab Manual' %}bg-purple-100 t...
                                {% elif item.resource_type == 'Assignment' %}bg-sky-100 text...
                                {% else %}bg-slate-100 text-slate-800{% endif %}">
                                <i class="fas fa-file-lines mr-1.5 text-xs"></i> {{ item.res...
                            </span>
                            <span class="text-slate-400 font-mono text-[11px]">{{ item.acade...
                        </div>

                        <!-- Department & Semester & Subject -->
                        <div class="text-xs font-semibold text-blue-600 mb-1 flex items-cent...
                            <span class="px-1.5 py-0.5 rounded bg-blue-50 text-blue-700 font...
                            <span>• Sem {{ item.semester }}</span>
                            <span>• {{ item.subject_code }}</span>
                        </div>
                        <div class="text-xs text-slate-500 font-medium mb-2">{{ item.subject...

                        <!-- Title -->
                        <h3 class="text-base font-bold text-slate-900 line-clamp-2 group-hov...
                            <a href="{{ url_for('main.resource_detail', resource_id=item.id)...
                                {{ item.title }}
                            </a>
                        </h3>

                        <!-- Description -->
                        <p class="text-xs text-slate-500 line-clamp-2 mb-4 leading-relaxed">
                            {{ item.description or 'Official verified study material submitt...
                        </p>

                        <!-- Exam Tag if any -->
                        {% if item.exam_type and item.exam_type != 'N/A' %}
                            <div class="mb-4">
                                <span class="inline-flex items-center px-2 py-0.5 rounded te...
                                    <i class="fas fa-pen-to-square mr-1 text-[10px]"></i> {{...
                                </span>
                            </div>
                        {% endif %}
                    </div>

                    <!-- Card Footer: Stats & Actions -->
                    <div class="border-t border-slate-100 pt-4">
                        <div class="flex items-center justify-between text-xs text-slate-500...
                            <div class="flex items-center space-x-3">
                                <span><i class="fas fa-download mr-1 text-slate-400"></i> {{...
                                {% if item.rating_count > 0 %}
                                    <span class="text-amber-500 font-bold flex items-center">
                                        <i class="fas fa-star mr-1"></i> {{ "%.1f"|format(it...
                                    </span>
                                {% else %}
                                    <span class="text-slate-400 text-[11px]">Unrated</span>
                                {% endif %}
                            </div>
                            <span class="text-[11px] text-slate-400 font-mono">
                                <i class="fas fa-file-pdf mr-1 text-rose-500"></i> {{ format...
                            </span>
                        </div>

                        <div class="flex items-center space-x-2">
                            <a href="{{ url_for('main.resource_detail', resource_id=item.id)...
                               class="flex-1 text-center py-2 rounded-xl text-slate-700 bg-s...
                                View Details
                            </a>
                            <a href="{{ url_for('main.download_resource', resource_id=item.i...
                               class="flex-1 text-center py-2 rounded-xl text-white bg-blue-...
                                <i class="fas fa-download text-[10px]"></i>
                                <span>Download</span>
                            </a>
                        </div>
                    </div>
                </div>
            {% endfor %}
        </div>

        <!-- Pagination Controls -->
        {% if total_pages > 1 %}
            <div class="mt-12 flex items-center justify-center space-x-2">
                {% if page > 1 %}
```

```
                    <a href="{{ url_for('main.browse', page=page-1, q=q, dept=selected_dept,...
                       class="px-4 py-2 rounded-xl border border-slate-200 bg-white text-xs ...
                        <i class="fas fa-chevron-left mr-1"></i> Previous
                    </a>
                {% endif %}

                {% for p in range(1, total_pages + 1) %}
                    <a href="{{ url_for('main.browse', page=p, q=q, dept=selected_dept, sem=...
                       class="w-9 h-9 rounded-xl text-xs font-semibold flex items-center jus...
                              {% if p == page %}bg-blue-600 text-white shadow-sm shadow-blue...
                        {{ p }}
                    </a>
                {% endfor %}

                {% if page < total_pages %}
                    <a href="{{ url_for('main.browse', page=page+1, q=q, dept=selected_dept,...
                       class="px-4 py-2 rounded-xl border border-slate-200 bg-white text-xs ...
                        Next <i class="fas fa-chevron-right ml-1"></i>
                    </a>
                {% endif %}
            </div>
        {% endif %}

    {% else %}

        <!-- Empty Results Placeholder -->
        <div class="bg-white rounded-3xl border border-slate-200 p-12 text-center max-w-xl m...
            <div class="w-16 h-16 rounded-2xl bg-blue-50 text-blue-600 flex items-center jus...
                <i class="fas fa-folder-open"></i>
            </div>
            <h3 class="text-lg font-bold text-slate-900 mb-2">No matching academic materials...
            <p class="text-xs text-slate-500 mb-6 leading-relaxed">
                We couldn't find any documents matching your current filter criteria. Try cl...
            </p>
            <div class="flex items-center justify-center space-x-3">
                <a href="{{ url_for('main.browse') }}"
                   class="px-4 py-2.5 rounded-xl bg-slate-100 hover:bg-slate-200 text-slate-...
                    Clear All Filters
                </a>
                <a href="{{ url_for('upload.upload') }}"
                   class="px-4 py-2.5 rounded-xl bg-blue-600 hover:bg-blue-700 text-white fo...
                    <i class="fas fa-cloud-arrow-up mr-1"></i> Upload this Resource
                </a>
            </div>
        </div>
    {% endif %}

</div>
{% endblock %}
```

■ File: templates/resource_detail.html

Resource Page: In-Browser PDF/Image Viewer, Download Counter, Peer Reviews, Bookmarks

```
{% extends 'base.html' %}

{% block title %}{{ resource.title }} | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

    <!-- Breadcrumb -->
    <nav class="flex items-center space-x-2 text-xs text-slate-500 mb-6 flex-wrap">
        <a href="{{ url_for('main.index') }}" class="hover:text-blue-600">Home</a>
        <i class="fas fa-chevron-right text-[10px] text-slate-400"></i>
        <a href="{{ url_for('main.browse') }}" class="hover:text-blue-600">Browse</a>
        <i class="fas fa-chevron-right text-[10px] text-slate-400"></i>
        <a href="{{ url_for('main.browse', dept=resource.department_id) }}" class="hover:tex...
        <i class="fas fa-chevron-right text-[10px] text-slate-400"></i>
        <span class="text-slate-800 font-semibold truncate max-w-xs">{{ resource.subject_nam...
    </nav>

    <!-- Header Card -->
    <div class="bg-white rounded-3xl p-6 sm:p-8 border border-slate-200 shadow-sm mb-8">
        <div class="flex flex-col lg:flex-row lg:items-start justify-between gap-6">
            <div class="space-y-3 flex-1">
                <div class="flex flex-wrap items-center gap-2">
                    <span class="inline-flex items-center px-3 py-1 rounded-lg text-xs font-...
                        {% if resource.resource_type == 'Question Paper' %}bg-amber-100 text...
                        {% elif resource.resource_type == 'Lecture Notes' %}bg-emerald-100 t...
                        {% elif resource.resource_type == 'Lab Manual' %}bg-purple-100 text-...
                        {% elif resource.resource_type == 'Assignment' %}bg-sky-100 text-sky...
                        {% else %}bg-slate-100 text-slate-800{% endif %}">
                        <i class="fas fa-file-lines mr-1.5"></i> {{ resource.resource_type }}
                    </span>
                    <span class="px-2.5 py-1 rounded-lg bg-blue-50 text-blue-700 text-xs fon...
                        {{ resource.department_code }} • Semester {{ resource.semester }}
                    </span>
                    <span class="px-2.5 py-1 rounded-lg bg-slate-100 text-slate-700 font-mon...
                        {{ resource.academic_year }}
                    </span>
                    {% if resource.exam_type and resource.exam_type != 'N/A' %}
                        <span class="px-2.5 py-1 rounded-lg bg-amber-50 text-amber-800 borde...
                            <i class="fas fa-pen-nib mr-1"></i> {{ resource.exam_type }}
                        </span>
                    {% endif %}
                </div>

                <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 leading-tight">
                    {{ resource.title }}
                </h1>

                <div class="flex items-center space-x-2 text-sm text-slate-600 font-medium">
                    <span class="text-blue-600 font-bold font-mono">{{ resource.subject_code...
                    <span>-</span>
                    <span>{{ resource.subject_name }}</span>
                </div>

                <!-- Contributor & Rating line -->
                <div class="flex flex-wrap items-center gap-4 text-xs text-slate-500 pt-2 bo...
                    <div class="flex items-center space-x-2">
                        <i class="fas fa-user-circle text-slate-400 text-sm"></i>
                        <span>Contributed by <strong class="text-slate-800">{{ resource.uplo...
                        <span class="px-2 py-0.5 rounded text-[10px] font-semibold uppercase...
                            {% if resource.uploader_role == 'admin' %}bg-rose-100 text-rose-700
                            {% elif resource.uploader_role == 'faculty' %}bg-amber-100 text-...
                            {% else %}bg-blue-100 text-blue-700{% endif %}">
                            {{ resource.uploader_role or 'faculty' }}
                        </span>
                    </div>

                    <div class="flex items-center space-x-1 text-amber-500 font-bold">
                        {% for i in range(1, 6) %}
                            <i class="fas fa-star text-xs {% if i <= resource.avg_rating %}t...
                        {% endfor %}
                        <span class="ml-1 text-slate-700 font-bold">{{ "%.1f"|format(resourc...
                        <span class="text-slate-400 font-normal">({{ resource.rating_count }...
                    </div>

```

```
                    <div class="text-slate-400">
                        <i class="far fa-clock mr-1"></i> Added on {{ resource.created_at|st...
                    </div>
                </div>
            </div>

            <!-- Quick Download & Bookmark CTA Box -->
            <div class="flex flex-col sm:flex-row lg:flex-col gap-3 min-w-[220px]">
                <a href="{{ url_for('main.download_resource', resource_id=resource.id) }}"
                   class="px-6 py-3.5 rounded-2xl bg-blue-600 hover:bg-blue-700 text-white f...

                    <i class="fas fa-cloud-arrow-down text-base"></i>
                    <span>Download Resource</span>
                </a>

                {% if session.get('user_id') %}
                    <form action="{{ url_for('main.toggle_bookmark', resource_id=resource.id...
                        <button type="submit"
                                class="w-full py-2.5 px-4 rounded-xl text-xs font-semibold b...
                                {% if is_bookmarked %}bg-amber-50 border-amber-300 text-ambe...
                            <i class="fas fa-bookmark {% if is_bookmarked %}text-amber-500{%...
                            <span>{% if is_bookmarked %}Saved in Bookmarks{% else %}Bookmark...
                        </button>
                    </form>
                {% else %}
                    <a href="{{ url_for('auth.login', next=request.url) }}"
                       class="py-2.5 px-4 rounded-xl text-xs font-semibold border border-sla...
                        <i class="far fa-bookmark text-slate-400"></i>
                        <span>Login to Bookmark</span>
                    </a>
                {% endif %}

                <!-- WhatsApp / Copy Share -->
                <button onclick="navigator.clipboard.writeText(window.location.href); alert(...
                        class="py-2 px-3 rounded-xl text-xs text-slate-600 bg-slate-100 hove...
                    <i class="far fa-clone text-slate-400"></i>
                    <span>Copy Share Link</span>
                </button>
            </div>
        </div>
    </div>

    <!-- Main Content Grid (Left Preview & Description, Right Metadata & Related) -->
    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">

        <!-- Left Column: Preview & Detailed Info -->
        <div class="lg:col-span-2 space-y-8">

            <!-- In-Browser Document Previewer -->
            <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
                <div class="flex items-center justify-between">
                    <h3 class="text-base font-bold text-slate-900 flex items-center space-x-2">
                        <i class="fas fa-eye text-blue-600"></i>
                        <span>In-Browser Document Preview</span>
                    </h3>
                    <a href="{{ url_for('main.preview_resource', resource_id=resource.id) }}...
                       class="text-xs font-semibold text-blue-600 hover:underline flex items...
                        <span>Open Full Screen</span>
                        <i class="fas fa-arrow-up-right-from-square text-[10px]"></i>
                    </a>
                </div>

                <!-- Preview container -->
                <div class="w-full bg-slate-900 rounded-2xl overflow-hidden border border-sl...
                    {% if resource.file_ext == 'pdf' %}
                        <iframe src="{{ url_for('main.preview_resource', resource_id=resourc...
                                class="w-full h-[520px] rounded-2xl border-0"
                                title="{{ resource.title }}">
                        </iframe>
                    {% elif resource.file_ext in ['jpg', 'jpeg', 'png'] %}
                        <div class="p-4 flex items-center justify-center min-h-[400px]">
                            <img src="{{ url_for('main.preview_resource', resource_id=resour...
                        </div>
                    {% else %}
                        <div class="py-16 text-center text-slate-400 p-6 space-y-3">
                            <i class="fas fa-file-lines text-5xl text-slate-600 mb-2"></i>
                            <h4 class="text-sm font-semibold text-white">Preview unavailable...
                            <p class="text-xs max-w-sm mx-auto">This file can be downloaded ...
                            <a href="{{ url_for('main.download_resource', resource_id=resour...
```

```
                                class="inline-flex items-center px-4 py-2 rounded-xl bg-blue-...
                                 <i class="fas fa-download mr-1.5"></i> Download File to View
                            </a>
                        </div>
                    {% endif %}
                </div>
            </div>

            <!-- Description & Academic Scope -->
            <div class="bg-white rounded-3xl p-6 sm:p-8 border border-slate-200 shadow-sm sp...
                <h3 class="text-base font-bold text-slate-900">About this Material</h3>
                <p class="text-sm text-slate-600 leading-relaxed whitespace-pre-line">
                    {{ resource.description or 'No extra description provided for this acade...
                </p>

                <!-- Tags -->
                {% if tag_list %}

                    <div class="pt-4 border-t border-slate-100">
                        <span class="block text-xs font-bold text-slate-400 uppercase tracki...
                        <div class="flex flex-wrap gap-1.5">
                            {% for tag in tag_list %}
                                <a href="{{ url_for('main.browse', q=tag) }}"
                                   class="px-2.5 py-1 rounded-lg bg-slate-100 hover:bg-slate...
                                    #{{ tag }}
                                </a>
                            {% endfor %}
                        </div>
                    </div>
                {% endif %}
            </div>

            <!-- Student Rating & Reviews Section -->
            <div class="bg-white rounded-3xl p-6 sm:p-8 border border-slate-200 shadow-sm sp...
                <div class="flex items-center justify-between">
                    <div>
                        <h3 class="text-base font-bold text-slate-900">Peer Reviews & Rating...
                        <p class="text-xs text-slate-500 mt-0.5">Verified feedback from stud...
                    </div>
                    <div class="text-right">
                        <div class="text-2xl font-extrabold text-slate-900">{{ "%.1f"|format...
                        <div class="text-[11px] text-slate-500">{{ resource.rating_count }} ...
                    </div>
                </div>

                <!-- Rating Form -->
                {% if session.get('user_id') %}
                    <div class="bg-slate-50 rounded-2xl p-5 border border-slate-200/80">
                        <h4 class="text-xs font-bold text-slate-800 uppercase tracking-wider...
                            {% if user_rating %}Update Your Rating & Feedback{% else %}Rate ...
                        </h4>
                        <form action="{{ url_for('main.rate_resource', resource_id=resource....
                            <div>
                                <label class="block text-xs text-slate-600 mb-1 font-medium"...
                                <div class="flex items-center space-x-3 text-sm">
                                    {% for r in range(5, 0, -1) %}
                                        <label class="flex items-center space-x-1 cursor-poi...
                                            <input type="radio" name="rating" value="{{ r }}...
                                                   {% if user_rating and user_rating.rating ...
                                                   class="text-amber-500 focus:ring-amber-400">
                                            <span class="font-bold text-slate-700">{{ r }} <...
                                        </label>
                                    {% endfor %}
                                </div>
                            </div>
                            <div>
                                <textarea name="review" rows="2" placeholder="Was this previ...
                                          class="w-full text-xs p-3 rounded-xl border border...
                            </div>
                            <button type="submit" class="px-4 py-2 bg-blue-600 hover:bg-blue...
                                Submit Feedback
                            </button>
                        </form>
                    </div>
                {% else %}
                    <div class="bg-slate-50 rounded-2xl p-4 border border-slate-200 text-cen...
                        <a href="{{ url_for('auth.login', next=request.url) }}" class="text-...
                    </div>
                {% endif %}
```

```
232 |
233 |                 <!-- Existing Reviews List -->
234 |                 <div class="space-y-4 pt-2">
235 |                     {% for rev in reviews %}
236 |                         <div class="p-4 rounded-2xl bg-white border border-slate-100 shadow-...
237 |                             <div class="flex items-center justify-between">
238 |                                 <div class="flex items-center space-x-2">
239 |                                     <div class="w-6 h-6 rounded-full bg-slate-200 text-slate...
240 |                                         {{ rev.full_name[:1].upper() }}
241 |                                     </div>
242 |                                     <span class="text-xs font-bold text-slate-800">{{ rev.fu...
243 |                                     <span class="text-[10px] text-slate-400 uppercase font-m...
244 |                                 </div>
245 |                                 <div class="flex items-center text-amber-400 text-xs">
246 |                                     {% for s in range(1, 6) %}
247 |                                         <i class="fas fa-star {% if s <= rev.rating %}text-a...
248 |                                     {% endfor %}
249 |                                 </div>
250 |                             </div>
251 |                             {% if rev.review %}
252 |                                 <p class="text-xs text-slate-600 leading-relaxed pl-8">
253 |                                     {{ rev.review }}
254 |                                 </p>
255 |                             {% endif %}

256 |                             <div class="text-[10px] text-slate-400 pl-8">
257 |                                 {{ rev.created_at|str_date }}
258 |                             </div>
259 |                         </div>
260 |                     {% else %}
261 |                         <p class="text-xs text-slate-400 text-center py-4">No reviews yet. B...
262 |                     {% endfor %}
263 |                 </div>
264 |             </div>
265 |
266 |         </div>
267 |
268 |         <!-- Right Column: Document Details, LAN Box, Related -->
269 |         <div class="space-y-6">
270 |
271 |             <!-- File Properties Box -->
272 |             <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
273 |                 <h3 class="text-xs font-bold text-slate-900 uppercase tracking-wider">Docume...
274 |
275 |                 <div class="divide-y divide-slate-100 text-xs">
276 |                     <div class="py-2.5 flex items-center justify-between">
277 |                         <span class="text-slate-500">File Type</span>
278 |                         <span class="font-bold text-slate-800 uppercase font-mono">{{ resour...
279 |                     </div>
280 |                     <div class="py-2.5 flex items-center justify-between">
281 |                         <span class="text-slate-500">File Size</span>
282 |                         <span class="font-bold text-slate-800 font-mono">{{ format_file_size...
283 |                     </div>
284 |                     <div class="py-2.5 flex items-center justify-between">
285 |                         <span class="text-slate-500">Total Downloads</span>
286 |                         <span class="font-bold text-blue-600">{{ resource.download_count }} ...
287 |                     </div>
288 |                     <div class="py-2.5 flex items-center justify-between">
289 |                         <span class="text-slate-500">Total Page Views</span>
290 |                         <span class="font-bold text-slate-800">{{ resource.view_count }}</span>
291 |                     </div>
292 |                     <div class="py-2.5 flex items-center justify-between">
293 |                         <span class="text-slate-500">Academic Year</span>
294 |                         <span class="font-bold text-slate-800">{{ resource.academic_year }}<...
295 |                     </div>
296 |                     <div class="py-2.5 flex items-center justify-between">
297 |                         <span class="text-slate-500">Verification Status</span>
298 |                         <span class="px-2 py-0.5 rounded-full text-[10px] font-bold bg-emera...
299 |                             <i class="fas fa-check-circle mr-1"></i> Verified & Approved
300 |                         </span>
301 |                     </div>
302 |                 </div>
303 |             </div>
304 |
305 |             <!-- Offline & Campus LAN QR / Link Card -->
306 |             <div class="bg-gradient-to-br from-indigo-900 to-slate-900 text-white rounded-3x...
307 |                 <div class="flex items-center space-x-2 text-emerald-400 text-xs font-bold u...
308 |                     <i class="fas fa-wifi"></i>
309 |                     <span>Offline Campus LAN</span>
```

```
                </div>
                <h4 class="text-sm font-bold text-white">Share with Classmates on Local Wi-F...
                <p class="text-xs text-slate-300 leading-relaxed">
                    Classmates on the same Wi-Fi router or mobile hotspot can directly open ...
                </p>
                <div class="p-2.5 bg-black/40 rounded-xl border border-white/10 text-xs font...
                    http://{{ lan_ip }}:{{ server_port }}/resource/{{ resource.id }}
                </div>
            </div>

            <!-- Related Materials in Subject -->
            {% if related %}
                <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space...
                    <h3 class="text-xs font-bold text-slate-900 uppercase tracking-wider">Mo...

                    <div class="space-y-3">
                        {% for rel in related %}
                            <a href="{{ url_for('main.resource_detail', resource_id=rel.id) ...
                               class="block p-3 rounded-2xl hover:bg-slate-50 border border-...
                                <div class="flex items-center justify-between text-[11px] mb...
                                    <span class="font-bold text-blue-600">{{ rel.resource_ty...
                                    <span class="text-slate-400 font-mono">{{ rel.academic_y...
                                </div>
                                <h4 class="text-xs font-bold text-slate-800 group-hover:text...
                                    {{ rel.title }}
                                </h4>
                                <div class="flex items-center space-x-3 text-[10px] text-sla...
                                    <span><i class="fas fa-download mr-1"></i> {{ rel.downlo...
                                    <span><i class="fas fa-star text-amber-400 mr-1"></i> {{...
                                </div>
                            </a>

                        {% endfor %}
                    </div>
                </div>
            {% endif %}

        </div>

    </div>

</div>
{% endblock %}
```

■ File: templates/upload.html

Upload Form: Interactive Dropzone, Cascading Department/Semester/Subject Dropdowns

```
{% extends 'base.html' %}

{% block title %}Upload Study Material or Question Paper | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-4xl mx-auto px-4 sm:px-6 lg:px-8 py-10">

    <div class="bg-white rounded-3xl p-6 sm:p-10 border border-slate-200 shadow-sm">

        <!-- Header -->
        <div class="border-b border-slate-100 pb-6 mb-8">
            <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Academic ...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-1">Upload Study...
            <p class="text-xs sm:text-sm text-slate-500 mt-1">Share lecture notes, question ...
        </div>

        <form action="{{ url_for('upload.upload') }}" method="POST" enctype="multipart/form-...

            <!-- Resource Title -->
            <div>
                <label for="title" class="block text-xs font-bold text-slate-700 uppercase t...
                    Resource Title <span class="text-rose-500">*</span>
                </label>
                <input type="text" name="title" id="title" required
                       placeholder="e.g. End-Semester Dec 2024 Question Paper with Solutions...
                       class="w-full px-4 py-3 bg-slate-50 focus:bg-white text-sm text-slate...
                <p class="text-[11px] text-slate-400 mt-1">Include the year, subject, and wh...
            </div>

            <!-- Resource Type & Academic Year & Exam Type in a row -->
            <div class="grid grid-cols-1 sm:grid-cols-3 gap-4">
                <div>
                    <label for="resource_type" class="block text-xs font-bold text-slate-700...
                        Category <span class="text-rose-500">*</span>
                    </label>
                    <select name="resource_type" id="resource_type" required
                            class="w-full px-3 py-2.5 bg-slate-50 text-sm text-slate-700 rou...
                        <option value="">Select Category</option>
                        <option value="Question Paper">Previous Year Question Paper (PYQ)</o...
                        <option value="Lecture Notes">Lecture Notes / PPT Slides</option>
                        <option value="Lab Manual">Laboratory Manual / Experiments</option>
                        <option value="Assignment">Assignment & Solutions</option>
                        <option value="Syllabus">Curriculum / Syllabus</option>
                        <option value="Reference Book">Reference Guide / Book</option>
                    </select>
                </div>

                <div>
                    <label for="academic_year" class="block text-xs font-bold text-slate-700...
                        Academic Year <span class="text-rose-500">*</span>
                    </label>
                    <select name="academic_year" id="academic_year" required
                            class="w-full px-3 py-2.5 bg-slate-50 text-sm text-slate-700 rou...
                        <option value="2025-2026" selected>2025-2026</option>
                        <option value="2024-2025">2024-2025</option>
                        <option value="2023-2024">2023-2024</option>
                        <option value="2022-2023">2022-2023</option>
                        <option value="2021-2022">2021-2022</option>
                    </select>
                </div>

                <div>
                    <label for="exam_type" class="block text-xs font-bold text-slate-700 upp...
                        Exam Type
                    </label>
                    <select name="exam_type" id="exam_type"
                            class="w-full px-3 py-2.5 bg-slate-50 text-sm text-slate-700 rou...
                        <option value="End-Semester">End-Semester University Exam</option>
                        <option value="Mid-Term 1">Mid-Term 1</option>
                        <option value="Mid-Term 2">Mid-Term 2</option>
                        <option value="Model Exam">Model / Pre-University Exam</option>
                        <option value="Supplementary">Supplementary / Arrear Exam</option>
                        <option value="N/A">Not Applicable (Notes/Manual)</option>
                    </select>
                </div>
```

```
            </div>

            <!-- Cascading Hierarchy: Department, Semester, Subject -->
            <div class="grid grid-cols-1 sm:grid-cols-3 gap-4 p-4 rounded-2xl bg-blue-50/50 ...
                <!-- Department -->
                <div>
                    <label for="department_id" class="block text-xs font-bold text-blue-900 ...
                        Department <span class="text-rose-500">*</span>
                    </label>
                    <select name="department_id" id="department_id" required

                            class="w-full px-3 py-2.5 bg-white text-sm text-slate-700 rounde...
                        <option value="">Select Department</option>
                        {% for d in departments %}
                            <option value="{{ d.id }}" {% if session.get('department_id') ==...
                                {{ d.code }} - {{ d.name }}
                            </option>
                        {% endfor %}
                    </select>
                </div>

                <!-- Semester -->
                <div>
                    <label for="semester" class="block text-xs font-bold text-blue-900 upper...
                        Semester <span class="text-rose-500">*</span>
                    </label>
                    <select name="semester" id="semester" required
                            class="w-full px-3 py-2.5 bg-white text-sm text-slate-700 rounde...
                        <option value="">Select Semester</option>
                        {% for s in range(1, 9) %}
                            <option value="{{ s }}">Semester {{ s }}</option>
                        {% endfor %}
                    </select>
                </div>

                <!-- Dynamic Subject Dropdown -->
                <div>
                    <label for="subject_id" class="block text-xs font-bold text-blue-900 upp...
                        Subject <span class="text-rose-500">*</span>
                    </label>
                    <select name="subject_id" id="subject_id" required
                            class="w-full px-3 py-2.5 bg-white text-sm text-slate-700 rounde...
                        <option value="">Select Dept & Semester first</option>
                    </select>
                </div>
            </div>

            <!-- Description -->
            <div>
                <label for="description" class="block text-xs font-bold text-slate-700 upper...
                    Description & Overview
                </label>
                <textarea name="description" id="description" rows="3"
                          placeholder="Briefly describe the contents, units covered, author/...
                          class="w-full px-4 py-3 bg-slate-50 focus:bg-white text-sm text-sl...
            </div>

            <!-- Tags -->
            <div>
                <label for="tags" class="block text-xs font-bold text-slate-700 uppercase tr...
                    Search Tags & Keywords
                </label>
                <input type="text" name="tags" id="tags"
                       placeholder="e.g. data structures, trees, pyq, end sem, solved questi...
                       class="w-full px-4 py-2.5 bg-slate-50 focus:bg-white text-sm text-sla...
            </div>

            <!-- Drag and Drop File Upload Area -->
            <div>
                <label class="block text-xs font-bold text-slate-700 uppercase tracking-wide...
                    Attach Academic Document <span class="text-rose-500">*</span>
                </label>
                <div id="dropzone"
                     class="border-2 border-dashed border-slate-300 hover:border-blue-500 bg...
                    <input type="file" name="file" id="file-input" required class="hidden"
                           accept=".pdf,.doc,.docx,.ppt,.pptx,.txt,.zip,.rar,.png,.jpg,.jpeg">
                    <div class="space-y-2">
                        <div class="w-12 h-12 rounded-2xl bg-blue-100 text-blue-600 flex ite...
                            <i class="fas fa-cloud-arrow-up"></i>
```

```
                        </div>
                        <div class="text-sm font-bold text-slate-800">
                            <span class="text-blue-600 hover:underline">Click to browse file...
                        </div>
                        <p class="text-xs text-slate-400">
                            Supported: PDF, Word (DOCX), PPT, TXT, ZIP, Images (Max 64 MB)
                        </p>
                        <div id="file-selected-name" class="hidden text-xs font-bold text-em...
                        </div>
                    </div>
                </div>
            </div>

            <!-- Review Notice Box -->
            <div class="p-4 rounded-xl bg-amber-50 border border-amber-200/80 text-xs text-a...
                <i class="fas fa-circle-info text-amber-600 text-base flex-shrink-0 mt-0.5">...
                <div class="leading-relaxed">

                    <strong>Quality & Moderation Notice:</strong>
                    {% if session.get('role') in ('admin', 'faculty') %}
                        As a faculty or administrative member, your resource will be automat...
                    {% else %}
                        Student uploads undergo quick peer/faculty verification to prevent s...
                    {% endif %}
                </div>
            </div>

            <!-- Submit Button -->
            <div class="flex items-center justify-end space-x-4 pt-4 border-t border-slate-1...
                <a href="{{ url_for('main.index') }}" class="px-5 py-2.5 rounded-xl text-sm ...
                    Cancel
                </a>
                <button type="submit"
                        class="px-8 py-3 rounded-xl bg-blue-600 hover:bg-blue-700 text-white...
                    <i class="fas fa-cloud-arrow-up"></i>
                    <span>Submit & Publish</span>
                </button>
            </div>

        </form>

    </div>

</div>
{% endblock %}

{% block extra_scripts %}
<script>
    // File drag and drop logic
    const dropzone = document.getElementById('dropzone');
    const fileInput = document.getElementById('file-input');
    const fileNameDisplay = document.getElementById('file-selected-name');

    dropzone.addEventListener('click', () => fileInput.click());

    dropzone.addEventListener('dragover', (e) => {
        e.preventDefault();
        dropzone.classList.add('border-blue-500', 'bg-blue-50/50');
    });

    dropzone.addEventListener('dragleave', () => {
        dropzone.classList.remove('border-blue-500', 'bg-blue-50/50');
    });

    dropzone.addEventListener('drop', (e) => {
        e.preventDefault();
        dropzone.classList.remove('border-blue-500', 'bg-blue-50/50');
        if (e.dataTransfer.files.length) {
            fileInput.files = e.dataTransfer.files;
            showFileName(fileInput.files[0].name);
        }
    });

    fileInput.addEventListener('change', () => {
        if (fileInput.files.length) {
            showFileName(fileInput.files[0].name);
        }
    });

```

```
    function showFileName(name) {
        fileNameDisplay.textContent = "Selected: " + name;
        fileNameDisplay.classList.remove('hidden');
    }

    // Dynamic Cascading Subject Dropdown
    const deptSelect = document.getElementById('department_id');
    const semSelect = document.getElementById('semester');
    const subjSelect = document.getElementById('subject_id');

    function updateSubjects() {
        const deptId = deptSelect.value;
        const sem = semSelect.value;

        if (!deptId) {
            subjSelect.innerHTML = '<option value="">Select Department first</option>';
            return;
        }

        subjSelect.innerHTML = '<option value="">Loading subjects...</option>';

        let url = `/api/subjects?dept_id=${deptId}`;
        if (sem) {
            url += `&semester=${sem}`;

        }

        fetch(url)
            .then(res => res.json())
            .then(data => {
                subjSelect.innerHTML = '';
                if (data.length === 0) {
                    subjSelect.innerHTML = '<option value="">No subjects found for this seme...
                    return;
                }
                subjSelect.innerHTML = '<option value="">Select Subject</option>';
                data.forEach(subj => {
                    const opt = document.createElement('option');
                    opt.value = subj.id;
                    opt.textContent = `${subj.code} - ${subj.name} (Sem ${subj.semester})`;
                    subjSelect.appendChild(opt);
                });
            })
            .catch(err => {
                console.error('Failed to load subjects', err);
                subjSelect.innerHTML = '<option value="">Error loading subjects</option>';
            });
    }

    deptSelect.addEventListener('change', updateSubjects);
    semSelect.addEventListener('change', updateSubjects);
    if (deptSelect.value) {
        updateSubjects();
    }
</script>
{% endblock %}
```

■ File: templates/my_uploads.html

Student Submission Dashboard: Approval Status Badges (Pending, Approved, Rejected)

```
{% extends 'base.html' %}

{% block title %}My Uploaded Study Materials | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

    <!-- Header -->
    <div class="flex flex-col sm:flex-row sm:items-center justify-between gap-4 mb-8">
        <div>
            <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Contribut...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-1">My Uploaded ...
            <p class="text-xs sm:text-sm text-slate-500 mt-1">Track the verification status,...
        </div>
        <a href="{{ url_for('upload.upload') }}"
           class="px-5 py-2.5 rounded-xl bg-blue-600 hover:bg-blue-700 text-white font-bold ...
            <i class="fas fa-cloud-arrow-up"></i>
            <span>Upload New Material</span>
        </a>
    </div>

    <!-- Stats Cards -->
    <div class="grid grid-cols-2 md:grid-cols-4 gap-4 mb-8">
        <div class="bg-white rounded-2xl p-5 border border-slate-200 shadow-sm">
            <div class="text-2xl font-extrabold text-slate-900 mb-1">{{ counts.total }}</div>
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Total...
        </div>
        <div class="bg-white rounded-2xl p-5 border border-slate-200 shadow-sm">
            <div class="text-2xl font-extrabold text-emerald-600 mb-1">{{ counts.approved }}...
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Appro...
        </div>
        <div class="bg-white rounded-2xl p-5 border border-slate-200 shadow-sm">
            <div class="text-2xl font-extrabold text-amber-500 mb-1">{{ counts.pending }}</div>
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Pendi...
        </div>
        <div class="bg-white rounded-2xl p-5 border border-slate-200 shadow-sm">
            <div class="text-2xl font-extrabold text-rose-600 mb-1">{{ counts.rejected }}</div>
            <div class="text-xs font-semibold text-slate-500 uppercase tracking-wider">Needs...
        </div>
    </div>

    <!-- Uploads List Table / Cards -->
    {% if resources %}
        <div class="bg-white rounded-3xl border border-slate-200 shadow-sm overflow-hidden">
            <div class="overflow-x-auto">
                <table class="w-full text-left border-collapse text-xs">
                    <thead>
                        <tr class="bg-slate-50/80 border-b border-slate-200 text-slate-500 f...
                            <th class="py-3.5 px-6">Material Title</th>
                            <th class="py-3.5 px-4">Category</th>
                            <th class="py-3.5 px-4">Subject & Dept</th>
                            <th class="py-3.5 px-4">Status</th>
                            <th class="py-3.5 px-4 text-center">Downloads</th>
                            <th class="py-3.5 px-4">Date</th>
                            <th class="py-3.5 px-6 text-right">Actions</th>
                        </tr>
                    </thead>
                    <tbody class="divide-y divide-slate-100">
                        {% for r in resources %}
                            <tr class="hover:bg-slate-50/70 transition-colors">
                                <!-- Title -->
                                <td class="py-4 px-6 max-w-xs">
                                    <div class="font-bold text-slate-900 line-clamp-1">
                                        {% if r.status == 'approved' %}
                                            <a href="{{ url_for('main.resource_detail', reso...
                                                {{ r.title }}
                                            </a>
                                        {% else %}
                                            {{ r.title }}
                                        {% endif %}
                                    </div>
                                    <div class="text-[11px] text-slate-400 font-mono mt-0.5">
                                        {{ r.original_file_name }} ({{ format_file_size(r.fi...
                                    </div>
                                    {% if r.status == 'rejected' and r.rejection_reason %}
```

```
                                        <div class="mt-1 text-[11px] text-rose-600 bg-rose-5...
                                            <strong>Moderator Feedback:</strong> {{ r.reject...
                                        </div>
                                    {% endif %}
                                </td>

                                <!-- Category -->
                                <td class="py-4 px-4 whitespace-nowrap">
                                    <span class="inline-flex items-center px-2 py-0.5 rounde...
                                        {% if r.resource_type == 'Question Paper' %}bg-amber...

                                        {% elif r.resource_type == 'Lecture Notes' %}bg-emer...
                                        {% elif r.resource_type == 'Lab Manual' %}bg-purple-...
                                        {% else %}bg-blue-100 text-blue-800{% endif %}">
                                        {{ r.resource_type }}
                                    </span>
                                </td>

                                <!-- Subject & Dept -->
                                <td class="py-4 px-4 whitespace-nowrap">
                                    <div class="font-bold text-slate-700">{{ r.department_co...
                                    <div class="text-[11px] text-slate-500 truncate max-w-[1...
                                </td>

                                <!-- Status Badge -->
                                <td class="py-4 px-4 whitespace-nowrap">
                                    {% if r.status == 'approved' %}
                                        <span class="inline-flex items-center px-2.5 py-1 ro...
                                            <i class="fas fa-circle-check mr-1 text-[10px]">...
                                        </span>
                                    {% elif r.status == 'pending' %}
                                        <span class="inline-flex items-center px-2.5 py-1 ro...
                                            <i class="fas fa-clock mr-1 text-[10px]"></i> In...
                                        </span>
                                    {% else %}
                                        <span class="inline-flex items-center px-2.5 py-1 ro...
                                            <i class="fas fa-circle-xmark mr-1 text-[10px]">...
                                        </span>
                                    {% endif %}
                                </td>

                                <!-- Downloads -->
                                <td class="py-4 px-4 text-center font-bold text-slate-700">
                                    {{ r.download_count }}
                                </td>

                                <!-- Date -->
                                <td class="py-4 px-4 whitespace-nowrap text-slate-500 font-m...
                                    {{ r.created_at|str_date }}
                                </td>

                                <!-- Actions -->
                                <td class="py-4 px-6 text-right whitespace-nowrap">
                                    <div class="flex items-center justify-end space-x-2">
                                        {% if r.status == 'approved' %}
                                            <a href="{{ url_for('main.resource_detail', reso...
                                               class="p-2 rounded-lg bg-slate-100 hover:bg-s...
                                                <i class="fas fa-eye text-xs"></i>
                                            </a>
                                        {% endif %}
                                        <a href="{{ url_for('main.download_resource', resour...
                                           class="p-2 rounded-lg bg-blue-50 hover:bg-blue-10...
                                            <i class="fas fa-download text-xs"></i>
                                        </a>

                                        <!-- Delete action -->
                                        <form action="{{ url_for('upload.delete_my_upload', ...
                                              onsubmit="return confirm('Are you sure you wan...
                                            <button type="submit" class="p-2 rounded-lg bg-r...
                                                <i class="fas fa-trash-can text-xs"></i>
                                            </button>
                                        </form>
                                    </div>
                                </td>
                            </tr>
                        {% endfor %}
                    </tbody>
                </table>
            </div>
```

```
        </div>
    {% else %}
        <div class="bg-white rounded-3xl border border-slate-200 p-12 text-center max-w-lg m...
            <div class="w-16 h-16 rounded-2xl bg-blue-50 text-blue-600 flex items-center jus...
                <i class="fas fa-cloud-arrow-up"></i>
            </div>
            <h3 class="text-base font-bold text-slate-900 mb-1">You haven't uploaded any stu...
            <p class="text-xs text-slate-500 mb-6 leading-relaxed">
                Contribute previous years' examination papers, lecture notes, or lab manuals...
            </p>
            <a href="{{ url_for('upload.upload') }}"
               class="px-5 py-2.5 rounded-xl bg-blue-600 hover:bg-blue-700 text-white font-b...
                Upload Your First Material
            </a>
        </div>
    {% endif %}

</div>
{% endblock %}
```

■ File: templates/bookmarks.html

User's Pinned Examination Revision Materials

```
{% extends 'base.html' %}

{% block title %}My Saved Study Bookmarks | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

    <div class="flex items-center justify-between mb-8">
        <div>
            <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Quick Acc...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-1">Saved Bookma...
            <p class="text-xs sm:text-sm text-slate-500 mt-1">Your pinned question papers, l...
        </div>
        <a href="{{ url_for('main.browse') }}" class="text-xs font-semibold text-blue-600 ho...
            Browse More <i class="fas fa-chevron-right ml-1"></i>
        </a>
    </div>

    {% if bookmarks %}
        <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {% for item in bookmarks %}
                <div class="bg-white rounded-2xl border border-slate-200 p-6 flex flex-col j...
                    <div>
                        <div class="flex items-center justify-between text-xs mb-3">
                            <span class="inline-flex items-center px-2.5 py-1 rounded-md fon...
                                {% if item.resource_type == 'Question Paper' %}bg-amber-100 ...
                                {% elif item.resource_type == 'Lecture Notes' %}bg-emerald-1...
                                {% else %}bg-blue-100 text-blue-800{% endif %}">
                                {{ item.resource_type }}
                            </span>
                            <span class="text-slate-400 font-mono text-[11px]">{{ item.acade...
                        </div>

                        <div class="text-xs font-semibold text-blue-600 mb-1">
                            {{ item.department_code }} • Sem {{ item.semester }} • {{ item.s...
                        </div>

                        <h3 class="text-sm font-bold text-slate-900 line-clamp-2 hover:text-...
                            <a href="{{ url_for('main.resource_detail', resource_id=item.id)...
                                {{ item.title }}
                            </a>
                        </h3>

                        <p class="text-xs text-slate-500 line-clamp-2 mb-4 leading-relaxed">
                            {{ item.description or 'Saved study material.' }}
                        </p>
                    </div>

                    <div class="border-t border-slate-100 pt-4 flex items-center justify-bet...
                        <span class="text-xs text-slate-400 font-mono">{{ format_file_size(i...

                        <div class="flex items-center space-x-2">
                            <form action="{{ url_for('main.toggle_bookmark', resource_id=ite...
                                <button type="submit" class="p-2 rounded-lg bg-rose-50 hover...
                                    <i class="fas fa-bookmark text-xs"></i>
                                </button>
                            </form>
                            <a href="{{ url_for('main.download_resource', resource_id=item.i...
                               class="px-3 py-1.5 rounded-lg bg-blue-600 hover:bg-blue-700 t...
                                <i class="fas fa-download text-[10px]"></i>
                                <span>Get</span>
                            </a>
                        </div>
                    </div>
                </div>
            {% endfor %}
        </div>
    {% else %}
        <div class="bg-white rounded-3xl border border-slate-200 p-12 text-center max-w-md m...
            <div class="w-14 h-14 rounded-2xl bg-amber-50 text-amber-600 flex items-center j...
                <i class="far fa-bookmark"></i>
            </div>
            <h3 class="text-base font-bold text-slate-900 mb-1">No bookmarked materials yet<...
            <p class="text-xs text-slate-500 mb-6">
                Click the "Bookmark" button on any question paper or note to quickly access ...
```

```
            </p>
            <a href="{{ url_for('main.browse') }}" class="px-4 py-2 rounded-xl bg-blue-600 t...
                Explore Study Materials
            </a>
        </div>
    {% endif %}

</div>
{% endblock %}
```

■ File: templates/offline.html

Campus LAN & Offline Access Guide with Wi-Fi Connection Instructions

```
{% extends 'base.html' %}

{% block title %}Offline & Campus LAN Access Guide | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-4xl mx-auto px-4 sm:px-6 lg:px-8 py-12">

    <div class="bg-white rounded-3xl p-8 sm:p-12 border border-slate-200 shadow-sm space-y-8">

        <div class="text-center max-w-xl mx-auto space-y-3">
            <div class="w-16 h-16 rounded-3xl bg-emerald-100 text-emerald-600 flex items-cen...
                <i class="fas fa-wifi"></i>
            </div>
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900">Offline & Local C...
            <p class="text-xs sm:text-sm text-slate-500 leading-relaxed">
                The Centralized College Portal is engineered to function 100% offline across...
            </p>
        </div>

        <!-- Connection Instructions Box -->
        <div class="p-6 rounded-2xl bg-slate-900 text-white space-y-4">
            <div class="flex items-center space-x-2 text-emerald-400 text-xs font-bold upper...
                <span class="w-2 h-2 rounded-full bg-emerald-400 animate-ping"></span>
                <span>Your Host Server Endpoint</span>
            </div>
            <div class="p-4 bg-black/40 rounded-xl border border-white/10 flex flex-col sm:f...
                <div>
                    <span class="text-xs text-slate-400 block mb-0.5">Campus Wi-Fi / Local N...
                    <span class="text-lg sm:text-xl font-mono font-bold text-amber-300">http...
                </div>
                <button onclick="navigator.clipboard.writeText('http://{{ lan_ip }}:{{ serve...
                        class="px-4 py-2 bg-blue-600 hover:bg-blue-500 text-white font-bold ...
                    Copy Link
                </button>
            </div>
            <p class="text-xs text-slate-400">
                Any phone, tablet, or laptop connected to the same Wi-Fi router as this comp...
            </p>
        </div>

        <!-- 3 Step Quick Guide -->
        <div class="grid grid-cols-1 md:grid-cols-3 gap-6 pt-4">
            <div class="p-5 rounded-2xl bg-slate-50 border border-slate-100 space-y-2">
                <div class="w-8 h-8 rounded-xl bg-blue-100 text-blue-700 font-bold text-sm f...
                <h3 class="text-sm font-bold text-slate-800">Connect to Same Wi-Fi</h3>
                <p class="text-xs text-slate-500 leading-relaxed">
                    Connect your smartphone or laptop to the college Wi-Fi or turn on a port...
                </p>
            </div>

            <div class="p-5 rounded-2xl bg-slate-50 border border-slate-100 space-y-2">
                <div class="w-8 h-8 rounded-xl bg-indigo-100 text-indigo-700 font-bold text-...
                <h3 class="text-sm font-bold text-slate-800">Open Browser</h3>
                <p class="text-xs text-slate-500 leading-relaxed">
                    Open Chrome, Safari, or Firefox and type <strong class="text-slate-700">...
                </p>
            </div>

            <div class="p-5 rounded-2xl bg-slate-50 border border-slate-100 space-y-2">
                <div class="w-8 h-8 rounded-xl bg-emerald-100 text-emerald-700 font-bold tex...
                <h3 class="text-sm font-bold text-slate-800">Study & Download</h3>
                <p class="text-xs text-slate-500 leading-relaxed">
                    Browse previous year question papers and faculty notes with zero interne...
                </p>
            </div>
        </div>

        <div class="pt-6 border-t border-slate-100 text-center">
            <a href="{{ url_for('main.browse') }}" class="px-6 py-3 bg-blue-600 hover:bg-blu...
                Browse Repository Now
            </a>
        </div>

    </div>

```

```
</div>
{% endblock %}
```

■ File: templates/error.html

404 / 500 Error Screen with Safe Redirection

```
{% extends 'base.html' %}

{% block title %}Error {{ error_code }} | Academix Portal{% endblock %}

{% block content %}
<div class="min-h-[70vh] flex items-center justify-center py-12 px-4 text-center">
    <div class="max-w-md w-full bg-white rounded-3xl p-8 sm:p-10 border border-slate-200 sha...
        <div class="w-16 h-16 rounded-2xl bg-rose-50 text-rose-600 flex items-center justify...
            <i class="fas fa-triangle-exclamation"></i>
        </div>
        <div class="text-4xl font-extrabold text-slate-900 font-mono">{{ error_code }}</div>
        <h2 class="text-lg font-bold text-slate-800">Something went wrong</h2>
        <p class="text-xs text-slate-500 leading-relaxed">
            {{ message or "We couldn't find the academic page or study resource you were loo...
        </p>
        <div class="pt-4 flex items-center justify-center space-x-3">
            <a href="{{ url_for('main.index') }}" class="px-5 py-2.5 bg-blue-600 hover:bg-bl...
                Back to Home
            </a>
            <a href="{{ url_for('main.browse') }}" class="px-5 py-2.5 bg-slate-100 hover:bg-...
                Browse Repository
            </a>
        </div>
    </div>
</div>
{% endblock %}
```

■ File: templates/auth/login.html

Sign In Page with Pre-Configured Test Credentials Hint

```
{% extends 'base.html' %}

{% block title %}Sign In | Academix Portal{% endblock %}

{% block content %}
<div class="min-h-[80vh] flex items-center justify-center py-12 px-4 sm:px-6 lg:px-8">
    <div class="max-w-md w-full space-y-8 bg-white p-8 sm:p-10 rounded-3xl border border-sla...

        <div class="text-center">
            <div class="w-12 h-12 rounded-2xl bg-gradient-to-tr from-blue-700 to-indigo-600 ...
                <i class="fas fa-book-open-reader text-xl"></i>
            </div>
            <h2 class="text-2xl font-extrabold text-slate-900 tracking-tight">Sign in to you...
            <p class="text-xs text-slate-500 mt-2">Access your study materials, bookmarks, a...
        </div>

        <form action="{{ url_for('auth.login', next=request.args.get('next')) }}" method="PO...
            <div>
                <label for="identifier" class="block text-xs font-bold text-slate-700 upperc...
                    Username or Email
                </label>
                <div class="relative">
                    <i class="fas fa-user absolute left-3.5 top-3.5 text-slate-400 text-xs">...
                    <input type="text" name="identifier" id="identifier" required
                           placeholder="e.g. admin, student1, or name@college.edu"
                           class="w-full pl-10 pr-4 py-2.5 bg-slate-50 text-slate-800 text-s...
                </div>
            </div>

            <div>
                <label for="password" class="block text-xs font-bold text-slate-700 uppercas...
                    Password
                </label>
                <div class="relative">
                    <i class="fas fa-lock absolute left-3.5 top-3.5 text-slate-400 text-xs">...
                    <input type="password" name="password" id="password" required
                           placeholder="••••••••"
                           class="w-full pl-10 pr-4 py-2.5 bg-slate-50 text-slate-800 text-s...
                </div>
            </div>

            <button type="submit"
                    class="w-full py-3 px-4 bg-blue-600 hover:bg-blue-700 text-white font-bo...
                Sign In
            </button>
        </form>

        <!-- Demo accounts quick hint box -->
        <div class="p-3.5 rounded-xl bg-slate-50 border border-slate-200/80 text-[11px] text...
            <span class="font-bold text-slate-700 block"><i class="fas fa-key text-amber-500...
            <div class="grid grid-cols-2 gap-1 text-[10px] text-slate-500 pt-1">
                <div>Admin: <code class="text-blue-600 font-bold">admin / admin123</code></div>
                <div>Student: <code class="text-blue-600 font-bold">student1 / student123</c...
                <div>Faculty: <code class="text-blue-600 font-bold">prof_sharma / faculty123...
                <div>Student 2: <code class="text-blue-600 font-bold">student2 / student123<...
            </div>
        </div>

        <div class="text-center text-xs text-slate-500">
            Don't have an account?
            <a href="{{ url_for('auth.register') }}" class="font-bold text-blue-600 hover:un...
        </div>

    </div>
</div>
{% endblock %}
```

■ File: templates/auth/register.html

Student and Faculty Registration Form with Department & Semester Selectors

```
{% extends 'base.html' %}

{% block title %}Create Account | Academix Portal{% endblock %}

{% block content %}
<div class="min-h-[85vh] flex items-center justify-center py-12 px-4 sm:px-6 lg:px-8">
    <div class="max-w-xl w-full space-y-8 bg-white p-8 sm:p-10 rounded-3xl border border-sla...

        <div class="text-center">
            <div class="w-12 h-12 rounded-2xl bg-gradient-to-tr from-blue-700 to-indigo-600 ...
                <i class="fas fa-user-plus text-xl"></i>
            </div>
            <h2 class="text-2xl font-extrabold text-slate-900 tracking-tight">Create your st...
            <p class="text-xs text-slate-500 mt-2">Join the centralized academic portal to s...
        </div>

        <form action="{{ url_for('auth.register') }}" method="POST" class="mt-8 space-y-4">

            <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
                <!-- Full Name -->
                <div>
                    <label for="full_name" class="block text-xs font-bold text-slate-700 upp...
                        Full Name <span class="text-rose-500">*</span>
                    </label>
                    <input type="text" name="full_name" id="full_name" required placeholder=...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>

                <!-- Username -->
                <div>
                    <label for="username" class="block text-xs font-bold text-slate-700 uppe...
                        Username <span class="text-rose-500">*</span>
                    </label>
                    <input type="text" name="username" id="username" required placeholder="a...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>
            </div>

            <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
                <!-- Email -->
                <div>
                    <label for="email" class="block text-xs font-bold text-slate-700 upperca...
                        College Email <span class="text-rose-500">*</span>
                    </label>
                    <input type="email" name="email" id="email" required placeholder="aarav@...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>

                <!-- Roll Number -->
                <div>
                    <label for="roll_number" class="block text-xs font-bold text-slate-700 u...
                        Roll / Reg Number
                    </label>
                    <input type="text" name="roll_number" id="roll_number" placeholder="21CS...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>
            </div>

            <!-- Role Selection -->
            <div>
                <label class="block text-xs font-bold text-slate-700 uppercase tracking-wide...
                    Account Type <span class="text-rose-500">*</span>
                </label>
                <div class="grid grid-cols-2 gap-3">
                    <label class="flex items-center p-3 rounded-xl border border-slate-200 c...
                        <input type="radio" name="role" value="student" checked class="text-...
                        <span class="ml-2.5 text-xs font-bold text-slate-800">Student Accoun...
                    </label>
                    <label class="flex items-center p-3 rounded-xl border border-slate-200 c...
                        <input type="radio" name="role" value="faculty" class="text-blue-600...
                        <span class="ml-2.5 text-xs font-bold text-slate-800">Faculty / Teac...
                    </label>
                </div>
            </div>

```

```
            <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
                <!-- Department -->
                <div>
                    <label for="department_id" class="block text-xs font-bold text-slate-700...
                        Department
                    </label>
                    <select name="department_id" id="department_id"
                            class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-700 text-xs r...
                        <option value="">Select Department</option>
                        {% for d in departments %}

                            <option value="{{ d.id }}">{{ d.code }} - {{ d.name }}</option>
                        {% endfor %}
                    </select>
                </div>

                <!-- Semester -->
                <div>
                    <label for="semester" class="block text-xs font-bold text-slate-700 uppe...
                        Current Semester
                    </label>
                    <select name="semester" id="semester"
                            class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-700 text-xs r...
                        <option value="">Select Semester</option>
                        {% for s in range(1, 9) %}
                            <option value="{{ s }}">Semester {{ s }}</option>
                        {% endfor %}
                    </select>
                </div>
            </div>

            <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
                <!-- Password -->
                <div>
                    <label for="password" class="block text-xs font-bold text-slate-700 uppe...
                        Password (Min 6 chars) <span class="text-rose-500">*</span>
                    </label>
                    <input type="password" name="password" id="password" required placeholde...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>

                <!-- Confirm Password -->
                <div>
                    <label for="confirm_password" class="block text-xs font-bold text-slate-...
                        Confirm Password <span class="text-rose-500">*</span>
                    </label>
                    <input type="password" name="confirm_password" id="confirm_password" req...
                           class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-sm ro...
                </div>
            </div>

            <button type="submit"
                    class="w-full py-3 px-4 bg-blue-600 hover:bg-blue-700 text-white font-bo...
                Complete Registration
            </button>
        </form>

        <div class="text-center text-xs text-slate-500">
            Already have an account?
            <a href="{{ url_for('auth.login') }}" class="font-bold text-blue-600 hover:under...
        </div>

    </div>
</div>
{% endblock %}
```

■ File: templates/auth/profile.html

User Account Profile, Academic Stats, Password Change Form

```
{% extends 'base.html' %}

{% block title %}User Profile & Settings | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-5xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

    <div class="mb-8">
        <span class="text-xs font-bold text-blue-600 uppercase tracking-wider">Account Overv...
        <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-1">Profile & Prefer...
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">

        <!-- Left: Profile Summary Card & Contribution Metrics -->
        <div class="space-y-6">
            <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm text-cent...
                <div class="w-20 h-20 rounded-3xl bg-gradient-to-tr from-blue-600 to-indigo-...
                    {{ user.full_name[:2].upper() }}
                </div>
                <h2 class="text-lg font-bold text-slate-900">{{ user.full_name }}</h2>
                <p class="text-xs text-slate-500 font-mono">@{{ user.username }}</p>

                <div class="mt-3">
                    <span class="inline-block px-3 py-1 rounded-full text-xs font-bold upper...
                        {% if user.role == 'admin' %}bg-rose-100 text-rose-700
                        {% elif user.role == 'faculty' %}bg-amber-100 text-amber-700
                        {% else %}bg-blue-100 text-blue-700{% endif %}">
                        {{ user.role }} Account
                    </span>
                </div>

                <div class="mt-6 border-t border-slate-100 pt-4 text-xs text-slate-600 space...
                    <div class="flex justify-between">
                        <span class="text-slate-400">Department:</span>
                        <span class="font-bold text-slate-800">{{ user.department_code or 'N...
                    </div>
                    <div class="flex justify-between">
                        <span class="text-slate-400">Semester:</span>
                        <span class="font-bold text-slate-800">{% if user.semester %}Semeste...
                    </div>
                    <div class="flex justify-between">
                        <span class="text-slate-400">Roll Number:</span>
                        <span class="font-bold font-mono text-slate-800">{{ user.roll_number...
                    </div>
                    <div class="flex justify-between">
                        <span class="text-slate-400">Member Since:</span>
                        <span class="text-slate-700 font-mono">{{ user.created_at|str_date }...
                    </div>
                </div>
            </div>

            <!-- Contribution Impact Card -->
            <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-3">
                <h3 class="text-xs font-bold text-slate-900 uppercase tracking-wider">Your I...
                <div class="grid grid-cols-2 gap-3 text-center">
                    <div class="p-3 rounded-2xl bg-slate-50 border border-slate-100">
                        <div class="text-xl font-extrabold text-blue-600">{{ stats.uploads_c...
                        <div class="text-[10px] text-slate-500 uppercase font-semibold">Uplo...
                    </div>
                    <div class="p-3 rounded-2xl bg-slate-50 border border-slate-100">
                        <div class="text-xl font-extrabold text-emerald-600">{{ stats.approv...
                        <div class="text-[10px] text-slate-500 uppercase font-semibold">Publ...
                    </div>
                    <div class="p-3 rounded-2xl bg-slate-50 border border-slate-100">
                        <div class="text-xl font-extrabold text-amber-500">{{ stats.bookmark...
                        <div class="text-[10px] text-slate-500 uppercase font-semibold">Book...
                    </div>
                    <div class="p-3 rounded-2xl bg-slate-50 border border-slate-100">
                        <div class="text-xl font-extrabold text-indigo-600">{{ stats.total_d...
                        <div class="text-[10px] text-slate-500 uppercase font-semibold">Down...
                    </div>
                </div>
            </div>
        </div>
```

```
76 |
77 |         <!-- Right: Edit Details & Password Tabs/Forms -->
78 |         <div class="lg:col-span-2 space-y-8">
79 |
80 |             <!-- Edit Profile Form -->
81 |             <div class="bg-white rounded-3xl p-6 sm:p-8 border border-slate-200 shadow-sm sp...
82 |                 <h3 class="text-base font-bold text-slate-900">Personal & Academic Details</h3>
83 |
84 |                 <form action="{{ url_for('auth.profile') }}" method="POST" class="space-y-4">
85 |                     <input type="hidden" name="action" value="update_info">

86 |
87 |                     <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
88 |                         <div>
89 |                             <label class="block text-xs font-bold text-slate-700 uppercase t...
90 |                             <input type="text" name="full_name" value="{{ user.full_name }}"...
91 |                                    class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 te...
92 |                         </div>
93 |                         <div>
94 |                             <label class="block text-xs font-bold text-slate-700 uppercase t...
95 |                             <input type="email" name="email" value="{{ user.email }}" required
96 |                                    class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 te...
97 |                         </div>
98 |                     </div>
99 |
100 |                    <div class="grid grid-cols-1 sm:grid-cols-3 gap-4">
101 |                        <div>
102 |                            <label class="block text-xs font-bold text-slate-700 uppercase t...
103 |                            <input type="text" name="roll_number" value="{{ user.roll_number...
104 |                                   class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 te...
105 |                        </div>
106 |                        <div>
107 |                            <label class="block text-xs font-bold text-slate-700 uppercase t...
108 |                            <select name="department_id" class="w-full px-3.5 py-2.5 bg-slat...
109 |                                <option value="">None</option>
110 |                                {% for d in departments %}
111 |                                    <option value="{{ d.id }}" {% if user.department_id == d...
112 |                                {% endfor %}
113 |                            </select>
114 |                        </div>
115 |                        <div>
116 |                            <label class="block text-xs font-bold text-slate-700 uppercase t...
117 |                            <select name="semester" class="w-full px-3.5 py-2.5 bg-slate-50 ...
118 |                                <option value="">None</option>
119 |                                {% for s in range(1, 9) %}
120 |                                    <option value="{{ s }}" {% if user.semester == s %}selec...
121 |                                {% endfor %}
122 |                            </select>
123 |                        </div>
124 |                    </div>
125 |
126 |                    <div class="pt-2 text-right">
127 |                        <button type="submit" class="px-5 py-2.5 bg-blue-600 hover:bg-blue-7...
128 |                            Save Changes
129 |                        </button>
130 |                    </div>
131 |                </form>
132 |            </div>
133 |
134 |            <!-- Change Password Form -->
135 |            <div class="bg-white rounded-3xl p-6 sm:p-8 border border-slate-200 shadow-sm sp...
136 |                <h3 class="text-base font-bold text-slate-900">Security & Password</h3>
137 |
138 |                <form action="{{ url_for('auth.profile') }}" method="POST" class="space-y-4">
139 |                    <input type="hidden" name="action" value="change_password">
140 |
141 |                    <div>
142 |                        <label class="block text-xs font-bold text-slate-700 uppercase track...
143 |                        <input type="password" name="current_password" required placeholder=...
144 |                               class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 text-s...
145 |                    </div>
146 |
147 |                    <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
148 |                        <div>
149 |                            <label class="block text-xs font-bold text-slate-700 uppercase t...
150 |                            <input type="password" name="new_password" required placeholder=...
151 |                                   class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 te...
152 |                        </div>
153 |                        <div>
```

```
                            <label class="block text-xs font-bold text-slate-700 uppercase t...
                            <input type="password" name="confirm_new_password" required plac...
                                   class="w-full px-3.5 py-2.5 bg-slate-50 text-slate-800 te...
                        </div>
                    </div>

                    <div class="pt-2 text-right">
                        <button type="submit" class="px-5 py-2.5 bg-slate-800 hover:bg-slate...
                            Update Password
                        </button>
                    </div>
                </form>
            </div>

        </div>

    </div>

</div>
{% endblock %}
```

■ File: templates/admin/dashboard.html

Admin Analytics Dashboard: KPI Cards, Quick Approval Queue, Resource Breakdown

```
{% extends 'base.html' %}

{% block title %}Admin Dashboard | Academix Portal{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Admin Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Administr...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">System Ove...
        </div>

        <!-- Navigation Tabs -->
        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-blue...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
                <span>Review Queue</span>
                {% if stats.pending_resources > 0 %}
                    <span class="px-1.5 py-0.2 bg-rose-500 text-white text-[10px] rounded-fu...
                {% endif %}
            </a>
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <!-- KPI Metric Cards -->
    <div class="grid grid-cols-2 lg:grid-cols-4 gap-4">
        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm flex items-ce...
            <div>
                <p class="text-xs font-bold text-slate-400 uppercase tracking-wider">Approve...
                <p class="text-3xl font-extrabold text-blue-600 mt-1">{{ stats.approved_reso...
                <p class="text-[11px] text-slate-400 mt-1">Total live documents</p>
            </div>
            <div class="w-12 h-12 rounded-2xl bg-blue-50 text-blue-600 flex items-center jus...
                <i class="fas fa-file-circle-check"></i>
            </div>
        </div>

        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm flex items-ce...
            <div>
                <p class="text-xs font-bold text-slate-400 uppercase tracking-wider">Pending...
                <p class="text-3xl font-extrabold text-amber-500 mt-1">{{ stats.pending_reso...
                <p class="text-[11px] text-slate-400 mt-1">Awaiting verification</p>
            </div>
            <div class="w-12 h-12 rounded-2xl bg-amber-50 text-amber-500 flex items-center j...
                <i class="fas fa-hourglass-half"></i>
            </div>
        </div>

        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm flex items-ce...
            <div>
                <p class="text-xs font-bold text-slate-400 uppercase tracking-wider">Total D...
                <p class="text-3xl font-extrabold text-emerald-600 mt-1">{{ stats.total_down...
                <p class="text-[11px] text-slate-400 mt-1">Campus wide access</p>
            </div>
            <div class="w-12 h-12 rounded-2xl bg-emerald-50 text-emerald-600 flex items-cent...
                <i class="fas fa-chart-line"></i>
            </div>
        </div>

        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm flex items-ce...
            <div>
                <p class="text-xs font-bold text-slate-400 uppercase tracking-wider">Registe...
                <p class="text-3xl font-extrabold text-purple-600 mt-1">{{ stats.total_users...
                <p class="text-[11px] text-slate-400 mt-1">Students & faculty</p>
            </div>
            <div class="w-12 h-12 rounded-2xl bg-purple-50 text-purple-600 flex items-center...
                <i class="fas fa-users"></i>
            </div>
```

```
        </div>
    </div>

    <!-- Review Queue Alert if items pending -->
    {% if pending_items %}
        <div class="bg-amber-50 rounded-3xl p-6 border border-amber-200">
            <div class="flex items-center justify-between mb-4">
                <div class="flex items-center space-x-2">
                    <span class="w-2.5 h-2.5 rounded-full bg-amber-500 animate-ping"></span>
                    <h3 class="text-sm font-bold text-amber-900 uppercase tracking-wider">It...

                </div>
                <a href="{{ url_for('admin.pending_queue') }}" class="text-xs font-bold text...
                    View all ({{ stats.pending_resources }}) <i class="fas fa-arrow-right ml...
                </a>
            </div>

            <div class="grid grid-cols-1 md:grid-cols-2 gap-4">
                {% for item in pending_items %}
                    <div class="bg-white rounded-2xl p-4 border border-amber-200/80 shadow-s...
                        <div>
                            <div class="flex items-center justify-between text-[11px] mb-1">
                                <span class="font-bold text-blue-600">{{ item.department_cod...
                                <span class="text-slate-400">{{ item.created_at|str_date }}<...
                            </div>
                            <h4 class="text-xs font-bold text-slate-900 line-clamp-1 mb-1">{...
                            <p class="text-[11px] text-slate-500 line-clamp-1">Uploaded by {...
                        </div>
                        <div class="flex items-center justify-between pt-3 mt-3 border-t bor...
                            <a href="{{ url_for('main.preview_resource', resource_id=item.id...
                               class="text-xs text-slate-600 hover:text-blue-600 font-semibo...
                                <i class="fas fa-eye text-[10px]"></i>
                                <span>Preview</span>
                            </a>
                            <div class="flex items-center space-x-2">
                                <form action="{{ url_for('admin.reject_resource', resource_i...
                                    <button type="submit" class="px-2.5 py-1 rounded-lg bg-r...
                                        Reject
                                    </button>
                                </form>
                                <form action="{{ url_for('admin.approve_resource', resource_...
                                    <button type="submit" class="px-3 py-1 rounded-lg bg-eme...
                                        Approve
                                    </button>
                                </form>
                            </div>
                        </div>
                    </div>
                {% endfor %}
            </div>
        </div>
    {% endif %}

    <!-- Resource Distribution & Activity Grid -->
    <div class="grid grid-cols-1 lg:grid-cols-2 gap-8">

        <!-- Breakdown by Category -->
        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
            <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Materials ...
            <div class="space-y-3">
                {% for t in type_breakdown %}
                    <div class="space-y-1">
                        <div class="flex justify-between text-xs font-semibold text-slate-700">
                            <span>{{ t.resource_type }}</span>
                            <span>{{ t.count }} items</span>
                        </div>
                        <div class="w-full bg-slate-100 h-2 rounded-full overflow-hidden">
                            <div class="bg-blue-600 h-full rounded-full" style="width: {{ (t...
                            </div>
                    </div>
                {% endfor %}
            </div>
        </div>

        <!-- Breakdown by Department -->
        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
            <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Materials ...
            <div class="space-y-3">
                {% for d in dept_breakdown %}
```

```
                    <div class="space-y-1">
                        <div class="flex justify-between text-xs font-semibold text-slate-700">
                            <span>{{ d.code }} - {{ d.name }}</span>
                            <span>{{ d.count }} items</span>
                        </div>
                        <div class="w-full bg-slate-100 h-2 rounded-full overflow-hidden">
                            <div class="bg-indigo-600 h-full rounded-full" style="width: {{ ...
                        </div>
                    </div>
                {% endfor %}
            </div>
        </div>

    </div>

    <!-- Recent Audit Logs -->
    <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">

        <div class="flex items-center justify-between">
            <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Recent Sys...
            <a href="{{ url_for('admin.activity_logs') }}" class="text-xs font-bold text-blu...
        </div>

        <div class="divide-y divide-slate-100 text-xs">
            {% for log in recent_logs %}
                <div class="py-2.5 flex items-center justify-between">
                    <div class="flex items-center space-x-3">
                        <span class="px-2 py-0.5 rounded font-mono font-bold text-[10px]
                            {% if log.action in ('APPROVE_RESOURCE', 'REGISTER') %}bg-emeral...
                            {% elif log.action in ('REJECT_RESOURCE', 'ADMIN_DELETE_RESOURCE...
                            {% elif log.action == 'UPLOAD_RESOURCE' %}bg-blue-100 text-blue-800
                            {% else %}bg-slate-100 text-slate-700{% endif %}">
                            {{ log.action }}
                        </span>
                        <span class="text-slate-700 font-medium">{{ log.details }}</span>
                        <span class="text-slate-400">by {{ log.full_name or log.username or ...
                    </div>
                    <span class="text-slate-400 font-mono text-[11px]">{{ log.created_at|str...
                </div>
            {% endfor %}
        </div>
    </div>

</div>
{% endblock %}
```

■ File: templates/admin/pending.html

Dedicated Moderation Queue with One-Click Approve and Reject with Reason

```
{% extends 'base.html' %}

{% block title %}Review Queue | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Moderatio...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">Pending Ap...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
                <span>Review Queue</span>
                {% if items|length > 0 %}
                    <span class="px-1.5 py-0.2 bg-white text-blue-600 text-[10px] rounded-fu...
                {% endif %}
            </a>
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    {% if items %}
        <div class="bg-white rounded-3xl border border-slate-200 shadow-sm overflow-hidden">
            <div class="overflow-x-auto">
                <table class="w-full text-left border-collapse text-xs">
                    <thead>
                        <tr class="bg-slate-50 border-b border-slate-200 text-slate-500 font...
                            <th class="py-3.5 px-6">Document Details</th>
                            <th class="py-3.5 px-4">Uploader</th>
                            <th class="py-3.5 px-4">Category & Exam</th>
                            <th class="py-3.5 px-4">Subject & Dept</th>
                            <th class="py-3.5 px-4">Date Submitted</th>
                            <th class="py-3.5 px-6 text-right">Moderation Actions</th>
                        </tr>
                    </thead>
                    <tbody class="divide-y divide-slate-100">
                        {% for item in items %}
                            <tr class="hover:bg-slate-50/70 transition-colors">
                                <td class="py-4 px-6 max-w-sm">
                                    <div class="font-bold text-slate-900 line-clamp-1 text-s...
                                    <p class="text-[11px] text-slate-500 line-clamp-2 mt-0.5...
                                    <div class="flex items-center space-x-2 mt-1.5">
                                        <a href="{{ url_for('main.preview_resource', resourc...
                                           class="inline-flex items-center px-2 py-0.5 round...
                                            <i class="fas fa-eye mr-1"></i> Preview File
                                        </a>
                                        <span class="text-slate-400 font-mono text-[10px]">
                                            {{ item.original_file_name }} ({{ format_file_si...
                                        </span>
                                    </div>
                                </td>

                                <td class="py-4 px-4 whitespace-nowrap">
                                    <div class="font-bold text-slate-800">{{ item.uploader_n...
                                    <div class="text-[11px] text-slate-400">{{ item.uploader...
                                    <span class="inline-block mt-0.5 px-2 py-0.2 rounded tex...
                                        {{ item.uploader_role }}
                                    </span>
                                </td>

                                <td class="py-4 px-4 whitespace-nowrap">
                                    <span class="inline-flex items-center px-2 py-0.5 rounde...
                                        {% if item.resource_type == 'Question Paper' %}bg-am...
                                        {% elif item.resource_type == 'Lecture Notes' %}bg-e...
                                        {% else %}bg-blue-100 text-blue-800{% endif %}">
                                        {{ item.resource_type }}
```

```
                                    </span>
                                    <div class="text-[11px] text-slate-500 mt-1 font-mono">{...
                                    {% if item.exam_type and item.exam_type != 'N/A' %}
                                        <div class="text-[10px] text-amber-700 font-semibold...
                                    {% endif %}
                                </td>

                                <td class="py-4 px-4 whitespace-nowrap">
                                    <div class="font-bold text-slate-800">{{ item.department...
                                    <div class="text-[11px] text-slate-500 truncate max-w-[1...

                                </td>

                                <td class="py-4 px-4 whitespace-nowrap text-slate-500 font-m...
                                    {{ item.created_at|str_time }}
                                </td>

                                <td class="py-4 px-6 text-right whitespace-nowrap">
                                    <div class="flex items-center justify-end space-x-2">
                                        <!-- Approve Form -->
                                        <form action="{{ url_for('admin.approve_resource', r...
                                            <button type="submit"
                                                    class="px-3.5 py-1.5 rounded-xl bg-emera...
                                                <i class="fas fa-check text-[10px]"></i>
                                                <span>Approve</span>
                                            </button>
                                        </form>

                                        <!-- Reject Form -->
                                        <form action="{{ url_for('admin.reject_resource', re...
                                              onsubmit="const r = prompt('Please enter rejec...
                                            <input type="hidden" name="reason" value="">
                                            <button type="submit"
                                                    class="px-3.5 py-1.5 rounded-xl bg-rose-...
                                                Reject
                                            </button>
                                        </form>
                                    </div>
                                </td>
                            </tr>
                        {% endfor %}
                    </tbody>
                </table>
            </div>
        </div>
    {% else %}
        <div class="bg-white rounded-3xl border border-slate-200 p-16 text-center max-w-lg m...
            <div class="w-16 h-16 rounded-2xl bg-emerald-50 text-emerald-600 flex items-cent...
                <i class="fas fa-circle-check"></i>
            </div>
            <h3 class="text-base font-bold text-slate-900 mb-1">Queue is Clear!</h3>
            <p class="text-xs text-slate-500">
                All submitted study materials and question papers have been reviewed and pub...
            </p>
        </div>
    {% endif %}

</div>
{% endblock %}
```

■ File: templates/admin/resources.html

Master Resource Manager: Search, Filter, Edit, Featured Toggle, Permanent Delete

```
{% extends 'base.html' %}

{% block title %}Manage Resources | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Repositor...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">All Academ...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <!-- Filter Bar -->
    <div class="bg-white rounded-2xl p-4 border border-slate-200 shadow-sm">
        <form action="{{ url_for('admin.manage_resources') }}" method="GET" class="flex flex...
            <div class="flex-1 relative">
                <i class="fas fa-search absolute left-3.5 top-3 text-slate-400 text-xs"></i>
                <input type="text" name="q" value="{{ q }}" placeholder="Search resources by...
                       class="w-full pl-9 pr-3 py-2 bg-slate-50 text-xs rounded-xl border bo...
            </div>
            <div class="w-full sm:w-44">
                <select name="status" onchange="this.form.submit()" class="w-full py-2 px-3 ...
                    <option value="">All Statuses</option>
                    <option value="approved" {% if status_filter == 'approved' %}selected{% ...
                    <option value="pending" {% if status_filter == 'pending' %}selected{% en...
                    <option value="rejected" {% if status_filter == 'rejected' %}selected{% ...
                </select>
            </div>
            <div class="w-full sm:w-48">
                <select name="dept" onchange="this.form.submit()" class="w-full py-2 px-3 bg...
                    <option value="">All Departments</option>
                    {% for d in departments %}
                        <option value="{{ d.id }}" {% if dept_filter|string == d.id|string %...
                    {% endfor %}
                </select>
            </div>
            <button type="submit" class="px-4 py-2 bg-slate-800 text-white rounded-xl text-x...
        </form>
    </div>

    <!-- Master Table -->
    <div class="bg-white rounded-3xl border border-slate-200 shadow-sm overflow-hidden">
        <div class="overflow-x-auto">
            <table class="w-full text-left border-collapse text-xs">
                <thead>
                    <tr class="bg-slate-50 border-b border-slate-200 text-slate-500 font-bol...
                        <th class="py-3.5 px-6">Title & Specs</th>
                        <th class="py-3.5 px-4">Subject & Dept</th>
                        <th class="py-3.5 px-4">Category</th>
                        <th class="py-3.5 px-4">Status</th>
                        <th class="py-3.5 px-4 text-center">Featured</th>
                        <th class="py-3.5 px-4 text-center">Downloads</th>
                        <th class="py-3.5 px-6 text-right">Actions</th>
                    </tr>
                </thead>
                <tbody class="divide-y divide-slate-100">
                    {% for r in resources %}
                        <tr class="hover:bg-slate-50/70 transition-colors">
                            <td class="py-4 px-6 max-w-xs">
                                <div class="font-bold text-slate-900 line-clamp-1">
                                    <a href="{{ url_for('main.resource_detail', resource_id=...
                                        {{ r.title }}
```

```
                                    </a>
                                </div>
                                <div class="text-[11px] text-slate-400 font-mono mt-0.5">
                                    {{ r.original_file_name }} ({{ format_file_size(r.file_s...
                                </div>
                            </td>

                            <td class="py-4 px-4 whitespace-nowrap">
                                <span class="font-bold text-slate-800">{{ r.department_code ...
                                <div class="text-[11px] text-slate-500 truncate max-w-[130px...

                            </td>

                            <td class="py-4 px-4 whitespace-nowrap">
                                <span class="inline-flex items-center px-2 py-0.5 rounded fo...
                                    {{ r.resource_type }}
                                </span>
                            </td>

                            <td class="py-4 px-4 whitespace-nowrap">
                                {% if r.status == 'approved' %}
                                    <span class="px-2 py-0.5 rounded-full text-[10px] font-b...
                                {% elif r.status == 'pending' %}
                                    <span class="px-2 py-0.5 rounded-full text-[10px] font-b...
                                {% else %}
                                    <span class="px-2 py-0.5 rounded-full text-[10px] font-b...
                                {% endif %}
                            </td>

                            <td class="py-4 px-4 text-center whitespace-nowrap">
                                <form action="{{ url_for('admin.toggle_featured', resource_i...
                                    <button type="submit" class="p-1 rounded text-xs transit...
                                        <i class="fas fa-star text-base"></i>
                                    </button>
                                </form>
                            </td>

                            <td class="py-4 px-4 text-center font-bold text-slate-700">
                                {{ r.download_count }}
                            </td>

                            <td class="py-4 px-6 text-right whitespace-nowrap">
                                <div class="flex items-center justify-end space-x-2">
                                    <a href="{{ url_for('main.resource_detail', resource_id=...
                                        <i class="fas fa-eye text-xs"></i>
                                    </a>
                                    <a href="{{ url_for('main.download_resource', resource_i...
                                        <i class="fas fa-download text-xs"></i>
                                    </a>
                                    <form action="{{ url_for('admin.delete_resource', resour...
                                          onsubmit="return confirm('Permanently delete this ...
                                        <button type="submit" class="p-2 rounded-lg bg-rose-...
                                            <i class="fas fa-trash text-xs"></i>
                                        </button>
                                    </form>
                                </div>
                            </td>
                        </tr>
                    {% endfor %}
                </tbody>
            </table>
        </div>
    </div>

</div>
{% endblock %}
```

■ File: templates/admin/users.html

User Directory: Role Promotion/Demotion, Account Activation/Deactivation

```
{% extends 'base.html' %}

{% block title %}Manage Users | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">User Dire...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">Students &...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-b...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <!-- Users Table -->
    <div class="bg-white rounded-3xl border border-slate-200 shadow-sm overflow-hidden">
        <div class="overflow-x-auto">
            <table class="w-full text-left border-collapse text-xs">
                <thead>
                    <tr class="bg-slate-50 border-b border-slate-200 text-slate-500 font-bol...
                        <th class="py-3.5 px-6">User Details</th>
                        <th class="py-3.5 px-4">Role</th>
                        <th class="py-3.5 px-4">Department & Sem</th>
                        <th class="py-3.5 px-4 text-center">Uploads</th>
                        <th class="py-3.5 px-4 text-center">Status</th>
                        <th class="py-3.5 px-6 text-right">Role & Account Actions</th>
                    </tr>
                </thead>
                <tbody class="divide-y divide-slate-100">
                    {% for u in users %}
                        <tr class="hover:bg-slate-50/70 transition-colors">
                            <td class="py-4 px-6 whitespace-nowrap">
                                <div class="font-bold text-slate-900">{{ u.full_name }}</div>
                                <div class="text-[11px] text-slate-400 font-mono">@{{ u.user...
                                {% if u.roll_number %}
                                    <span class="text-[10px] text-slate-500 font-mono">Roll:...
                                {% endif %}
                            </td>

                            <td class="py-4 px-4 whitespace-nowrap">
                                <span class="px-2.5 py-1 rounded-full text-[10px] font-bold ...
                                    {% if u.role == 'admin' %}bg-rose-100 text-rose-800
                                    {% elif u.role == 'faculty' %}bg-amber-100 text-amber-800
                                    {% else %}bg-blue-100 text-blue-800{% endif %}">
                                    {{ u.role }}
                                </span>
                            </td>

                            <td class="py-4 px-4 whitespace-nowrap">
                                <span class="font-bold text-slate-700">{{ u.department_code ...
                                {% if u.semester %}
                                    <span class="text-slate-400">• Sem {{ u.semester }}</span>
                                {% endif %}
                            </td>

                            <td class="py-4 px-4 text-center font-bold text-slate-700">
                                {{ u.upload_count }}
                            </td>

                            <td class="py-4 px-4 text-center whitespace-nowrap">
                                {% if u.is_active %}
                                    <span class="px-2 py-0.5 rounded-full text-[10px] font-b...
                                {% else %}
                                    <span class="px-2 py-0.5 rounded-full text-[10px] font-b...
```

```
                                {% endif %}
                            </td>

                            <td class="py-4 px-6 text-right whitespace-nowrap">
                                {% if u.id != session.get('user_id') %}
                                    <div class="flex items-center justify-end space-x-2">
                                        <!-- Role change form -->
                                        <form action="{{ url_for('admin.update_user_role', u...
                                            <select name="role" onchange="this.form.submit()...
                                                <option value="student" {% if u.role == 'stu...

                                                <option value="faculty" {% if u.role == 'fac...
                                                <option value="admin" {% if u.role == 'admin...
                                            </select>
                                        </form>

                                        <!-- Toggle Active Form -->
                                        <form action="{{ url_for('admin.toggle_user_active',...
                                            <button type="submit" class="px-2.5 py-1 rounded...
                                                {% if u.is_active %}bg-rose-50 text-rose-700...
                                                {% if u.is_active %}Deactivate{% else %}Acti...
                                            </button>
                                        </form>
                                    </div>
                                {% else %}
                                    <span class="text-[11px] text-slate-400 italic">Current ...
                                {% endif %}
                            </td>
                        </tr>
                    {% endfor %}
                </tbody>
            </table>
        </div>
    </div>

</div>
{% endblock %}
```

■ File: templates/admin/departments.html

Department CRUD: Add/Edit Academic Departments

```
{% extends 'base.html' %}

{% block title %}Manage Departments | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Academic ...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">Department...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">
        <!-- Add Department Form -->
        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
            <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Add New De...
            <form action="{{ url_for('admin.manage_departments') }}" method="POST" class="sp...
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <input type="text" name="name" required placeholder="e.g. Chemical Engin...
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <input type="text" name="code" required placeholder="e.g. CHEM" class="w...
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <input type="text" name="icon" value="fas fa-flask" class="w-full px-3 p...
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <textarea name="description" rows="3" placeholder="Brief department desc...
                </div>
                <button type="submit" class="w-full py-2.5 bg-blue-600 hover:bg-blue-700 tex...
                    Create Department
                </button>
            </form>
        </div>

        <!-- Department List -->
        <div class="lg:col-span-2 bg-white rounded-3xl border border-slate-200 shadow-sm ove...
            <div class="p-6 border-b border-slate-100 flex items-center justify-between">
                <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Active...
            </div>
            <div class="divide-y divide-slate-100 text-xs">
                {% for d in departments %}
                    <div class="p-4 flex items-center justify-between hover:bg-slate-50/70 t...
                        <div class="flex items-center space-x-3">
                            <div class="w-10 h-10 rounded-xl bg-blue-50 text-blue-600 flex i...
                                <i class="{{ d.icon }}"></i>
                            </div>
                            <div>
                                <div class="font-bold text-slate-900 text-sm">
                                    {{ d.name }} <span class="text-xs font-mono font-bold te...
                                </div>
                                <p class="text-[11px] text-slate-500 line-clamp-1 mt-0.5">{{...
                            </div>
                        </div>
                        <div class="flex items-center space-x-4 text-xs font-semibold">
                            <span class="text-slate-500">{{ d.subject_count }} subjects</span>
                            <span class="text-emerald-600">{{ d.resource_count }} resources<...
                            <a href="{{ url_for('admin.manage_subjects', dept=d.id) }}" clas...
```

```
                                <i class="fas fa-arrow-right"></i>
                            </a>
                        </div>
                    </div>
                {% endfor %}
            </div>
        </div>
    </div>

</div>

{% endblock %}
```

■ File: templates/admin/subjects.html

Subject CRUD: Add/Edit Courses Mapped to Departments and Semesters

```
{% extends 'base.html' %}

{% block title %}Manage Subjects | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Academic ...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">Subject & ...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">
        <!-- Add Subject Form -->
        <div class="bg-white rounded-3xl p-6 border border-slate-200 shadow-sm space-y-4">
            <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Add New Su...
            <form action="{{ url_for('admin.manage_subjects') }}" method="POST" class="space...
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <input type="text" name="name" required placeholder="e.g. Distributed Co...
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <input type="text" name="code" required placeholder="e.g. CS702" class="...
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <select name="department_id" required class="w-full px-3 py-2 bg-slate-5...
                        <option value="">Select Department</option>
                        {% for d in departments %}
                            <option value="{{ d.id }}" {% if selected_dept|string == d.id|st...
                        {% endfor %}
                    </select>
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <select name="semester" required class="w-full px-3 py-2 bg-slate-50 tex...
                        {% for s in range(1, 9) %}
                            <option value="{{ s }}">Semester {{ s }}</option>
                        {% endfor %}
                    </select>
                </div>
                <div>
                    <label class="block text-xs font-bold text-slate-700 uppercase tracking-...
                    <textarea name="description" rows="2" placeholder="Syllabus overview..."...
                </div>
                <button type="submit" class="w-full py-2.5 bg-blue-600 hover:bg-blue-700 tex...
                    Add Subject
                </button>
            </form>
        </div>

        <!-- Subjects List with Department Filter -->
        <div class="lg:col-span-2 bg-white rounded-3xl border border-slate-200 shadow-sm ove...
            <div class="p-6 border-b border-slate-100 flex flex-col sm:flex-row sm:items-cen...
                <h3 class="text-sm font-bold text-slate-900 uppercase tracking-wider">Subjec...

                <form action="{{ url_for('admin.manage_subjects') }}" method="GET" class="fl...
                    <select name="dept" onchange="this.form.submit()" class="py-1.5 px-3 bg-...
                        <option value="">Filter by Department (All)</option>
                        {% for d in departments %}
                            <option value="{{ d.id }}" {% if selected_dept|string == d.id|st...
```

```
                        {% endfor %}
                    </select>
                </form>
            </div>

            <div class="divide-y divide-slate-100 text-xs">
                {% for s in subjects %}
                    <div class="p-4 flex items-center justify-between hover:bg-slate-50/70 t...
                        <div>
                            <div class="font-bold text-slate-900 text-sm flex items-center s...

                                <span class="text-blue-600 font-mono font-bold">{{ s.code }}...
                                <span>{{ s.name }}</span>
                            </div>
                            <div class="text-[11px] text-slate-500 mt-0.5">
                                {{ s.department_name }} ({{ s.department_code }}) • <strong ...
                            </div>
                        </div>
                        <div class="flex items-center space-x-3">
                            <span class="px-2.5 py-1 rounded-full text-[11px] font-semibold ...
                                {{ s.resource_count }} study materials
                            </span>
                            <a href="{{ url_for('main.browse', subj=s.id) }}" class="text-bl...
                                Browse
                            </a>
                        </div>
                    </div>
                {% else %}
                    <div class="p-8 text-center text-slate-400">No subjects found for this s...
                {% endfor %}
            </div>
        </div>
    </div>

</div>
{% endblock %}
```

■ File: templates/admin/logs.html

Security Audit Trail: Real-Time Activity Log with IP Addresses and Timestamps

```
{% extends 'base.html' %}

{% block title %}Audit Logs | Academix Portal Admin{% endblock %}

{% block content %}
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8 space-y-8">

    <!-- Header & Nav Ribbon -->
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b bor...
        <div>
            <span class="text-xs font-bold text-rose-600 uppercase tracking-wider">Security ...
            <h1 class="text-2xl sm:text-3xl font-extrabold text-slate-900 mt-0.5">System Act...
        </div>

        <div class="flex items-center space-x-1 overflow-x-auto pb-2 md:pb-0 text-xs font-se...
            <a href="{{ url_for('admin.dashboard') }}" class="px-3.5 py-2 rounded-xl bg-whit...
            <a href="{{ url_for('admin.pending_queue') }}" class="px-3.5 py-2 rounded-xl bg-...
            <a href="{{ url_for('admin.manage_resources') }}" class="px-3.5 py-2 rounded-xl ...
            <a href="{{ url_for('admin.manage_users') }}" class="px-3.5 py-2 rounded-xl bg-w...
            <a href="{{ url_for('admin.manage_departments') }}" class="px-3.5 py-2 rounded-x...
            <a href="{{ url_for('admin.manage_subjects') }}" class="px-3.5 py-2 rounded-xl b...
            <a href="{{ url_for('admin.activity_logs') }}" class="px-3.5 py-2 rounded-xl bg-...
        </div>
    </div>

    <div class="bg-white rounded-3xl border border-slate-200 shadow-sm overflow-hidden">
        <div class="overflow-x-auto">
            <table class="w-full text-left border-collapse text-xs">
                <thead>
                    <tr class="bg-slate-50 border-b border-slate-200 text-slate-500 font-bol...
                        <th class="py-3.5 px-6">Timestamp</th>
                        <th class="py-3.5 px-4">Action</th>
                        <th class="py-3.5 px-4">Actor</th>
                        <th class="py-3.5 px-4">Details</th>
                        <th class="py-3.5 px-6 text-right">IP Address</th>
                    </tr>
                </thead>
                <tbody class="divide-y divide-slate-100 font-mono">
                    {% for l in logs %}
                        <tr class="hover:bg-slate-50/70 transition-colors">
                            <td class="py-3 px-6 whitespace-nowrap text-slate-500 text-[11px]">
                                {{ l.created_at|str_time }}
                            </td>
                            <td class="py-3 px-4 whitespace-nowrap">
                                <span class="px-2 py-0.5 rounded text-[10px] font-bold font-...
                                    {% if 'APPROVE' in l.action or 'REGISTER' in l.action %}...
                                    {% elif 'REJECT' in l.action or 'DELETE' in l.action %}b...
                                    {% elif 'UPLOAD' in l.action %}bg-blue-100 text-blue-800
                                    {% else %}bg-slate-100 text-slate-700{% endif %}">
                                    {{ l.action }}
                                </span>
                            </td>
                            <td class="py-3 px-4 whitespace-nowrap font-sans">
                                {% if l.username %}
                                    <span class="font-bold text-slate-800">{{ l.full_name }}...
                                    <span class="text-slate-400 text-[11px]">(@{{ l.username...
                                {% else %}
                                    <span class="text-slate-400 italic">Anonymous/System</span>
                                {% endif %}
                            </td>
                            <td class="py-3 px-4 text-slate-700 font-sans max-w-md">
                                {{ l.details }}
                            </td>
                            <td class="py-3 px-6 text-right text-slate-400 text-[11px]">
                                {{ l.ip_address }}
                            </td>
                        </tr>
                    {% endfor %}
                </tbody>
            </table>
        </div>
    </div>

</div>
{% endblock %}
```

■ File: static/js/main.js

Client JavaScript: Live Autocomplete Search, Alert Dismissals, Dynamic DOM Handlers

```
// Academix Portal - Client Side JavaScript

document.addEventListener('DOMContentLoaded', () => {
    // 1. Live Global Quick Search Autocomplete
    const searchInput = document.getElementById('global-search-input');
    const searchResults = document.getElementById('quick-search-results');
    let debounceTimer;

    if (searchInput && searchResults) {
        searchInput.addEventListener('input', () => {
            clearTimeout(debounceTimer);
            const query = searchInput.value.trim();

            if (query.length < 2) {
                searchResults.classList.add('hidden');
                searchResults.innerHTML = '';
                return;
            }

            debounceTimer = setTimeout(() => {
                fetch(`/api/quick-search?q=${encodeURIComponent(query)}`)
                    .then(res => res.json())
                    .then(data => {
                        searchResults.innerHTML = '';
                        if (data.length === 0) {
                            searchResults.innerHTML = `
                                <div class="p-3 text-xs text-slate-400 text-center">
                                    No matching study materials found
                                </div>
                            `;
                        } else {
                            data.forEach(item => {
                                const a = document.createElement('a');
                                a.href = `/resource/${item.id}`;
                                a.className = 'block p-3 hover:bg-slate-50 transition-colors';
                                a.innerHTML = `
                                    <div class="flex items-center justify-between text-[11px...
                                        <span class="font-bold text-blue-600">${item.dept} •...
                                        <span class="text-slate-400 font-mono">${item.year}<...
                                    </div>
                                    <div class="text-xs font-bold text-slate-800 line-clamp-...
                                    <div class="text-[11px] text-slate-500">${item.subject}<...
                                `;
                                searchResults.appendChild(a);
                            });
                        }
                        searchResults.classList.remove('hidden');
                    })
                    .catch(err => {
                        console.error('Quick search error:', err);
                    });
            }, 250);
        });

        // Close on outside click
        document.addEventListener('click', (e) => {
            if (!searchInput.contains(e.target) && !searchResults.contains(e.target)) {
                searchResults.classList.add('hidden');
            }
        });
    }

    // 2. Auto-dismiss alerts after 5 seconds
    const alerts = document.querySelectorAll('[role="alert"]');
    alerts.forEach(alert => {
        setTimeout(() => {
            alert.style.transition = 'opacity 0.5s ease-out, transform 0.5s ease-out';
            alert.style.opacity = '0';
            alert.style.transform = 'translateY(-10px)';
            setTimeout(() => alert.remove(), 500);
        }, 5000);
    });
});
```

■ File: static/js/sw.js

Progressive Web App (PWA) Service Worker: Offline Caching Engine

```
// Service Worker for Academix Offline Academic Portal
const CACHE_NAME = 'academix-cache-v1';
const ASSETS_TO_CACHE = [
    '/',
    '/browse',
    '/offline',
    '/static/js/main.js',
    '/static/manifest.json'
];

self.addEventListener('install', (event) => {
    event.waitUntil(
        caches.open(CACHE_NAME).then((cache) => {
            return cache.addAll(ASSETS_TO_CACHE).catch(err => {
                console.warn('Some offline assets failed to cache:', err);
            });
        })
    );
    self.skipWaiting();
});

self.addEventListener('activate', (event) => {
    event.waitUntil(
        caches.keys().then((keys) => {
            return Promise.all(
                keys.map((key) => {
                    if (key !== CACHE_NAME) {
                        return caches.delete(key);
                    }
                })
            );
        })
    );
    self.clients.claim();
});

self.addEventListener('fetch', (event) => {
    if (event.request.method !== 'GET') return;

    event.respondWith(
        fetch(event.request)
            .then((networkResponse) => {
                // Cache a copy of fresh GET responses
                if (networkResponse && networkResponse.status === 200 && networkResponse.typ...
                    const responseClone = networkResponse.clone();
                    caches.open(CACHE_NAME).then((cache) => {
                        cache.put(event.request, responseClone);
                    });
                }
                return networkResponse;
            })
            .catch(() => {
                // If offline, serve from cache, else fallback to offline page
                return caches.match(event.request).then((cachedResponse) => {
                    if (cachedResponse) {
                        return cachedResponse;
                    }
                    if (event.request.headers.get('accept').includes('text/html')) {
                        return caches.match('/offline');
                    }
                });
            })
    );
});
```

■ File: static/manifest.json

Web App Manifest for Mobile / Desktop App Installation

```
{
  "name": "Centralized College Study Material & Question Paper Portal",
  "short_name": "AcademixPortal",
  "description": "Centralized repository for college study materials, notes, assignments, an...
  "start_url": "/",
  "display": "standalone",
  "background_color": "#0f172a",
  "theme_color": "#2563eb",
  "icons": [
    {
      "src": "https://cdn-icons-png.flaticon.com/512/2997/2997295.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "https://cdn-icons-png.flaticon.com/512/2997/2997295.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

■ File: static/css/custom.css

Custom Scrollbars, Print Styles for Examination Revision, Glassmorphism

```
/* Custom styles for Academix Portal */
html {
    scroll-behavior: smooth;
}

/* Custom modern scrollbars */
::-webkit-scrollbar {
    width: 8px;
    height: 8px;
}

::-webkit-scrollbar-track {
    background: #f1f5f9;
}

::-webkit-scrollbar-thumb {
    background: #cbd5e1;
    border-radius: 4px;
}

::-webkit-scrollbar-thumb:hover {
    background: #94a3b8;
}

/* Glassmorphism helpers */
.glass-panel {
    background: rgba(255, 255, 255, 0.85);
    backdrop-filter: blur(12px);
    -webkit-backdrop-filter: blur(12px);
}

/* Print styles for offline revision */
@media print {
    header, footer, aside, .no-print {
        display: none !important;
    }
    body {
        background: white !important;
        color: black !important;
    }
}
```

■ File: requirements.txt

Project Dependencies Specification

```
1 | Flask>=3.0.0
2 | Werkzeug>=3.0.0
```

■ File: start.bat

One-Click Windows Execution Script (Auto-seeds DB and launches Flask server)

```
1 | @echo off
2 | title Centralized College Study Material & Question Paper Portal
3 | cd /d "%~dp0"
4 |
5 | echo ========================================================================
6 | echo   CENTRALIZED COLLEGE STUDY MATERIAL & QUESTION PAPER PORTAL
7 | echo ========================================================================
8 | echo.
9 |
10 | python --version >nul 2>&1
11 | if errorlevel 1 (
12 |     echo [ERROR] Python is not installed or not in system PATH!
13 |     echo Please install Python 3.9+ from https://www.python.org/
14 |     pause
15 |     exit /b 1
16 | )
17 |
18 | if not exist portal.db (
19 |     echo [INFO] First time initialization: Seeding database and sample materials...
20 |     python seed_data.py
21 |     echo [OK] Database seeded successfully!
22 | )
23 |
24 | echo.
25 | echo [STARTING] Launching Academix Portal Web Server...
26 | echo [INFO] Press CTRL+C at any time to stop the server.
27 | echo.
28 |
29 | python app.py
30 |
31 | pause
```